

Chapter 2 :

Semiconductor and PN-Junction Diode

2.1 Introduction

We people are most fortunate that we live in such an era of developed engineering and technology mainly Electrical and Electronics Engineering. In today's world, people cannot think of his life without mobile phone, television, computer and other devices that have made our life easy and enjoyable. Well, these devices are the gift of Electrical and Electronics Engineering. All of them work based on the flow of charge. Charge is found in electron, a fundamental particle of atom. A complete path of flowing charge is called circuit. There are two types of circuits – electrical circuits and electronic circuits.

2.2 Electrical and Electronic Circuits

Electrical Circuits: An electric circuit is a complete path in which charge (i.e. electrons) flows from a voltage or current source. This type of circuit (electrical circuit) consists of three elements – resistor (R), capacitor (C) and inductor (L) [Fig.2-1(a)]. They have no processing capability of signals. That is, a circuit consisting of only R , L , and C is called **electrical circuit**.

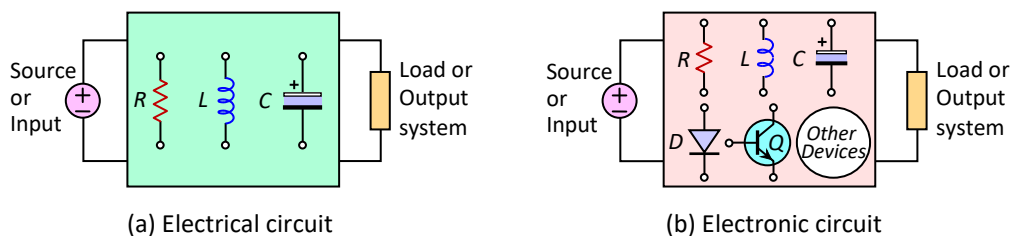


Fig.2-1: Electrical and Electronic circuits

Electronic Circuits: An electronic circuit is also a complete path in which charge (i.e. electrons) flows from a voltage or current source, but here, they have processing capability of signals. That is, electronic circuits can amplify, rectify or attenuate a signal and so on. Electronic circuits are also consisting of resistor, capacitor and inductor, but they must have at least one semiconductor device- like a diode, a transistor etc. [Fig.2-1(b)].

Whether it is electrical circuit or electronic circuit, current flows due to the movement of charge or electrons. **Sir Joseph John Thomson**, a British Physicist, discovered the electron in 1897. He found that the electron has one negative charge ($= 1.60217662 \times 10^{-19}$ Coulombs) and a mass of $9.10938356 \times 10^{-31}$ kg. We find electron in materials, i.e., in atoms. Atom is the smallest element that makes the materials. Here, we will discuss the structure of the atoms.

2.3 Atomic Structure

J. J. Thomson, who discovered the electron in 1897, proposed the **plum-pudding model** of the atom in 1904 before the discovery of the atomic nucleus. In Thomson's model, the atom is composed of electrons surrounded by a positive charge to balance the electrons' negative charge.

He compared the atomic structure with **Christmas pudding**. In an atom, the negatively charged electrons are like the raisins in the pudding and the pudding itself is like the positive charge [Fig.2-2]. He also compared an atom with a watermelon. The red fleshy part of the melon is like the positive charge while the electrons are embedded in the atom like the seeds in the melon [Fig.2-3]. The oppositely charged particles remain together by the electrical force of attraction. Further, the negative and positive charges are equal in number, making the atom electrically neutral. Thomson concluded that an atom is a positively charged sphere and electrons are set within the sphere.

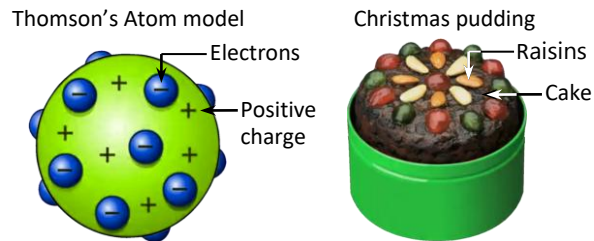


Fig.2-2: Thomson's Atom Model (Comparison with pudding)

Later in 1911, **Rutherford**, along with his assistants, **H. Geiger** and **E. Marsden**, performed the Alpha Particle scattering experiment, which led to the birth of **nuclear model** of an atom.

They took a thin gold foil having a thickness of 2.1×10^{-7} m and placed it in the centre of a rotatable detector made of zinc sulfide and a microscope [Fig.2-4]. Then, they directed a beam of 5.5 MeV (Mega electron Volt) alpha particles (α -particles) emitted from a radioactive source at the foil. Lead bricks collimated these alpha particles as they passed through them. [**Electron volt** (eV) is a unit to measure small amount of energy and discussed in Section 2.5.

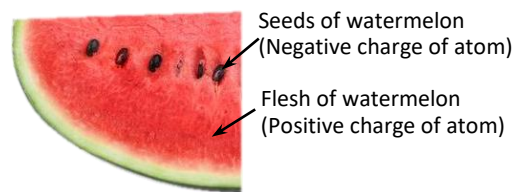


Fig.2-3: Thomson's Atom Model (Comparison with watermelon)

Rutherford had Three Observations

- Most of the fast, highly charged alpha particles went whizzing straight without being deflected. This was the expected result for all of the particles if the plum-pudding model was correct.
- Some of the alpha particles were deflected back through large angles. This was not expected.
- A very small number of alpha particles were deflected backwards! This was definitely not as expected. Rutherford later remarked "***It was as incredible as if you fired a 15 inch shell at a piece of tissue paper and it bounce back to you!***"

These observations led to many arguments and conclusions which laid down the structure of the **nuclear model** of an atom.

The results of this experiment were not in sync with the plum-pudding model of the atom as suggested by Thomson. Rutherford concluded that, since alpha particles are positively charged, for them to be deflected back, they needed a large repelling force. He further argued that for this to happen, the positive charge of the atom needs to be concentrated in the centre, unlike scattered in the earlier accepted model of Thomson.

Hence, when the incident alpha particle came very close to the positive mass in the centre of the atom, it would repel leading to a deflection. On the other hand, if it passes through a fair distance from this mass, then there would be no deflection and it would simply pass through.

He then suggested the **nuclear model** of an atom wherein the entire positive charge and most of the mass of the atom is concentrated in the centre of the atom and is called **nucleus**. Also, the electrons are moving in orbits around the nucleus similar to the planets around the sun. Further, Rutherford concluded from his experiments that the size of the nucleus is between 10^{-15}m to 10^{-14}m .

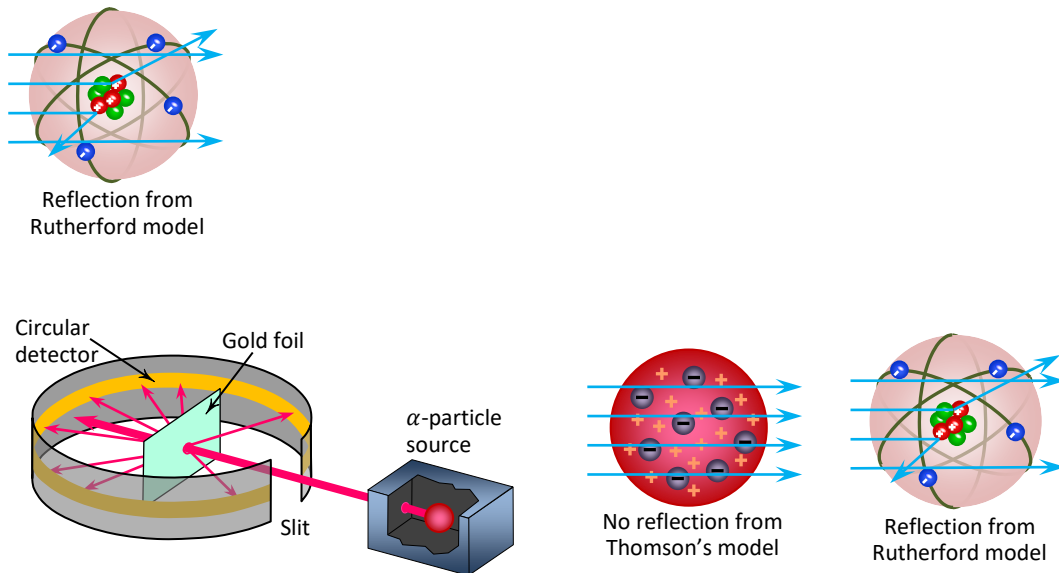


Fig.2-4: Rutherford's α -particle scattering experiment

According to **Kinetic theory**, the size of an atom is around 10^{-10}m or around 10,000 to 100,000 times the size of the nucleus proposed by Rutherford. Hence, the distance of the electrons from the nucleus should be around 10,000 to 100,000 times the size of the nucleus.

This eventually implies that most of the atom is empty space and explains why most alpha particles went right through the foil. And, some particles are deflected or scattered through a large angle on coming close to the nucleus. Also, the electrons having negligible mass, do not affect the trajectory of these incident alpha particles.

The trajectory traced by an alpha particle depends on the impact parameter of the collision. The **impact parameter** is simply the perpendicular distance of each alpha particle from the centre of the nucleus. Since in a beam all alpha particles have the same kinetic energy, the scattering of these particles depends solely on the impact parameter.

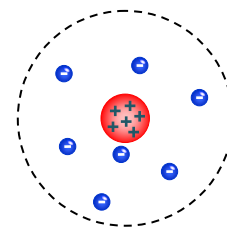


Fig.2-5: Rutherford's nuclear model of the atom

Hence, the particles with a small impact parameter or the particles closer to the nucleus, experience large angle of scattering. On the other hand, those with a large impact parameter suffer

no deflection or scattering at all. Finally, those particles having almost zero impact parameter or a head-on collision with the nucleus bounce back.

Coming to the experiment, Rutherford and his team observed that a really small fraction of the incident alpha particles was bouncing back. Hence, only a small number of particles were colliding head-on with the nucleus. This, subsequently, led them to believe that the mass of the atom is concentrated in a very small volume [Fig.2-5].

In a nutshell, Rutherford's nuclear model of the atom describes it as:

- An atom is an electrically neutral sphere with equal number of positive and negative charges.
- A small and positively charged nucleus at the centre called nucleus
- Surrounded by revolving electrons in their dynamically stable orbits but the orbits have no definite paths

The centripetal force that keeps the electrons in their orbits, against the centrifugal force, is an outcome of:

- The electrostatic force of attraction between:
 - i) The positively charged nucleus and
 - ii) The negatively charged revolving electrons.

As Rutherford model did not assume fixed definite orbital paths, this model failed to explain the electromagnetic theory, the stability of atom and the existence of definite lines in hydrogen spectrum. For these limitations of Rutherford model, **Niels Bohr** in 1922, proposed a new model (actually some corrections to Rutherford's model) of atom.

In this model, Bohr mentioned that

'the most of the atomic mass lies in the central nucleus that contains protons, and electrons are arranged in definite energy levels (i.e., fixed definite orbital paths) and revolve around the nucleus [Fig.2-6]'. The model also proposed electronic configuration, which explains the arrangement of electrons in circular orbits designated as K, L, M, N, etc. Atoms with complete electron configurations are not active. Electron configuration determines the reactivity of the atoms.

Bohr's model is able to explain the spectrum of the hydrogen atom, but it cannot fully explain the reactivity of multielectron atoms. Moreover, it has some other problems. In 1932, **James Chadwick** discovered the neutron a new elemental particle, having no charge, in the nucleus. Although the neutron solves some nuclear problems, there have some other limitations of Bohr's model. Recently many atomic models come up to overcome these limitations. But to understand the operation of semiconductor devices, Bohr's model will be sufficient.

2.3.1 The Atom in a Nutshell

The atom of any material can be thought of as consisting of a central nucleus surrounded by some orbiting electrons (Fig.2-6). It may be compared to the solar system. Nucleus is as like the sun and the orbiting electrons are like as the planets. In the solar system, the planets remain in their

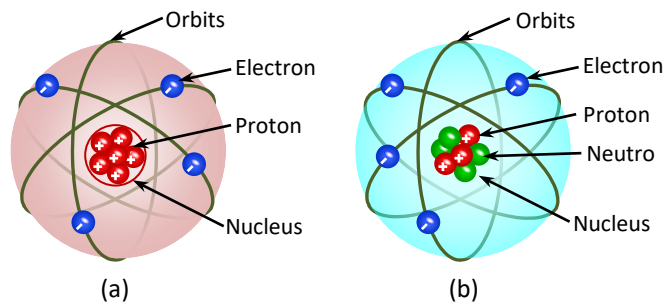


Fig.2-6: Niels Bohr's Atom Model:
(a) without neutron, and (b) With neutron

corresponding orbit by the gravitational force. Similarly in the atom, the electrons remain in their orbits by the electrostatic force of attraction between the negatively charged electrons and positively charged proton.

Each electron has a negative electrical charge of $1.60217662 \times 10^{-19}$ Coulombs (C) and a mass of $9.10938356 \times 10^{-31}$ kg. Similarly proton has same amount of positive charge but its mass is $1.6726219 \times 10^{-27}$ kg. Because unlike charges attract each other, a force of attraction exists between the oppositely charged electrons and nucleus. For the stability of the nucleus, there have some electrically neutral particles called neutrons. Neutron has same amount of mass as the proton, but not electrical charge. An atom has same number of electrons and protons. So, they are as a whole electrically neutral. The number of neutrons may differ from that of protons. The number of protons (or electrons) of an atom is called **atomic number** and the sum of the number of protons and neutrons is called **mass number**.

The paths through which the electrons move are called **orbits**. According to the Bohr's model, only those orbits, with the length of the circumference exactly equal to the integer multiple of the **de Broglie wavelength** of the electrons moving in those orbits, are possibly stable orbits. The number of electrons in the n^{th} orbit is given by $2n^2$, n being an integer. For example:

The first orbit contains $2 \times 1^2 = 2$ electrons.

The second orbit contains $2 \times 2^2 = 8$ electrons.

The third orbit contains $2 \times 3^2 = 18$ electrons and so on.

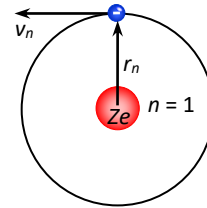


Fig.2-7: Atom model used to calculate the energy of an electron

2.3.2 The Electron at a Glance

Since electronics deals with the electrons, these small particles require detailed study and memorizing. As discussed before, an electron is a negatively charged particle having negligible mass. The important properties of the electrons are:

Charge of an electron, $e = 1.602 \times 10^{-19}$ Coulombs (C)

Mass of an electron, $m = 9.1 \times 10^{-31}$ kg

Radius of an electron, $r = 1.9 \times 10^{-15}$ m

The charge to mass ratio, $e/m = 1.76 \times 10^{11}$ C/kg

2.4 Energy of an Electron

An electron revolving around the nucleus possesses two types of energies, like **kinetic energy** due to its motion and **potential energy** due to the charge on the nucleus.

The kinetic energy of an electron in the n^{th} orbit is given by:

$$KE = \frac{1}{2} m v_n^2 \quad (2-1)$$

where, m is the mass of electron and v_n is the velocity of an electron in the n^{th} orbit. The velocity of the electron is given by:

$$v_n = \sqrt{\frac{Ze^2}{4\pi\epsilon_0 m r_n}} \quad (2-2)$$

where, Z is the atomic number, e is the charge of an electron and a proton and r_n is the radius of the n^{th} orbit.

Combining Equ.(2-1) and (2-2) we can find the kinetic energy as:

$$KE = \frac{Ze^2}{8\pi\epsilon_0 r_n} \quad (2-3)$$

Similarly, potential energy is equal to the potential due to the nucleus at a distance r_n multiplied by the charge of the electron. That is:

$$PE = (\text{Potential at a distance of } r_n) \times (-e) = \frac{Ze}{4\pi\epsilon_0 r_n} \times (-e)$$

$$\text{or,} \quad PE = -\frac{Ze^2}{4\pi\epsilon_0 r_n} \quad (2-4)$$

The total energy of an electron at the n^{th} orbit, can be found combining Equ.(2-3) and (2-4) as,

$$E_n = KE + PE = \frac{Ze^2}{8\pi\epsilon_0 r_n} - \frac{Ze^2}{4\pi\epsilon_0 r_n} = -\frac{Ze^2}{8\pi\epsilon_0 r_n} \quad (2-5)$$

The value of r_n can be found from Bohr's postulates as:

$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2} \quad (2-6)$$

where, h is Planck's constant and putting the value of r_n into Equ.(2-5), the total energy of an electron in the n^{th} orbit can be found as,

$$E_n = -\frac{Ze^2}{8\pi\epsilon_0} \times \frac{\pi m Z e^2}{n^2 h^2 \epsilon_0}$$

$$E_n = -\frac{m Z^2 e^4}{8 \epsilon_0^2 n^2 h^2} \quad (2-7)$$

The values of the parameters in Equ.(2-7) are:

- $m = 9.1 \times 10^{-31} \text{ kg}$
- $e = 1.6 \times 10^{-19} \text{ C}$
- $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$
- $h = 6.62 \times 10^{-34} \text{ J} \cdot \text{s}$

and we know, $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

Putting all these values into Equ.(2-7) we get:

$$E_n = -\frac{13.6 Z^2}{n^2} \text{ eV} \quad (2-8)$$

Thus, the energy of an electron in the hydrogen atom ($n = 1$ and $Z = 1$) is -13.6 eV . This is the minimum energy an electron can possess in hydrogen atoms. The state of this energy is called **ground state**. The energy of an electron increases as its distance from the nucleus increases (as negative energy decreases). Thus, an electron in the second orbit possesses more energy than the

electron in the first orbit. An electron in the third orbit has even higher energy than in the second orbit. It is clear that electrons in the last orbit possess very high energy as compared to the electrons in the inner orbits. These last orbit electrons play an important role in determining the physical, chemical and electrical properties of a material. These electrons are called **valence electrons**. The outer orbits of a material may be completely filled or partially filled up with electrons.

Example 2-1

The ground state energy of a hydrogen atom is -13.6 eV. Calculate the kinetic energy (KE) and potential energy (PE) of electrons in this energy state.

Solution:

Using Equ.(2-3),
$$KE = \frac{Ze^2}{8\pi\epsilon_0 r_n} = \frac{e^2}{8\pi\epsilon_0 r_n} [\because \text{for hydrogen, } Z = 1]$$

The net energy of a hydrogen electron is given by Equ.(2-5) as,

	$E_n = -\frac{Ze^2}{8\pi\epsilon_0 r_n} = -\frac{e^2}{8\pi\epsilon_0 r_n} [\because \text{for hydrogen, } Z = 1]$	(2-9)
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Comparing these two equations, we find that,

$$KE = -E_n = -(-13.6 \text{ eV}) = 13.6 \text{ eV [Ans.]}$$

Using Equ.(2-4),	$PE = -\frac{Ze^2}{4\pi\epsilon_0 r_n} = -\frac{e^2}{4\pi\epsilon_0 r_n} [Z = 1]$	(2-10)
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Comparing Equ.(2-9) and Equ.(2-10), we can easily find that,

$$PE = 2E_n = 2 \times (-13.6 \text{ eV}) = -27.2 \text{ eV [Ans.]}$$

Comments: The *PE* is negative, this energy tries to keep the electron bound to the nucleus. The *KE* is positive, i.e., it tries to make the electrons free from the nucleus. As the net energy is negative, the electron remains bound to nucleus.

2.5 eV Unit

As mentioned earlier, eV is a unit of energy that is used to measure very small amount of energy. In electronics eV is frequently used. Hence, let us explain eV a little more.

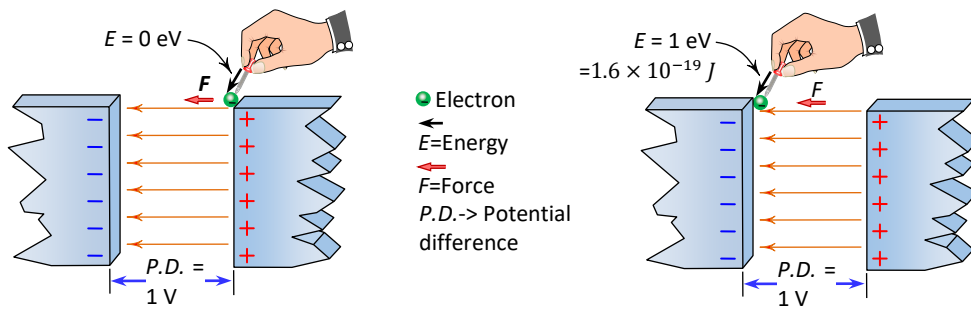


Fig.2-8: 1 eV: An electron is moved (without acceleration) across 1 V potential difference

Well, we know that the electrical power is calculated as: Power (Watt) = Potential difference (Volt) × Current (Ampere)

or,	$W = V \times I$ But, $I = Q/t$	
$\therefore$	$W = V \times \frac{Q}{t}$	(2-11)

Multiplying both sides of Equ.(2-11) by t , we get,

$$Wt = V \times Q$$

But, we know Power×time = Energy (Joule)

$$\therefore V \times Q = E \quad (2-12)$$

The significance of this equation is- we need E Joule of energy to move Q Coulomb charge from one point to another point having an electric potential difference of V Volt. Now, if we move 1 Coulomb of charge through a potential difference of 1 Volt the required energy (work) is 1 Joule ($1 \text{ V} \times 1 \text{ C} = 1 \text{ J}$). But, instead of moving 1 Coulomb, if we move an electron through a potential difference of 1 Volt the required energy is called 1 eV.

Thus,

$$1 \text{ eV} = (1 \text{ V}) \times (\text{Charge of an electron})\text{J}$$

$$\text{or,} \quad 1 \text{ eV} = (1 \text{ V}) \times (1.6022 \times 10^{-19}) \text{ J}$$

$$\therefore \quad \boxed{1 \text{ eV} = (1.6022 \times 10^{-19})\text{J}} \quad (2-13)$$

The concept of 1 eV is illustrated in Fig.2-8.

Example 2-2

At a certain time, an electron acquires a speed of 10^6 ms^{-1} . Calculate its energy in eV unit at that instant of time.

Solution:

The energy of the electron will be,

$$E = \frac{1}{2}mv^2 = \frac{1}{2}(9.1 \times 10^{-31} \text{ kg})(10^6 \text{ ms}^{-1})^2$$

$$\text{or,} \quad E = \frac{1}{2}(9.1 \times 10^{-31} \text{ kg})(10^{12} \text{ m}^2 \text{ s}^{-2})$$

$$\text{or,} \quad E = 4.55 \times 10^{-19} \text{ J}$$

From Equ.(2-13), we know, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$.

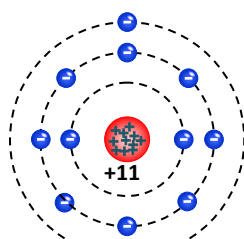
$$\therefore E = \frac{4.55 \times 10^{-19} \text{ J}}{1.6 \times 10^{-19} \text{ J/eV}} \approx 2.84 \text{ eV [Ans.]}$$

Comments: The energy of the electron is very small when measured in Joule, but it is a moderate value when measured in eV. For this reason, the eV unit is used to measure very small energy.

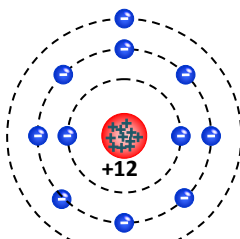
2.6 Valence Electrons

The electrons in the outermost orbit of an atom are known as valence electrons.

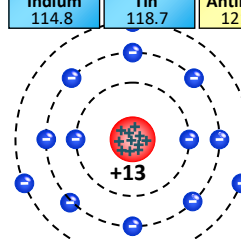
Every atom has a tendency to get eight electrons (Except H_2 and He) at its outermost orbit to get the structure of an inert gas (octet rule). The valence electrons determine the physical and chemical properties of a material. These electrons determine whether the material is chemically active or not, metal or non-metal, or, a gas or solid. These electrons also determine the electrical properties of a material.



(a) Sodium atom



(b) Magnesium atom



(c) Aluminium atom

Fig.2-9: Rutherford's nuclear model of some metal atoms

Table 2-1: A portion of the periodic table

5 B Boron 10.81	6 C Carbon 12.01	7 N Nitrogen 14.01
13 Al Aluminum 26.98	14 Si Silicon 28.09	15 P Phosphorus 30.97
31 Ga Gallium 69.72	32 Ge Germanium 72.64	33 As Arsenic 74.92
49 In Indium 114.8	50 Sn Tin 118.7	51 Sb Antimony 121.8

On the basis of electrical conductivity, materials are generally classified into conductors, insulators and semi-conductors. As a rough rule, one can determine the electrical behavior of a material from the number of valence electrons as given below:

- When the number of valence electrons of an atom is less than 4 (i.e., half of the maximum eight electrons), the material is usually a **metal** and a **conductor**. Examples are sodium (Na), magnesium (Mg) and aluminum (Al) which have 1, 2 and 3 valence electrons, respectively (See Fig.2-9).

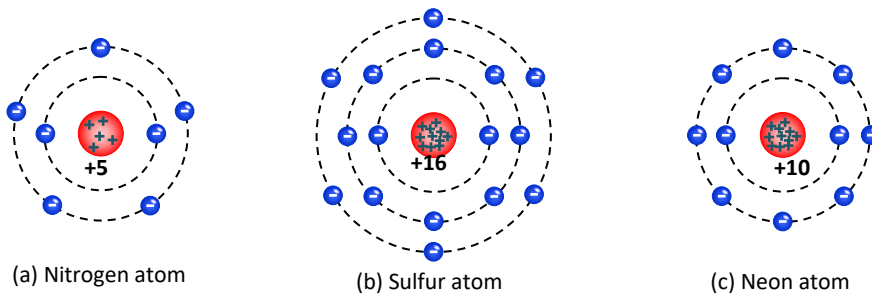


Fig.2-10: Rutherford's nuclear model of some gaseous atoms

- When the number of valence electrons of an atom is more than 4, the material is usually a **non-metal** and an **insulator**. Examples are nitrogen (N), sulfur (S), and neon (Ne) which have 5, 6, and 8 valence electrons respectively (See Fig.2-10).

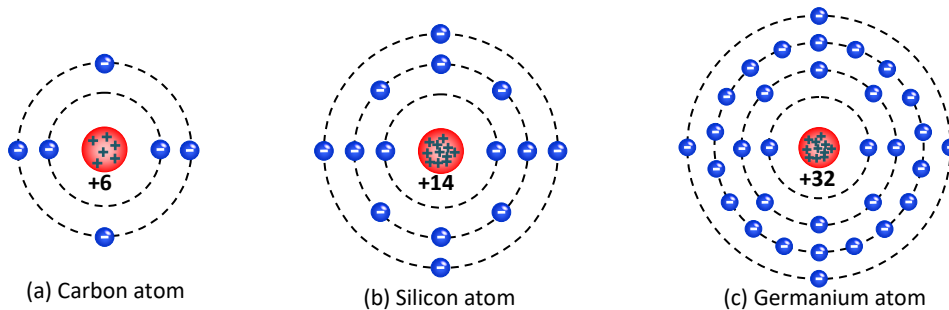


Fig.2-11: Rutherford's nuclear model of some semiconductor atoms

- When the number of valence electrons of an atom is exactly 4, (*i.e.*, exactly half of the maximum eight electrons) the materials have the properties of both the metals and non-metal and these materials are called **semiconductors**. Examples are carbon (C), silicon (Si), and germanium (Gee) all of which have 4 valence electrons (See Fig.2-11). For reference, a portion of the Periodic Table is shown in Table 2-1.

2.7 Energy Levels or States

As described earlier, according to Rutherford's atom model, electrons cloud can travel in an orbit of any radius, provided its velocity had the right value. But modern physics, tells this is not possible due to the continuous radiation of electromagnetic energy. According to Bohr's atom model, only some specific orbits are permitted from where electrons do not radiate energy (as described earlier). For instance, as shown in Fig.2-12(a), electrons can travel in the 1st, 2nd or 3rd orbits, but they cannot travel in any intermediate orbits. As described in Section 2.4, every electron in a particular orbit will have a certain fixed amount of energy given by Equ.(2-8). The energy levels of different electrons are shown in Fig.2-12. The 1st orbit represents the 1st energy level; the 2nd orbit represents the 2nd energy level, and so on. The larger the orbit of an electron, the higher is its energy level or potential energy with respect to the nucleus.

It takes energy to move an electron from a smaller orbit to a larger orbit, because work has to be done to overcome the attraction of the nucleus [Fig.2-13(a)]. The energy may come from external source, like- heat, light, or other radiation. When an electron moves from the lower energy state to higher energy state, the atom is then said to be in a state of **excitation**. Similarly when an electron moves from a higher orbit to a lower orbit it radiates the additional energy in different form like- photon or phonon as shown in Fig.2-13(b).

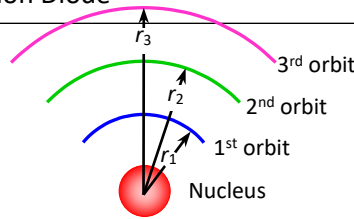


Fig.2-12: (a) Orbits of Bohr's model

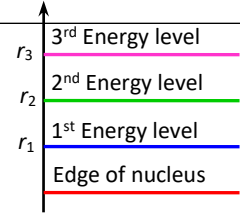


Fig.2-12: (b) Energy levels of orbits

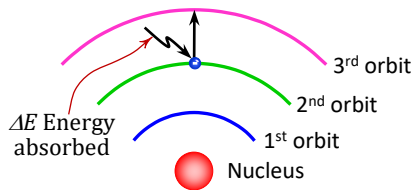


Fig.2-13: (a) Energy absorbed by an electron

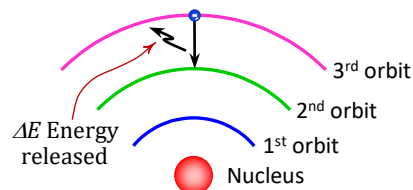


Fig.2-13: (b) Energy released by an electron

2.8 Energy Bands

The energy levels described in Section 2.7 (Fig.2-12) is possible when the atoms are isolated, i.e., the atoms are separated from one another. Let us now see what happens to the allowed energy levels in a system of atoms, arranged periodically in a crystal. The outer electrons of each atom in the crystal are now affected by the outer electrons of the neighboring atoms. The whole crystal becomes a single system in which Pauli Exclusion Principle holds: **no two electrons in the same system can have the same quantum state**. The discrete energy states of the solitary atom, therefore, split in the crystal to form a virtual band of permitted energy states. Fig.2-14 compares the solution of Schrödinger's equation for the potential field of a solitary atom and the same for a hypothetical crystal of four atoms. Fig.2-14(a) shows the three lowest states of an isolated atom. They are obtained from the wave equation solution to the potential field $V(x)$ of that atom only. If, instead of a single atom, we consider a crystal of four atoms in a line, as shown in Fig.2-14 (b), the resulting potential field is the sum of the contributions of all four atoms and this must now be used in the wave equation.

The new solutions for the allowed states $n = 1$ and $n = 2$ are essentially unchanged, since the form of $V(x)$ in their vicinity, very near the nucleus, is not affected by the neighboring atoms. But the third state $n = 3$ has an orbit far enough from the nucleus for $V(x)$ to be strongly influenced by the potential fields of the neighbors. This state can no longer be thought of as belonging to one specific

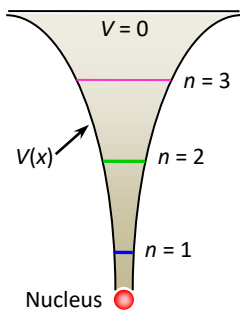


Fig.2-14: (a) Allowed states for a single isolated atom

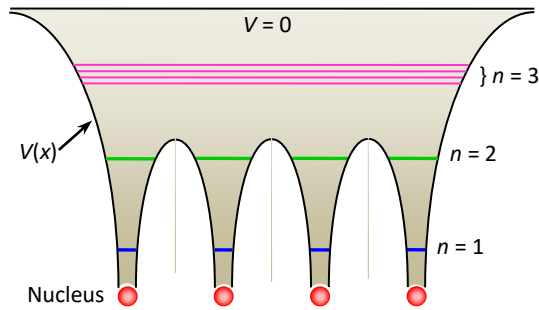


Fig.2-14: (b) Allowed states for a crystal of four atoms

atom only, but it now belongs to the whole crystal (consisting of four atoms). By Pauli's exclusion principle this level must split up into four sublevels, or four states, so that the four electrons in them, all belonging to the same system, will have different quantum numbers.

In a real crystal, containing many more atoms, that third level would have split up into a practically continuous band of allowed sublevels, and the electrons in them belong to the whole crystal and not to this or that atom. Those states are said to be **unlocalized** and called **energy band** instead of energy state. Thus for a crystal, the energy band diagram will be like Fig.2-15 instead of Fig.2-14. Among the various energy bands we are interested in the last energy band which is called **valence** band. As the electrons of a particular orbit of all atoms of the crystal form an energy band, all of the energy levels will be completely filled up by electrons. Similarly the valence band is also completely filled up by electrons. To explain the operation of electronic devices, we frequently use another energy band which is called **conduction** band. Electrons in this energy band are no longer bound to the atom, i.e., they act like free electrons. Current flows through a material with the help of these free electrons. The energy difference between the conduction band and the valence band is called **forbidden energy gap or simply Energy gap** (E_g).

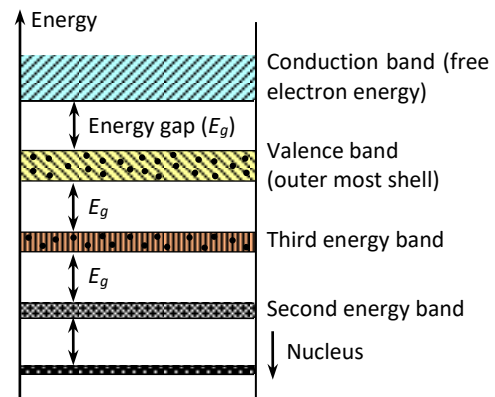


Fig.2-15: Energy band diagram of a crystal

2.9 Insulator Conductor and Semiconductor

As discussed in the previous topic the electrons in the conduction energy band are free electrons and can carry electrical current. In a material the maximum energy of an electron may be that of the valence band. That is, electrons can exist at most in the valence band. The difference between the lowest level of conduction band and the highest level of valence band is called **Energy gap** (E_g). The current carrying capacity of a material depends on the position of the valence band and the conduction band, i.e., on the value of E_g of materials. Depending on the current carrying capacity materials are divided into three categories as described below.

2.9.1 Insulator

The materials that cannot carry electricity at all are called **insulators**. The energy gaps (E_g) of these materials are very high, so no electrons can get sufficient energy from external sources (light, temperature etc.) to go to the conduction band [Fig.2-16(a)]. Generally the E_g values of these materials are greater than 5 eV (electron volt). Examples of insulators are rubber, plastic, glass, wood, cloths etc. Among these, rubbers and plastics are used as insulating cover of electrical wire.

2.9.2 Conductor

The materials that can carry electrical current easily are called **conductors**. In these materials there is no energy gap between the conduction band and the valence band. In fact, there is an overlap between the conduction band and the valence band [Fig.2-16(b)]. We know that the valence band is completely filled up by electrons. For the overlap of the conduction band and the valence band, some electrons of the valence band also possess the energy of the conduction band. These electrons behave as free electrons and can carry electrical current easily. Examples of conductors are silver, gold, copper, aluminum etc. There are many conductor materials but their current carrying capacity is not same. The current carrying capacity of a material is measured by **conductivity**. Reciprocal of conductivity is called **resistivity**. The conductivity of a material depends on the degree of overlap of the valence band and the conduction band, i.e., the number of free electrons of that material. The higher the number of free electrons the higher is the conductivity. The conductivity order of some common materials are: silver (Ag) > copper (Cu) > gold (Au) > aluminum (Al) > zinc (Zn) > nickel (Ni) > brass > bronze > iron (Fe). Although silver is the best conductor, in some sophisticated devices like microprocessor's leads gold coating is used as gold does not form oxide in any weather condition.

2.9.3 Semiconductor

In between the conductor and the insulator, there are some materials called semiconductors. They are neither good conductor nor insulator. The materials that have E_g values less than 5 eV are called **semiconductors** [Fig.2-16(c)]. At absolute zero temperature the conduction bands of these materials are completely empty, i.e., they have no free electrons. Hence, they cannot carry electrical current. But at room temperature some electrons of valence band can gain sufficient

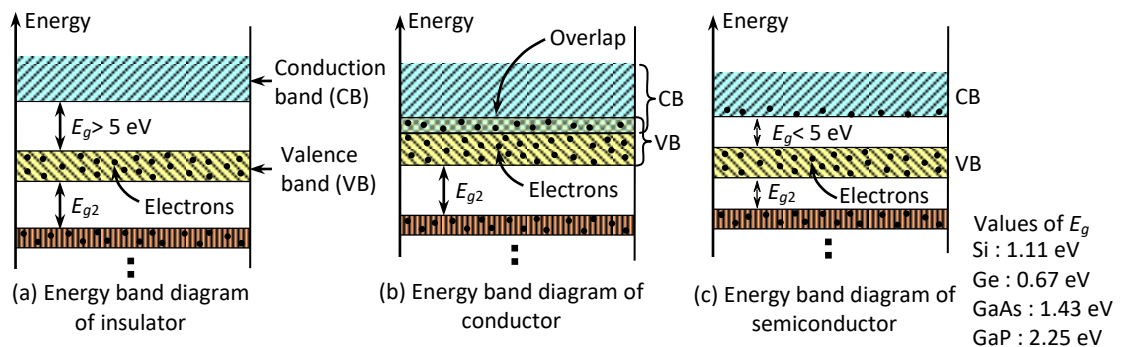


Fig.2-16: Energy band diagrams of different materials

energy to jump to the conduction band. Thus, at room temperature they have some free electrons and carry electricity, but their resistivity is very high compared to that of conductor. Examples of semiconductors are: silicon (Si), germanium (Ge), boron (B), phosphorus (P) etc. At room temperature, the values of energy gaps of some common semiconductors are given in Fig.2-16(c).

Typical resistance values for a 1-cm cubic sample of insulator, conductor and semiconductor are $10^{14}\Omega$, $10^{-6}\Omega$, and 10Ω , respectively.

Semiconductor materials are divided into two categories. These are *intrinsic* semiconductor and *extrinsic* semiconductor.

2.10 Intrinsic semiconductor

Semiconductor in its purest form is called *intrinsic* semiconductor. Semiconductors used for different purpose have different purity levels. Metallurgical-grade semiconductor has the purity level of 98 – 99%, whereas the purity level of electronic-grade silicon is as high as 99.9999%. The impurity level of the electronic-grade semiconductor will be in the order of few hundred ppm (part per million). As mentioned in the previous topic, at room temperature some electrons can absorb sufficient amount of thermal energy to move to the conduction band leaving behind a vacancy of electron which is called *hole*. Thus, an intrinsic semiconductor has some free electrons in the conduction band and an equal number of holes in the

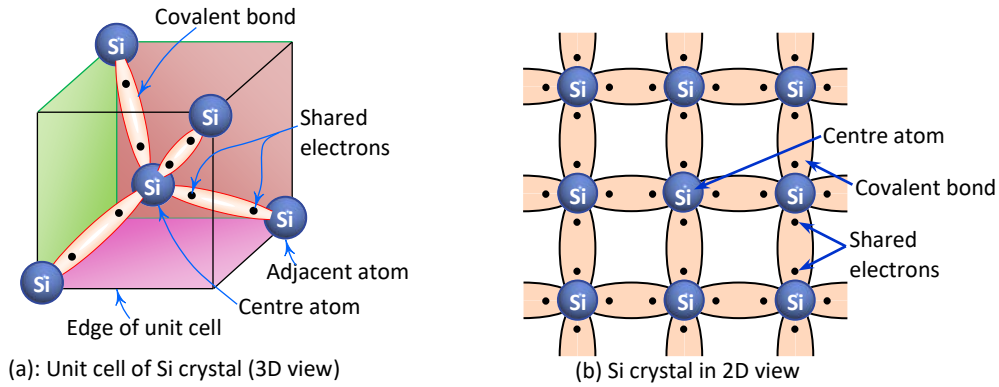


Fig.2-17: Crystalline structure of Silicon (Si)

valence band. The intrinsic carrier concentration is the number of electrons in the conduction band (represented by n_i) or the number of holes in the valence band (represented by p_i). At room temperature (300 K), the intrinsic carrier concentration (value of n_i or p_i) per cubic centimeter is 1.5×10^{10} for silicon, 2.4×10^{13} for germanium and 2.1×10^6 for gallium-arsenide. On the other hand, Cu contains 8.4×10^{22} free charge carriers (electrons) per cubic centimeter. For intrinsic semiconductor $n_i = p_i$.

Moreover, the intrinsic carrier concentration can be calculated using the following equation.

$$n_i = BT^{3/2}e^{(-E_g/2kT)} \quad (2-14)$$

where B is a coefficient related to the specific semiconductor material, E_g is the bandgap energy (in eV), T is the absolute temperature (K), k is Boltzmann's constant (86×10^{-6} eV/K), and e , in this context, represents the exponential function. The values for B and E_g for several semiconductor materials are given in Table 2-2. The band gap energy E_g and coefficient B are not strong functions of temperature. The intrinsic concentration n_i is a parameter that appears often in the current-voltage equations for semiconductor devices.

Table 2-2: Energy Gap (E_g) and Semiconductor Constant of some materials

Material	E_g (eV)	B ($\text{cm}^{-3}\text{K}^{-3/2}$)
Germanium (Ge)	0.66	1.66×10^{15}
Silicon (Si)	1.1	5.23×10^{15}
Gallium arsenide (GaAs)	1.4	2.10×10^{14}
Gallium phosphide (GaP)	2.26	? $7.30 \times 10^{-4} \text{ eV/K}$

Example 2-3

Calculate the intrinsic carrier concentration in Si at a temperature of $T = 300 \text{ K}$.

Solution:

The Semiconductor Constant (B) and Energy Gap (E_g) for Si can be found from Table 2-2, which are $5.23 \times 10^{15} \text{ cm}^{-3}\text{K}^{-3/2}$, and 1.1 eV, respectively. The value of intrinsic carrier concentration of a material at any temperature can be calculated using Equ.(2-14).

$$n_i = BT^{3/2}e^{(-E_g/2kT)} = (5.23 \times 10^{15})(300)^{3/2}e^{(-1.1)/2 \times (86 \times 10^{-6})(300)}$$

or,
$$n_i = 1.5 \times 10^{10} \text{ cm}^{-3} \text{ [Ans.]}$$

Comments: The value of intrinsic electron concentration of $1.5 \times 10^{10} \text{ cm}^{-3}$ may appear to be large, but it is relatively small compared to the concentration of silicon atoms, which is $5 \times 10^{22} \text{ cm}^{-3}$.

2.11 Fermi Level

The name Fermi Level comes from its inventor-**Enrico Fermi** who was an Italian physicist (1901-1954). In solid state physics, to explain the energy band diagrams, Fermi Level is commonly used.

The highest energy level that an electron can occupy at the absolute zero temperature is known as the **Fermi Level**. The Fermi level lies between the valence band and conduction band because at absolute zero temperature the electrons reside all in the lowest energy state. Due to the lack of sufficient energy at 0 Kelvin, the Fermi level can be considered as the sea of electrons (or fermions) above which no electron exist, that is, at 0 Kelvin there exists an energy level such that no electron will have energy more than this. The Fermi level changes as the solids are warmed and as electrons are added to or withdrawn from the solid. Fermi level can also be defined as:

The Fermi Level is the energy level in the energy band diagram for which the probability of occupancy (in other words we can say the presence of majority charge carriers) becomes half.

So according to the Fermi Energy definition, it is the highest occupied energy level of any material at absolute zero. To be more specific, all the electrons in the material will occupy the energy level at or below that material's Fermi Level at 0 K.

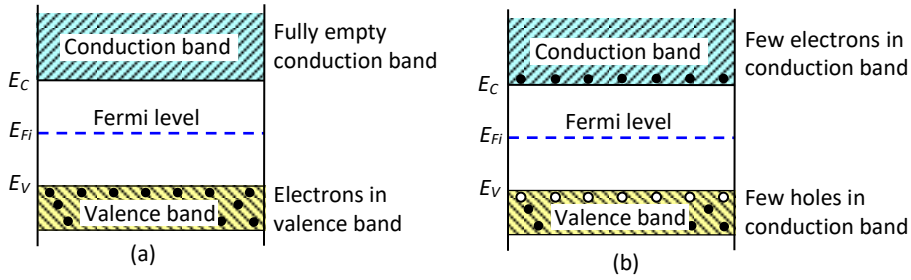


Fig.2-18: Energy band diagram of intrinsic semiconductor:
(a) at 0 K and (b) at room temperature

The carrier concentration, i.e., the free electron concentration in the conduction band of an intrinsic semiconductor is given by,

$$n_i = n_C e^{-(E_C - E_F)/kT} \quad (2-15)$$

where, n_i is the electron concentration in conduction band, E_C = the minimum energy in conduction band, E_F = Fermi energy, k = Boltzmann's constant, T = absolute temperature, and n_C = **effective density of states function** in conduction band and is given by,

$$n_C = 2 \left(\frac{2\pi m_e kT}{h^2} \right)^{\frac{3}{2}} \times (1.602 \times 10^{-19})^{\frac{3}{2}} = 2 \left(\frac{2\pi m_e k'T}{h^2} \right)^{\frac{3}{2}} \quad (2-16)$$

Here,

m_e = effective mass of electron in kg = 9.107×10^{-31} kg

k' = Boltzmann constant in J/K = $k \times (1.602 \times 10^{-19})$

For Si and Ge semiconductors, at room temperature (300 K), the values of n_C are 1.02×10^{25} and 2.8×10^{25} , respectively.

Similarly, the concentration of holes in the valence band is given by,

$$n_i = n_V e^{-(E_F - E_V)/kT} \quad (2-17)$$

Here, E_V = the maximum energy in valence band, and n_V = **effective density of states function** in valence band and is given by,

$$n_V = 2 \left(\frac{2\pi m_h k'T}{h^2} \right)^{\frac{3}{2}} \quad (2-18)$$

m_h = effective mass of holes in kg = 9.107×10^{-31} kg

The Fermi level of an intrinsic semiconductor is given by,

$$E_{Fi} = \frac{E_C + E_V}{2} - \frac{kT}{2} \ln \left(\frac{n_C}{n_V} \right) \quad (2-19)$$

If the effective mass of an electron and a hole are the same, that is, $n_C = n_V$, the above equation reduces to,

$$E_F = \frac{E_C + E_V}{2} \quad (2-20)$$

From this equation, we find that the Fermi Level lies exactly at the centre of forbidden energy gap (E_g).

2.12 Extrinsic semiconductor

Well, we all want pure things in our daily life, but to fabricate electronic devices we need impure semiconductor not pure one! To increase the conductivity of semiconductor, a few amount of impurity is mixed with the pure semiconductor. A semiconductor when mixed with some suitable impurity is called **extrinsic** semiconductor. The process of mixing the impurity is called **doping** and the impurity itself is called **dopant** atom. The order of mixed impurity is 1 to 100 ppm. These mixed impurity atoms produce some additional energy levels in the energy gap of the semiconductor and hence decrease the energy gap. Now, at room temperature, more electrons can go to the conduction band from the added impurity bands. Thus, the conductivity of the semiconductor increases.

Extrinsic semiconductors are of two types- **N-type semiconductor** and **P-type semiconductor**.

According to the **Mass Action Law**, at thermal equilibrium, at a constant temperature, the product of the number of electrons in the conduction band of a semiconductor material and the number of holes in the valence band will be constant regardless the level of doping concentration. It is expressed as,

$$n_e n_h = \text{constant} = n_i^2 (\text{function of temperature}) \quad (2-21)$$

Example2-4

A Si crystal is doped by arsenic (As) with a doping concentration of 1 ppm. If the crystal contains $5 \times 10^{28} \text{m}^{-3}$ atoms and $n_i = 1.5 \times 10^{16} \text{m}^{-3}$. Calculate the number of electrons and holes in the crystal.

Solution:

The doping concentration of semiconductor material is 1 ppm, means 1 part per million (10^6).

That is, 1 atom of As is doped in 10^6 atoms of Si. Therefore, total number of dopant atoms will be,

$$n_{As} = \frac{5 \times 10^{28} \text{m}^{-3}}{10^6} = 5 \times 10^{22} \text{m}^{-3}$$

We know, one As atom produce one free electron. Thus, the total number of electrons created by doping is,

$$n_e = 5 \times 10^{22} \text{m}^{-3} [\text{Ans.}]$$

Again we know,

$$n_e n_h = n_i^2$$

$$\text{or, } n_h = \frac{n_i^2}{n_e} = \frac{(1.5 \times 10^{16} \text{m}^{-3})^2}{5 \times 10^{22} \text{m}^{-3}} = 4.5 \times 10^9 \text{m}^{-3} [\text{Ans.}]$$

Comments: Here, $n_e > n_h$. Thus, the type of the semiconductor is N-type.

2.12.1 N-type semiconductor

Both the N-type and P-type semiconductor materials are formed by adding a suitable amount of impurity atoms into a pure semiconductor like- germanium or silicon. The added impurity is called the **dopants** or **dopant atoms**. The dopant atoms are added during the crystal fabrication process from the molten semiconductor or are introduced into the already fabricated crystal by the method of **diffusion**. The N-type is created by introducing those impurity elements that have five valence electrons (pentavalent elements), such as **phosphorus** (P), **arsenic** (As), **antimony** (Sb) and so on. The effect of added impurity elements (phosphorus in silicon crystal) is depicted in Fig.2-19. Although, a silicon crystal is a three dimensional arrangement of Si atoms as shown in Fig.2-17 (a), for simplicity it is represented as a two dimensional arrangement of Si atoms [Fig.2-19(a)]. There are five valence electrons at the outer most shell of phosphorus atoms. Note that the four valence electrons of phosphorus atom form four covalent bonds with four adjacent silicon atoms. However, the 5th electron of the impurity (P) atom remains unused, as it cannot form covalent bond. Every impurity P atoms added to the Si crystal has such an unused electron. These electrons are loosely bound to the parent atoms and can jump to the conduction band at room temperature and behave as free electrons. In this way, by adding suitable amount of pentavalent impurity atoms, the number of free electrons can be increased to a desired level. Now, as the crystal has many free electrons, it is called **N-type semiconductor**. Since the inserted impurity atom has donated a relatively free electron to the structure, the impurities with five valence electrons are called **donor atoms**.

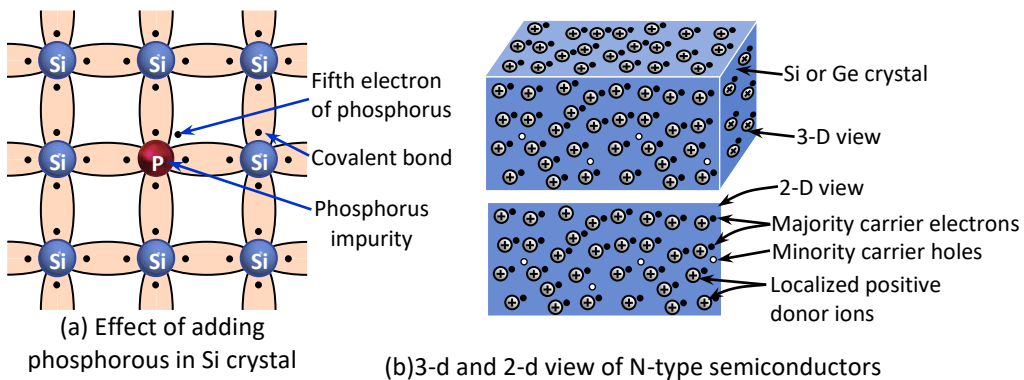


Fig.2-19:Preparation of N-type semiconductor doping with pentavalent impurities

It is important to realize that even though a large number of free carriers have been established in the N-type material, it is still electrically neutral since ideally the number of positively charged protons in the nuclei is still equal to the number of free and orbiting negatively charged electrons in the structure.

Note that a discrete energy level, called the **donor level**, is developed (in between the conduction band and the valence band) closed

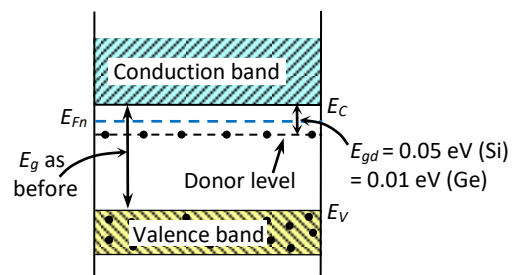


Fig.2-20: Energy band diagram of N-type semiconductor (at 0 K)

to the conduction band with an E_{gd} (energy gap of donor level) significantly less than that of the intrinsic material (E_g) as shown in Fig.2-20. Those unused electrons of the added impurity atoms sit at this energy level and can absorb sufficient thermal energy to move into the conduction band at room temperature. The result is that at room temperature, there are a large number of carriers (electrons) in the conduction band and the conductivity of the material increases significantly. Although, electrons are the majority charge carriers in N-type semiconductor, at room temperature some electrons move from the valence band to the conduction band and produce some holes in the valence band as in intrinsic semiconductors. These holes are called minority carries. At 300 K, the majority carrier (electrons) concentration is $\sim 10^{16} \text{ cm}^{-3}$ and that of minority holes is $\sim 10^4 \text{ cm}^{-3}$.

An important relationship between the electron and hole densities in most practical semiconductor materials is given by Equ.(2-21).

In summary, we can say an N-type semiconductor has electrons as majority carriers, holes as minority carriers as shown in Fig.2-19(b). As every donor atom donates one electron, electrically it becomes positive. They cannot move rather remain as **localized positive ions** in the crystal. Fermi level of the N-type semiconductor lies in between the donor energy level and the lower energy level of conduction band [Fig.2-20].

Example 2-5

At 300 K temperature, a sample of intrinsic silicon has electron and hole density each (n_i) equal to $1.5 \times 10^{10} \text{ cm}^{-3}$. One of every 1.0×10^6 atoms is replaced by a phosphorous atom. Assuming all the donor electrons are in the conduction band, calculate the ratio of the charge carrier density in the doped silicon to that for the intrinsic silicon. Given: density of silicon, $\rho = 2.3290 \text{ g/cm}^3$, Avogadro's number, $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$, and Molar mass of Si, $M = 28.09 \text{ g/mol}$.

Solution:

The number of Si atoms per cm^3 can be calculated as,

$$N = \frac{N_A \rho}{M} = \frac{6.022 \times 10^{23} \text{ mol}^{-1} \times 2.3290 \text{ g/cm}^3}{28.09 \text{ g/mol}} = 499.119 \times 10^{20} \text{ cm}^{-3}$$

Carriers density added by the phosphorous (P) atoms is,

$$n_{Pe} = \frac{499.119 \times 10^{20} \text{ cm}^{-3}}{1.0 \times 10^6} = 499.119 \times 10^{14} \text{ cm}^{-3}$$

The ratio of carrier concentration in doped Si to intrinsic Si will be,

$$\frac{n_{Pe}}{n_e + n_h} = \frac{n_{Pe}}{2n_i} = \frac{499.119 \times 10^{20} \text{ cm}^{-3}}{1.5 \times 10^{10} \text{ cm}^{-3}} = 3.32746 \times 10^{12} \text{ [Ans.]}$$

Comments: The very high value of the ratio implies that the conductivity of the extrinsic silicon has highly increased.

2.12.2 P-type semiconductor

The P-type semiconductor material is developed in the same process as the N-type semiconductor. This time a pure germanium or silicon crystal is doped with impurity atoms having three valence electrons. The elements most frequently used for this purpose are **boron** (B), **gallium** (Ga), and

indium (In). The effect of adding boron atoms in a base material of silicon is indicated in Fig.2-21. Here, the three

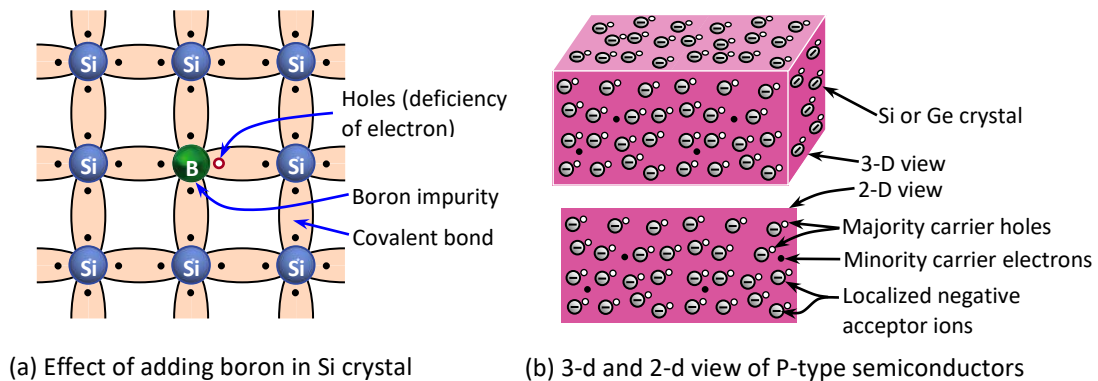


Fig.2-21: Preparation of P-type semiconductor doping with trivalent impurities

electrons of the outer most energy shell of boron form three covalent bonds with three adjacent Si atoms. But the fourth adjacent Si atom cannot form a complete covalent bond. Here the Si gives an electron but the B atom has no more electrons to share with the Si atom. Hence, this covalent bond is incomplete with the deficiency of one electron. This deficiency of electron is called a **hole** and is represented by a circle or a plus sign. If any randomly moving free electrons or even electron from other covalent bonds come close contact to this incomplete covalent bond, the hole can capture that free electron. Thus, the hole behaves like a positive charge. Every trivalent impurity will produce a hole and will capture electrons. Since, the trivalent impurities capture (accept) electrons, they are called **acceptors**. Every acceptor added in the silicon crystal introduces an energy state in the energy band diagram. This energy level is called **acceptor level**. The acceptor level is produced in between the conduction band and the valence band but close to the valence band [Fig.2-22]. The energy difference between the valence band and the acceptor level is very small and is represented by E_{ga} . At room temperature electrons can easily absorb sufficient heat energy to jump from the valence band to the acceptor level. The electrons that move to the acceptor level leave behind holes in the valence band. Thus, many holes are developed in the valence band. As holes behave like positive charges, the semiconductor is now called **P-type semiconductor**. The holes are the majority charge carriers in P-type semiconductor. As in the case of intrinsic semiconductors, some electrons will absorb sufficient thermal energy and will jump from valence band to the conduction band. These electrons are called minority charge carriers.

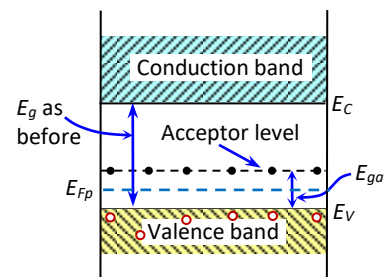


Fig.2-22: Energy band diagram of P-type semiconductor (at 0 K)

We know the impurity Boron atom is electrically neutral. When it accepts an electron it becomes electrically negative. Thus, in P-type semiconductor the acceptor atoms can be considered as negatively charged ions. As they cannot move, they are called **localized positive ions**. Although the acceptors are positive, the whole P-type material is electrically neutral, as the total number of positively charged protons and negatively charged electrons are same.

In summary, we can say a P-type semiconductor has holes as majority carriers, electrons as minority carriers and negatively charge localized acceptor ions as shown in Fig.2-21(b). The Fermi level of the P-type semiconductor lies in between the acceptor energy level and the upper level of valence band.

Example 2-6

In a process of P-type silicon preparation, trivalent impurity atoms have been mixed in such an amount that its hole density becomes 8.5×10^{21} holes/m³. The intrinsic electron density of silicon is $n_i = 1.5 \times 10^{16}$ electrons/m³. Find the (minority carrier) electron density of the developed P-type silicon sample. Also calculate the ratio of minority carrier and majority carriers.

Solution:

From Equ.(2-21) we know, $n_e \times n_h = (n_i)^2$

$$\text{or, } n_e = \frac{n_i^2}{n_h} = \frac{(1.5 \times 10^{16})^2}{8.5 \times 10^{21}} = 2.7 \times \frac{10^{10} \text{ electrons}}{\text{m}^3} \text{ [Ans.]}$$

The ratio of the minority carriers and majority carrier concentration is

$$n_e : n_h = \frac{2.7 \times 10^{10}}{8.5 \times 10^{21}} = 1 : (3.15 \times 10^{-11}) \text{ [Ans.]}$$

Comments: The concentration of minority carriers is very small compared to majority carriers.

2.13 Drift and Diffusion of Charge Carriers

Diffusion

The movement of charge carriers (in fact any particles) from higher concentration region to lower concentration region is called **diffusion**. We know that if a spoon of sugar is dropped into a glass of water, after a long time the whole water of the glass will be sweet. At first the sugar will be melt and as the concentration of sugar atom is higher at the bottom of the glass, it will move upward and gradually the sugar concentration will be same at every point of the glass water. This movement is due to the concentration gradient. Similarly, if we spray some

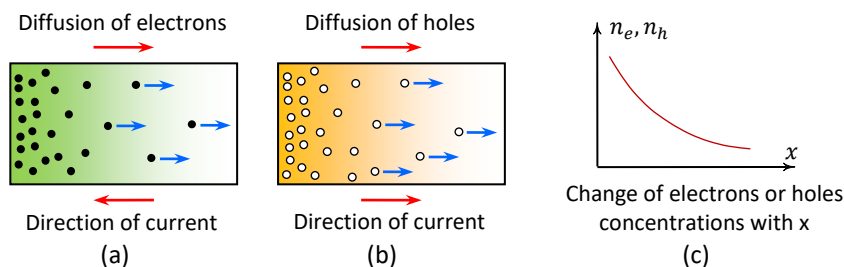


Fig.2-23: Diffusion of electrons, and holes from lower concentration to higher concentration

perfume at a corner of a closed room, after some time we can smell it at the other corners of the room. The density of perfume concentration will be higher in the air where we have sprayed it than the air of other places of the room. Due to this variation of concentration of perfume particles, they

move from higher concentration to the lower concentration region and we can smell it anywhere in the room. Although, many moveable particles can diffuse, in electronics we are highly concerned with the diffusion of holes and electrons. When electrons or holes move from one place to another by the process of diffusion electrical current is produced. This current is called **diffusion current**. The conventional current direction will be the same as the direction of holes diffusion. On the other hand, the conventional current direction will be opposite to the direction of electrons diffusion. The mechanisms are illustrated in Fig.2-23.

The diffusion current density due to the diffusion of electrons can be written as(for one dimension)

$$J_e = q_e D_e \frac{dn_e}{dx} \quad (2-22)$$

where q_e is the magnitude of the electronic charge, dn_e/dx is the gradient of the electron concentration, and D_e is the **diffusion coefficient** of electron.

Example 2-7

Calculate the diffusion current density for Si semiconductor at $T = 300$ K. The electron concentration varies linearly from $n_e = 10^{12} \text{ cm}^{-3}$ to $n_e = 10^{15} \text{ cm}^{-3}$ over a distance from $x = 0$ to $x = 5 \text{ } \mu\text{m}$. Assume that the diffusion constant for electrons is $D_e = 30 \text{ cm}^2/\text{s}$.

Solution:

Using Equ.(2-22),
$$J_e = q_e D_e \frac{dn_e}{dx} = (1.6 \times 10^{-19})(30) \left(\frac{10^{15} - 10^{12}}{5 \times 10^{-4} - 0} \right)$$

or,
$$J_e = (48 \times 10^{-19})(199.8 \times 10^{16}) \approx 9.6 \text{ A/cm}^2 \text{ [Ans.]}$$

Comments: In this calculation, all distances have been converted into cm.

Diffusion current densities can be as high as few hundred amperes per square centimeter in semiconductor materials.

The variations of electrons and holes are shown in Fig.2-23(a,b) and Fig.2-23(c) shows the graph of the concentration variation for both electrons and holes which are functions of distance (x). The diffusion of holes from a high-concentration region to a low-concentration region produces a flow of holes in the negative x direction. (Conventional current is in the direction of the flow of positive charge.). The diffusion current density due to the diffusion of holes can be written as (for one dimension),

$$J_h = -q_h D_h \frac{dn_h}{dx} \quad (2-23)$$

Where, q_h is the magnitude of the hole charge (same as electron but +ve), dn_h/dx is the gradient of the hole concentration, and D_h is the **diffusion coefficient** of holes. Note the change in sign between the two diffusion current equations. This change in sign, in Equ.(2-23), is due to the difference in sign of the electronic charge between electrons and the positively charged holes. In case of holes, the $-ve$ sign indicates that the conventional current flows from higher density to lower density

$(-dn_h/dx)$. Similarly, electrons flow from higher density to lower density but the current flows to the opposite direction.

Drift

The movement of charge carriers (in fact any particles) with the help of any force is called **drift**. We know that if a free electron is placed in an electric field, it moves to opposite direction of the electric field due to the force exerted by the electric field on the electron. Similarly, if a positive charge (E.g., holes) is placed in an electric field, it moves to the same direction of the electric field due to the force produced by the electric field on the holes. This type of motion of electrons and holes is called **drift** and the current produced by this process is called **drift current**.

There is a relation between the electric field (E) and the potential difference (V) [potential difference will be sometimes referred as simply **potential**]. If the distance between two points in a uniform electric field is d and the potential difference

between the points is V then, $E = \frac{V}{d}$ or, $Ed = V$. It means that if a potential difference (or simply voltage) is applied in a device an electric field will also be applied. Thus, if a voltage is applied across a material that has free electrons and/or holes, drift current will be produced due to the motion of the electrons and/or holes. Drift can take place from lower concentration of carriers to higher concentration [Fig.2-24].

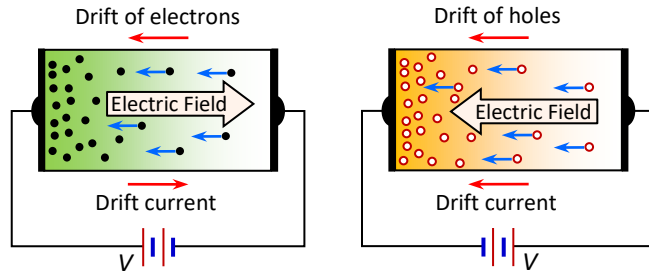


Fig.2-24: Drift of electrons and holes due to the electric field or voltage

Although, both electrons and holes can move with the help of applied field, their moving speeds are not same. The moving speed is represented by **mobility** (μ) and is defined as the distance traveled by holes or electrons per unit time and unit field-intensity. As the unit of electric field intensity is V/m , the unit of mobility is $\frac{m/s}{V/m} = m^2/(V \cdot s)$. In silicon, the mobility of electron is $\mu_e = 0.14 m^2/(V \cdot s)$ and the mobility of hole is $\mu_h = 0.05 m^2/(V \cdot s)$. That is, the mobility of electrons is higher than that of holes. Using this mobility, the drift velocity of electrons and holes can be represented as,

$$v_{de} = -\mu_e E \quad (2-24)$$

and,

$$v_{dh} = \mu_h E \quad (2-25)$$

The $-ve$ sign indicates the movement of electrons to opposite direction of electric field, E . Similarly, the $+ve$ sign in Equ.(2-25), indicates that the holes move in the same direction of E .

The resistivity of semiconductor materials depends on the concentrations of carriers, and their mobilities. The conductivity (σ) can be calculated using the following equation.

$$\sigma = n_e \mu_e q_e + n_h \mu_h q_h \quad (2-26)$$

Here, n_e and n_h is the concentration of electrons and holes, respectively.

2.14 Mechanism of Holes Movement

Electrons are very small particles and we easily understand the movement of electrons. As holes are not physical entities their movement is not the same as that of electrons. To understand the movement of holes, let us consider a semiconductor material that has higher hole concentration at the left-hand side as shown in Fig.2-25. Although there are many holes in this semiconductor slab, for simplicity only one hole is shown at point x . We know that the electrons of the covalent bonds of materials are not stationary rather they oscillate. With the increase of temperature the degree of oscillation also increases. Due to this oscillation, if an electron (at point a) comes close to the hole at point x , it will be captured by the hole. But the covalent bond at point a will lose an electron. This incomplete covalent bond will behave as a hole.

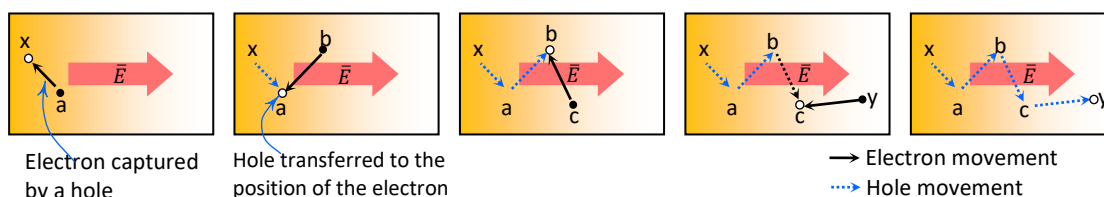


Fig.2-25: Mechanism of hole movement

We can say that the hole has moved from point x to point a . Again, the hole can capture an electron from point b and the hole will be transferred to position b . As shown in the figure the process will be continued and the hole will move from point x to point y . Since the point x is at higher concentration region of holes and y is at lower concentration end, the hole movement will continue until the density becomes uniform. Here, the movement of holes takes place from left to right or from higher concentration to the lower concentration. Even in a material having uniform holes and electrons concentration random movements of holes and electrons will occur. But in this case the net holes and electrons transfer will be zero.

Example 2-8

Suppose a silicon bar is made using the P-type silicon sample of Example 2-6. If the mobilities of electrons and holes are $\mu_e = 0.14 \text{ m}^2/(\text{V} \cdot \text{s})$ and $\mu_h = 0.05 \text{ m}^2/(\text{V} \cdot \text{s})$, respectively, calculate the conductivity and resistivity of the silicon bar.

Solution:

Conductivity depends on the carrier concentrations and the mobility of carriers. The conductivity of the P-type silicon can be calculated using Equ.(2-26).

	$\sigma = n_e \mu_e q_e + n_h \mu_h q_h$	
--	--	--

From Example 2-6, we get $n_e = 2.7 \times 10^{10} \text{ electrons/m}^3$ and $n_h = 8.5 \times 10^{21} \text{ holes/m}^3$

$\therefore$	$\sigma = (2.7 \times 10^{10})(0.14)(1.6 \times 10^{-19}) + (8.5 \times 10^{21})(0.05)(1.6 \times 10^{-19})$	
$or,$	$\sigma = 6.1 \times 10^{-10} + 68 \approx 68 \text{ S/m(Ans.)}$	

The resistivity of this bar will be,

$$\rho = \frac{1}{\sigma} = \frac{1}{68} \Omega \cdot \text{m} = \frac{100}{68} \Omega \cdot \text{cm} = 1.4 \Omega \cdot \text{cm} \text{ (Ans.)}$$

Comments: The resistivity of this Si is very large compared to that of Cu ($1.72 \mu\Omega \cdot \text{cm}$).

2.15 PN Junction Diode

In Section 2-12, we have discussed how P-type and N-type semiconductor is developed. What are inside a P-type and an N-type semiconductors are shown in Fig.2-19 and Fig.2-21, respectively. Now if one half of a semiconductor slab is made P-type and the other half is made N-type or some portion of a semiconductor is made P-type and the rest of it is made N-type a PN junction will be developed (Fig.2-26). But if we electrically connect a P-type semiconductor with an N-type semiconductor PN junction will not be produced due to the crystal mismatch. Some changes take place at the interface of P-type and N-type material and a depletion layer and a barrier voltage is developed.

2.15.1 Development of Depletion Layer and Barrier Voltage

As shown in Fig.2-26, there are majority of holes in the P-type semiconductor and majority of electrons in the N-type semiconductor. Due to this concentration variation, electrons diffuse from N-type to P-type and holes diffuse to N-type. If an electron comes to P-type material it will neutralize a hole, similarly if a hole moves to N-type it will neutralize an electron. In this way, due to the neutralization of holes and electrons a region will be developed that has no free carrier as shown in Fig.2-27. This region at the interface of P-type and N-type material is called **depletion region** or **depletion layer** as the region has been depleted off

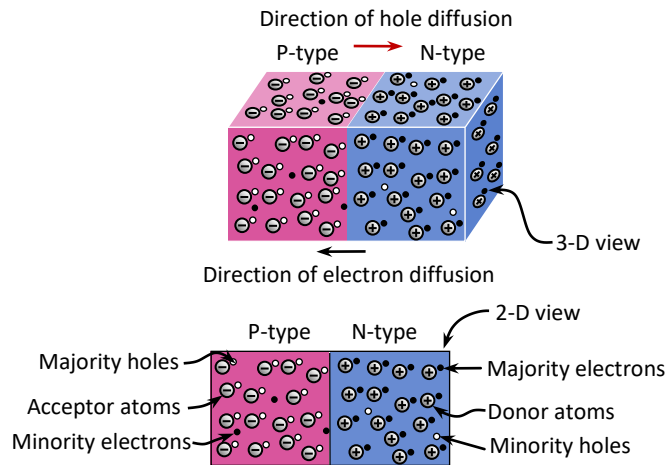


Fig.2-26: PN-junction before equilibrium

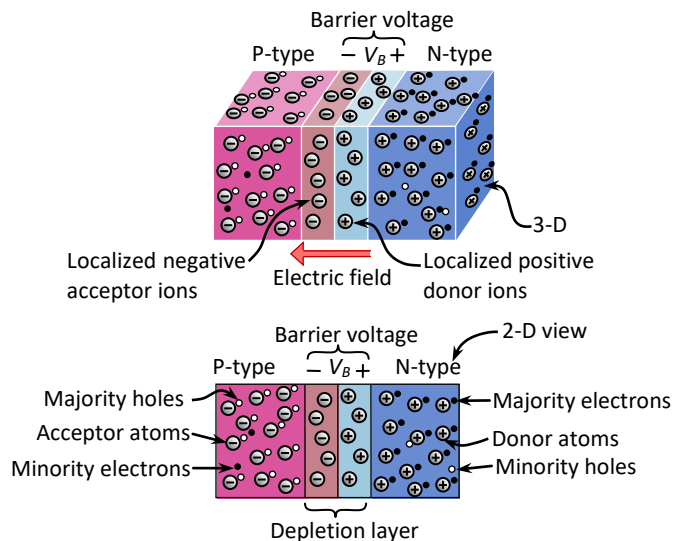


Fig.2-27: PN-junction at equilibrium (Barrier voltage and depletion layer are formed by diffusion of carriers)

free charge carriers. This region is also called the **space charge region**.

Note that there are localized acceptor ions in the depletion layer of P-type and there are localized donor ions in the N-type material. Since, acceptor ions are negative and the donor ions are positive an **electric field** will be developed in the depletion layer and its direction will be from positive to negative, i.e., from N-type to P-type as shown in Fig.2-27. The direction of the electric field will be such that it will oppose the diffusion of majority carriers. At a certain value of electric field strength the diffusion of the majority carriers will stop and we say that the PN junction has reached to equilibrium. In other way, we can say that, the negative ions in the P-type and the positive ions in the N-type will produce a potential difference. Actually this potential difference will produce the electric field. As this potential difference opposes the flow of majority carriers it is called **barrier voltage** (V_B). For silicon PN junction, the typical value of the barrier voltage is 0.7 V and for the germanium PN junction its value is 0.3 V.

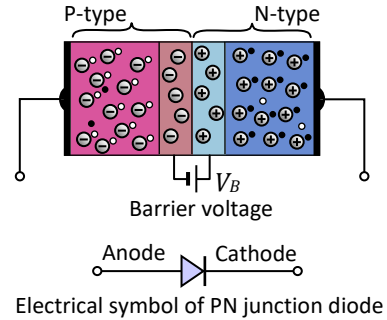


Fig.2-28: PN junction

The positively charged region and the negatively charged region comprise the space-charge region, or depletion region, of the PN junction, in which there are essentially no mobile electrons or holes. Because of the electric field in the space charge region, there is a potential difference across that region (Fig.2-28). This potential difference is also called the built-in potential barrier, or built-in voltage, and is given by,

$$V_B = \frac{kT}{q_e} \ln \left(\frac{N_a N_d}{n_i^2} \right) = V_T \ln \left(\frac{N_a N_d}{n_i^2} \right) \quad (2-27)$$

where $V_T \equiv kT/e$, k = Boltzmann's constant, T = absolute temperature, e = the magnitude of the electronic charge, and N_a and N_d are the net acceptor and donor concentrations in the P- and N-regions, respectively. The parameter V_T is called the **thermal voltage** and is approximately $V_T = 0.026$ V at room temperature, $T = 300$ K.

When metal contacts are made to the P-type and N-type semiconductor as shown in Fig.2-28, a PN junction diode or simply a semiconductor diode is produced. The electrical symbol of a diode is shown in the same figure. The terminal connected to P-type material is called **anode** and that connected to N-type material terminal is called **cathode**. The arrow-head in the symbol indicates the direction of the conventional current flow and will be described later.

Example 2-9

Calculate the potential barrier of a silicon PN junction, at junction temperature $T = 300$ K. Consider that the doping concentration in P-type region is $N_a = 10^{16} \text{ cm}^{-3}$ and that in N-type region is $N_d = 10^{17} \text{ cm}^{-3}$ and the intrinsic carrier concentration in silicon is $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$.

Solution:

Using Equ.(2-27),
$$V_B = V_T \ln \left(\frac{N_a N_d}{n_i^2} \right) = 26 \times 10^{-3} \text{V} \ln \left[\frac{(10^{16} \text{cm}^{-3})(10^{17} \text{cm}^{-3})}{(1.5 \times 10^{10} \text{cm}^{-3})^2} \right]$$

or,
$$V_B = 26 \times 10^{-3} \text{V} \ln \left[\frac{10^{13}}{1.5 \times 1.5} \right] \approx 757.12 \times 10^{-3} \text{V} = 0.757 \text{V} [\text{Ans.}]$$

Comment: Because of the $\ln$ function, the magnitude of V_B is not a strong function of the doping concentrations. Therefore, this calculated value may vary by a small amount (0.1 to 0.2 V).

2.15.2 PN Junction at Equilibrium

When the barrier voltage (V_B) of a PN junction becomes 0.7 V (for Si) or 0.3 V (for Ge) the diffusion of majority carriers stops and the junction comes to an electrical equilibrium condition. However, at room temperature, the movements of carriers through the junction do not completely stop. We know that, at room temperature, there are some minority carrier electrons in the P-type semiconductor and some minority carrier holes in the N-type semiconductor. Although the electric field, produced by the barrier voltage, opposes the flow of majority carriers, it will force minority carriers to move through the PN junction.

At room temperature, both the majority and minority carriers produce random motions. Hence, if a minority carrier electron of P-type semiconductor comes within the depletion region (where the electric field exists), due to random movement, the electric field will push it to the N-side of the PN junction. As the electron moves with the help of the force of electric field, it is called drift. Similarly the minority carriers of N-type material drift to the P-side. In this way, two drift currents will be produced (Fig.2-29). Since, the direction of current is just opposite to the direction of electron flow, the two drift currents will add up and will produce a resultant **drift current** as shown by a thicker arrow in the figure.

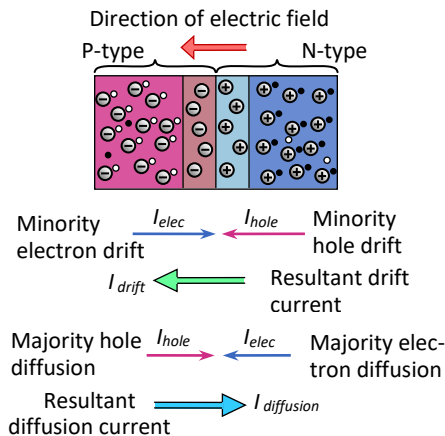


Fig.2-29: Current components of PN junction in equilibrium

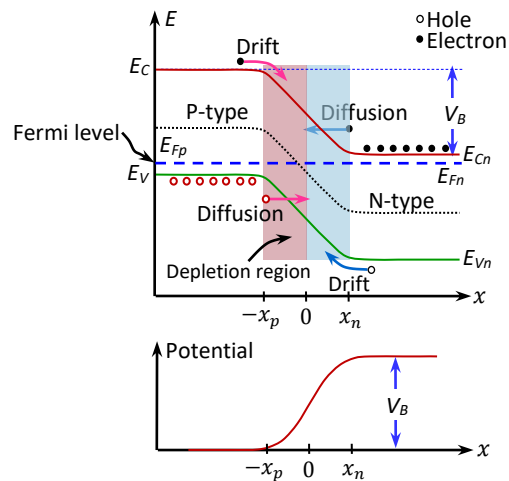


Fig.2-30: Energy band diagram of PN junction

As, due to the drift of minority carriers, the number of majority carriers has increased (both in P-side and N-side) than that without drift, now the diffusion pressure will increase. Hence, majority carrier holes will diffuse from P-type material to the N-type material and the majority carrier electrons will diffuse from N-type material to P-type material. In this way, two diffusion currents will be developed. As the direction of current will be opposite to the direction of electron diffusion, the two diffusion currents will add up and will produce a resultant current which is called **diffusion current**. As shown in Fig.2-29, at equilibrium condition, the magnitudes of the drift current and the diffusion current will be same. As the direction of drift current is just opposite to the direction of diffusion current, the net current through the PN-junction will be zero at equilibrium.

Thus we can say, in the absence of any applied bias voltage across a semiconductor diode, the net flow of charge is zero or the net current is zero.

Fig.2-30 shows the energy band diagram and the potential difference graphically. As discussed before, the Fermi level lies near the conduction band in N-type material, while in P-type material, it lies near the valence band. But in a PN junction, the Fermi level must lie in the same level. To adjust the Fermi level in the same energy, the every levels in N-type materials move down and bend. This bending to the energy level, results in the barrier voltage.

2.16 I-V Characteristic of PN Junction Diode

We have already discussed the behavior of a PN junction diode with no bias condition (i.e., at equilibrium). What is biasing? **Biasing** is the process of applying an external potential (voltage) to a device. The voltage applied for biasing is called **bias voltage**. No net current will flow in any direction of the diode without bias. The *I-V* characteristic of a diode is described in two conditions. One is forward biased condition and the other is reverse biased condition.

2.16.1 Characteristics in Reverse Biased Condition

When an external source of potential (i.e., a battery) of V_D Volts is connected across a PN junction diode such that the positive terminal of the battery is connected to the N-type material and the negative terminal is connected

to the P-type material of the diode as shown in Fig.2-31, the situation is called **reverse bias**. The number of positive ions in N-type material of the depletion region will increase due to the large number of free electrons drawn by the positive terminal of the applied potential. For similar reasons, the number of negative ions will increase in the P-type material. The net effect, therefore, is that the width of the depletion region will

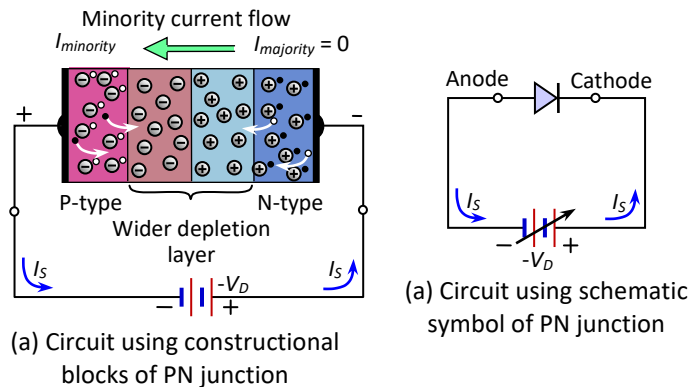


Fig.2-31: Reverse biased PN junction diode

increase. This widening of the depletion region will establish a greater barrier for the majority carriers to overcome, effectively reducing the majority carrier flow to zero, as shown in Fig.2-31. The drift of the minority carriers, however, will continue. As the number of minority carriers is very small both in the P-type and N-type materials, the drift current will saturate for a very small value

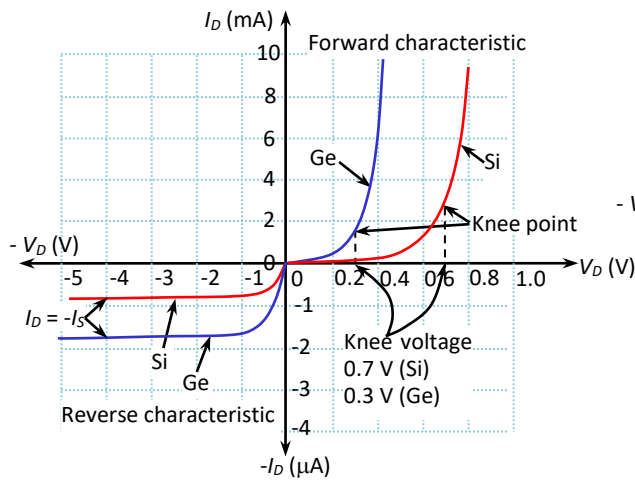


Fig.2-32: I - V Characteristic curve of PN junction diode

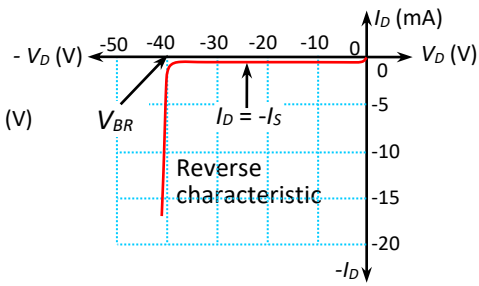


Fig.2-33: Breakdown voltage of reverse characteristic

of reverse bias voltage as shown in Fig.2-32. The current that exists under reverse-bias condition is called **reverse current** and is represented by I_R . Actually, the reverse current is the sum of the **reverse saturation current** (I_S) and the **surface leakage current** (I_{SL}). As the value of I_{SL} is very small, in most cases, we approximate $I_R \approx I_S$. The reverse saturation current increases with the increase of thermal energy or external light energy. But as the diode consists of a PN junction covered in a package, the junction of the diode always remains in dark condition. That's why I_S is also called **dark saturation current**. The value of I_S depends on the temperature. At 25 °C the value of a low power diode is typically 1 μ A. The characteristic curve of the diode under reverse biased condition is shown on the third quadrant of the graph of Fig.2-32. If we increase the reverse bias voltage the value of I_S remains constant, as the number of minority carriers is constant at a fixed temperature. For a very large value of reverse voltage, which is beyond the tolerance voltage of the diode, the reverse current increases suddenly and if the current is not controlled, the diode may be destroyed. The value of the reverse bias voltage for which the current increases rapidly is called **reverse breakdown voltage** V_{BR} as shown in Fig.2-33. This breakdown takes place due to a mechanism called **Zener breakdown** and is described in Zener Diode section.

2.16.2 Characteristics in Forward Biased Condition

A forward bias (or ON condition of diode) is established by applying the positive potential to the P-type material and the negative potential to the N-type material, i.e., by connecting the positive terminal of the battery to the anode and the negative terminal to the cathode of the diode as shown in Fig.2-34. The application of a forward bias potential V_D will pressure electrons in the N-type material and holes in the P-type material to recombine with the ions near the boundary and reduce the width of the depletion region as shown in Fig.2-34. The minority-

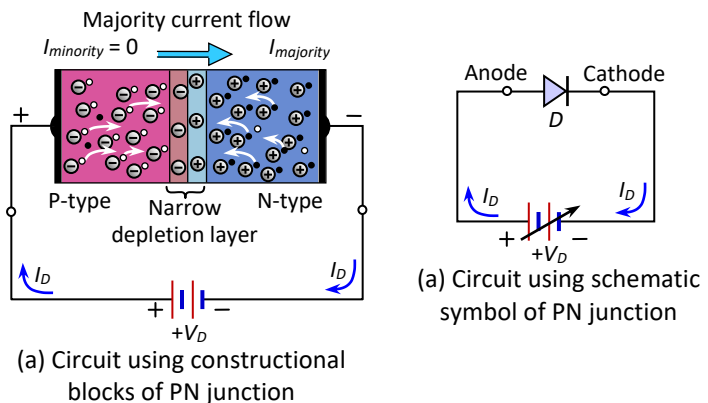


Fig.2-34 Forward biased PN junction diode

carrier (electrons) flow from P-type material to N-type material (and of holes from N-type material to P-type material) will decrease with the increase in applied forward bias voltage. As the resistance of the depletion layer is the maximum the applied voltage will almost completely drop across the depletion layer. This forward voltage will produce an electric field which is in the opposite direction of original electric field of the barrier voltage. Thus, the resultant electric field will decrease and the current due to minority carriers (minority drift current) will decrease and the majority current (diffuse current) will increase. If the applied forward bias voltage is equal to the barrier voltage V_B , the resultant field in the depletion layer will be zero. Now the minority carrier current will be zero and the current due to the majority carriers will be high. In the forward bias condition, therefore, the forward current increases rapidly when the applied voltage is near about the barrier voltage V_B . This point of the I - V curve where the forward current increases rapidly is called **knee point** and the applied voltage corresponding to this point is called **knee voltage**. After the knee point the current will increase rapidly with a very small increase in the forward voltage. In fact, the voltage across a forward-biased diode will be less than 1 V.

From the theory of solid-state physics, the general characteristics of a semiconductor diode can be defined by the following equation [Equ.2-29]. This equation is referred to as **Shockley's equation**.

$$I_D = I_S \left(e^{\frac{q_e V_D}{nkT}} - 1 \right) \quad (2-28)$$

$$\text{or,} \quad I_D = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right) \quad (2-29)$$

where,

I_S = reverse saturation current or dark saturation current

q_e = magnitude of charge of an electron = 1.6×10^{-19} C

V_D = applied forward bias voltage to the diode

n = diode ideality factor, which is a function of the operating conditions and physical construction; its value has the range between 1 and 2 depending on a wide variety of factors (for simplicity, in this book, we will consider $n = 1$ unless otherwise stated).

k = Boltzmann's constant = 1.38×10^{-23} J/K

T = Absolute temperature

$V_T = kT/q_e$, which is called thermal voltage. At 27° C or 300 K the value of V_T is ≈ 26 mV.

Example2-10

APN junction diode is operated at a temperature of $T = 300$ K. Assuming $I_S = 10^{-14}$ A, and $n = 1$, determine the diode current for i) $V_D = +0.72$ V and ii) $V_D = -0.72$ V.

Solution:

i) For $V_D = +0.72$ V the diode is forward biased. Using the Shockley's equation [Equ.(2-29)],

$$I_D = I_S [e^{V_D/nV_T} - 1] = 10^{-14} \text{ A} [e^{0.72 \text{ V}/0.026 \text{ V}} - 1]$$

$$\text{or,} \quad I_D = 10^{-14} \text{ A} \times [e^{27.69} - 1] \approx 9.63 \text{ mA [Ans.]}$$

ii) For $V_D = -0.72$ V the diode is reverse biased. Using the same equation,

$$I_D = I_S [e^{V_D/nV_T} - 1] = 10^{-14} \text{ A} \times [e^{-0.72 \text{ V}/0.026 \text{ V}} - 1]$$

or,
$$I_D = 10^{-14} \text{ A} \times [e^{-27.69} - 1] \approx -10^{-14} \text{ A [Ans.]}$$

Comments: Although I_S is quite small, even a relatively small value of forward-bias voltage can induce a moderate junction current. With a reverse-bias voltage, the junction current is virtually zero.

Example 2-11

Determine the diode voltage (V_D) and current (I_D) for the circuit shown in Fig. 2-35. Consider a diode with a given reverse-saturation current of $I_S = 10^{-13} \text{ A}$.

Solution:

Using KVL,
$$V_S = I_D R + V_D$$

Using Equ.(2-29),
$$12 \text{ V} = I_S [e^{V_D/V_T} - 1] R + V_D$$

or,
$$12 \text{ V} = (10^{-13} \text{ A}) [e^{V_D/0.026 \text{ V}} - 1] (4.7 \times 10^3 \Omega) + V_D \quad (2-30)$$

This expression contains only one unknown variable (V_D). Due to the exponential part and the presence of V_D on both sides, it will be difficult to solve this equation analytically. Let's solve this equation using iteration method.

If we first try $V_D = 0.60 \text{ V}$, the right side of Equ.(2-30) is 5.5463 V, so the equation is not balanced and the value is lower than 12 V. If we next try $V_D = 0.65 \text{ V}$, the right side of Equ.(2-30) is 34.4923 V which is very larger than 12 V. Now let's try $V_D = 0.62 \text{ V}$, the result is 11.29458 V. So the result should be little higher than $V_D = 0.62 \text{ V}$. If we continue refining our guesses, we will be able to show that, when $V_D = 0.62166 \text{ V}$, the right side of Equ.(2-30) is 12.00 V, which is essentially equal to the value of the left side of the equation. Thus, the exact voltage drop across the diode is: $V_D = 0.62166 \text{ V [Ans.]}$

The current in the circuit can then be determined by dividing the voltage difference across the resistor by its resistance as,

$$I_D = \frac{V_S - V_D}{R} = \frac{12 \text{ V} - 0.62166 \text{ V}}{4.7 \times 10^3} = \frac{11.37834 \text{ V}}{4.7 \times 10^3} = 2.4209 \text{ mA [Ans.]}$$

Comment: Once the diode voltage is known, the current can also be determined from the ideal diode equation [Equ.(2-29)]. However, dividing the voltage difference across a resistor by the resistance is usually easier, and this approach is used extensively in the analysis of diode and transistor circuits.

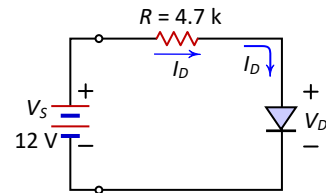


Fig. 2-35: Circuit for Example 2-11

2.17 Effect of Temperature on the Performance of Diode

The values of different parameters of semiconductor devices are mentioned at the temperature of 25 °C. But the devices are not always operated at 25 °C. The operating temperature may be higher

or lower than 25 °C. This increase and decrease in temperature affects various parameters of the semiconductor devices. The mostly affected diode parameters are discussed below.

2.17.1 Effect of Temperature on Barrier Voltage

The characteristics of a semiconductor diode are highly dependent on (operating) temperature. The value of barrier voltage V_B decreases by 2.5 mV per °C temperature increase. As a result the forward current increases (if the input voltage is fixed) and the forward characteristic curve shifts to the left as shown in Fig.2-36(a). For example, if the barrier voltage of a Si diode is 0.7 V at room temperature (25 °C), at 70 °C temperature its value will be $0.7 \text{ V} - 2.5 \text{ mV} \times (70 - 25) = 0.588 \text{ V}$.

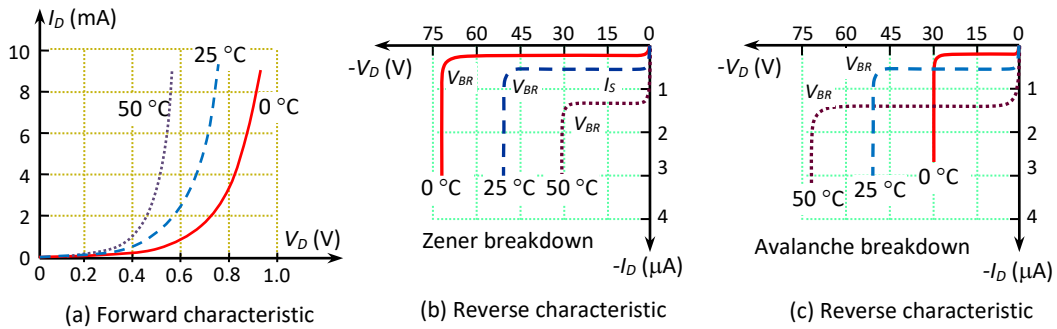


Fig.2-36: Effect of temperature on PN junction

Example 2-12

At ambient temperature of 25 °C, the barrier voltage of a Si diode is 0.7 V. What will be the barrier-voltage when the junction temperature is i) 0 °C, and ii) 50 °C ?

Solution:

i) When the junction temperature is 0 °C, the change in barrier voltage will be,

$$\Delta V_B = (-2.5 \text{ mV/}^\circ\text{C})\Delta T = (-2.5 \text{ mV/}^\circ\text{C})(0^\circ\text{C} - 25^\circ\text{C})$$

or,

$$\Delta V_B = 62.5 \text{ mV}$$

This positive value means that the barrier voltage will increase by 62.5 mV from its room temperature value. Thus, the barrier voltage at 0 °C will be,

$$V_B = 0.7 \text{ V} + 0.0625 \text{ mV} = 0.7625 \text{ V [Ans.]}$$

ii) When the junction temperature is 50 °C, the change in barrier voltage will be,

$$\Delta V_B = (-2.5 \text{ mV/}^\circ\text{C})(50^\circ\text{C} - 25^\circ\text{C})$$

or,

$$\Delta V_B = -62.5 \text{ mV}$$

This negative sign means that the barrier voltage will decrease by 62.5 mV from its room temperature value. Thus, the barrier voltage at 25 °C will be,

$$V_B = 0.7 \text{ V} - 0.0625 \text{ mV} = 0.6375 \text{ V [Ans.]}$$

Comments: Barrier voltage increases with the decrease in junction temperature and vice versa.

2.17.2 Effect of Temperature on Reverse Saturation Current

As discussed earlier, forward-biasing a diode decreases the width of the depletion layer and allows majority carriers to cross the junction. Reverse bias has the opposite effect: It widens the depletion layer by moving holes and free electrons away from the junction and helps to move the minority carriers.

Suppose that thermal energy creates a hole and free electron inside the depletion layer of a reverse-biased diode, as shown in Fig.2-37. The free electron at A and the hole at B can now contribute to reverse current. Because of the reverse bias, the free electron will move to the right, effectively pushing an electron out of the right end of the diode. Similarly, the hole will move to the left. This extra hole on the P-side lets an electron enter the left end of the crystal. Thus a reverse current will be produced, for the movement of minority carriers.

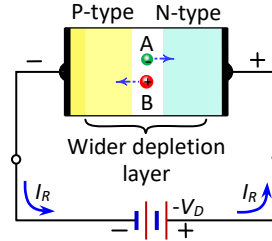


Fig.2-37: Thermal energy produces holes and electrons in depletion layer

The higher the junction temperature, the greater the saturation current [Fig.2-36(b)]. In the reverse bias condition, the reverse saturation current of silicon diode becomes double for every 10 °C rise in temperature. For example, at 25 °C temperature, the typical value of I_S of Si diode is 20 nA. At 75 °C the value of I_S will be $20 \text{ nA} \times (75-25)/10 = 100 \text{ nA}$. Mathematically,

$$\% \Delta I_S = 100\% \text{ for a } 10^\circ \text{C increase in temperature} \quad (2-31)$$

If the changes in temperature are less than 10 °C, we can use this equivalent rule:

$$\% \Delta I_S = 7\% \text{ for a } 1^\circ \text{C increase in temperature} \quad (2-32)$$

The approximate value of reverse saturation current can easily be calculated as,

$$I_{S(\text{at } T^\circ\text{C})} = 2^{(T-25)/10} \times I_{S(\text{at } 25^\circ\text{C})} \quad (2-33)$$

The value of I_S for Ge diode is as high as 1 μA. Hence, Ge diode is not used for high temperature applications. On the other hand, the value of I_S of GaAs diode is so low that it can be used at temperature as high as 400 °C. Generally speaking, we need a diode with lower I_S value for high temperature applications.

Example 2-13

A silicon diode has a saturation current of 6.5 nA at an ambient temperature of 25 °C. What will be the value of the saturation current at 80 °C?

Solution:

The change in temperature is from 25 °C to 80 °C. Now, let's see how many 10° are there. Starting from 25 °C the steps are: 35 °C, 45 °C, 55 °C, 65 °C, 75 °C → 5 steps. So, the saturation current will be multiplied by 2, five times. That is,

$$I_{S(\text{at } 75^\circ\text{C})} = (((((6.5 \text{ nA} \times 2) \times 2) \times 2) \times 2) \times 2) \times 2$$

or,

$$I_{S(\text{at } 75^\circ\text{C})} = 6.5 \text{ nA} \times 2^5 = 6.5 \text{ nA} \times 32$$

or,

$$I_{S(\text{at } 75^\circ\text{C})} = 208 \text{ nA}$$

But, temperature has increased 5°C more (75°C to 80°C). To calculate that we can use Equ.(2-32). After 75°C , 7% increase means $1.07 \times I_{S(\text{at } 75^\circ\text{C})} = 1.07 \times 208 \text{ nA}$. This will happen 5 times for 5°C increase. Thus,

$$I_{S(\text{at } 80^\circ\text{C})} = (1.07)^5 \times 208 \text{ nA} = 291.73 \text{ nA [Ans.]}$$

Comments: As the saturation current doubles for every 10°C rise in temperature, it will increase exponentially for every 10°C rise in temperature.

2.17.3 Effect of Temperature on Breakdown Voltage

The temperature variation has also some effect on the reverse breakdown voltage V_{BR} . For Zener breakdown mechanism (below 5 V) the value of V_{BR} decreases with the increase in temperature. As we increase the temperature the energy gap between the valence band and conduction band decreases. So, the less strong electric field is required to move valance electrons to the conduction band. Consequently, the required breakdown voltage decreases as the temperature increases. For avalanche breakdown mechanism (above 5 V), the value of V_{BR} increases with temperature. Due to increase in temperature, vibrations of atoms increase that decreases the mean free path for electrons (lower kinetic energy). Hence, for avalanche breakdown mechanism, breakdown voltage increases with increase in temperature as shown in Fig.3-36(c). The effects of temperature on different parameters are shown in Fig.2-36.

2.18 Surface-Leakage Current

The **surface leakage current** is basically a type of surface recombination current caused by surface defects. Surface defects can form the defect energy levels on the semiconductor surface, further generating surface charges or recombination centers, increase the surface recombination. Suppose that the atoms at the top and bottom of Fig.2-38(a) are on the surface of the crystal. Since these atoms have no neighbors (to upper and lower sides), they have only six electrons in the valence orbits, implying two holes in each surface atom. These holes, along the surface of the crystal, are shown in Fig. 2-37(b). Then

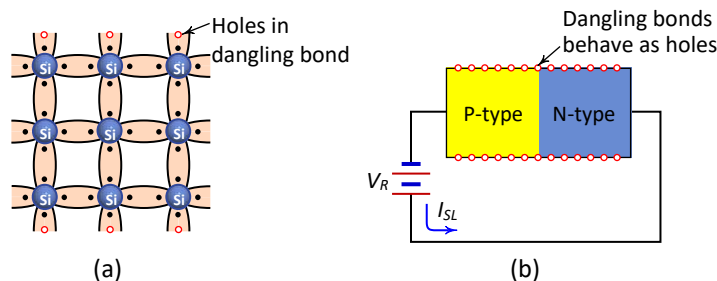


Fig.2-38: (a) Atoms on the surface of a crystal have no neighbors, (b) holes on the surface of a crystal

we can see that the skin of a crystal is like a P-type semiconductor. Because of this, traveling through the surface, electrons can leave the right end of the crystal and enter the left end. In this way, we get a small reverse current along the surface.

The surface-leakage current is directly proportional to the reverse voltage. For instance, if we double the reverse voltage, the surface-leakage current I_{SL} doubles, i.e., it follows the Ohm's Law. We can define the surface-leakage resistance as follows,

$$R_{SL} = \frac{V_R}{I_{SL}} \quad (2-34)$$

Example 2-14

Given a Si diode with a reverse current of 5 μA at 25 °C and 100 μA at 100 °C, calculate the surface leakage current.

Solution:

The reverse current (I_R) in a diode is the sum of reverse saturation current (I_S) and the surface leakage current (I_{SL}), i.e.

$$I_R = I_S + I_{SL} \quad (2-35)$$

Here, only the temperature is changed, not the biasing voltage. So, the surface leakage current will remain constant.

For the 1st case (25 °C temperature),

$$I_{R1} = I_{S1} + I_{SL}$$

$$\text{or,} \quad 5 \mu\text{A} = I_{S1} + I_{SL}$$

$$\text{or,} \quad I_{SL} = 5 \mu\text{A} - I_{S1} \quad (2-36)$$

For the 2nd case (100 °C temperature),

$$I_{R2} = I_{S2} + I_{SL}$$

$$\text{or,} \quad 100 \mu\text{A} = \left(2^{\frac{100-25}{10}} \times I_{S1}\right) + I_{SL} \quad [\text{using Equ. (2-33)}]$$

Putting the value of I_{SL} from Equ.(2-36),

$$100 \mu\text{A} = \left(2^{\frac{100-25}{10}} \times I_{S1}\right) + 5 \mu\text{A} - I_{S1}$$

$$\text{or,} \quad 95 \mu\text{A} = (2^{7.5} - 1) \times I_{S1}$$

$$\text{or,} \quad I_{S1} = \frac{95 \mu\text{A}}{180.019} \approx 0.53 \mu\text{A}$$

Putting this value of I_{S1} into Equ.(2-36),

$$I_{SL} = 5 \mu\text{A} - 0.53 \mu\text{A} = 4.47 \mu\text{A} [\text{Ans.}]$$

Comments: At lower temperature, the value of surface leakage current is higher than the reverse saturation current.

2.19 Resistance of PN Junction Diode

If we look at the diode characteristic curve, we find that the curve is nonlinear (in fact exponential). At different points of the curve the rate of change of current is different. The resistance corresponding to a particular point is inversely proportional to the rate of change of current. Thus, we can say that every point of the characteristic curve has different resistance. However, if we apply a DC voltage across a diode, it will operate at a fixed point and if we apply an AC voltage the operating point will change (oscillate). For this reason, we have to study diode resistance in two conditions – DC condition and AC condition.

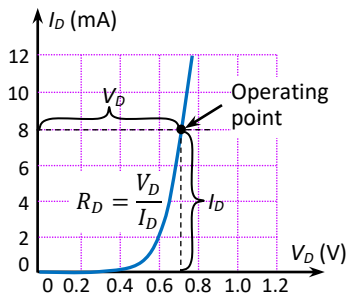


Fig.2-39: DC or static resistance from diode characteristics

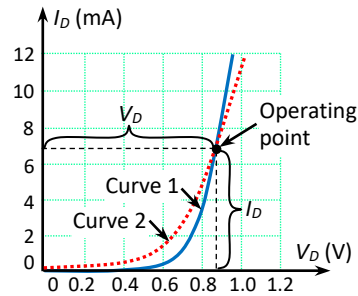


Fig.2-40: Two different curves give same DC resistance

2.19.1 DC or Static Resistance

If we apply a forward DC voltage, V_D , across a diode a DC current, I_D , will flow through the diode. If we put the values of voltage and current on the I - V characteristic curve of the diode, a point will be found. This point is called the **operating point** of the diode [Fig.2-39]. In a simple method, if we divide the voltage, V_D , by the current, I_D , a resistance will be found as

$$R_D = \frac{V_D}{I_D} \quad (2-37)$$

This resistance is called **DC resistance** or **static resistance**. Actually, this resistance is not merely the junction resistance of the operating point, but the total resistance offered by the diode for this operating point. For higher values of DC voltage the operating point will shift upward and the static resistance will decrease. From the graph of Fig.2-39, it is found that for $V_D = 0.75$ V, $I_D = 8$ mA. Thus, the values of $R_D = \frac{0.75 \text{ V}}{8 \times 10^{-3} \text{ A}} = 87.5 \Omega$.

2.19.2 AC or Dynamic Resistance

The DC resistance of a diode does not properly depends on the shape of the characteristic curve it only depends on the values of voltage and current at the operating point. For example, as shown in the Fig.2-40, the given operating point the two characteristic curves will give the same value of DC resistance (as the operating point is same) despite the different shape of the curves, the AC resistance will depend exactly on the shape of the curve. If we want to pass an AC signal through a PN junction, we have to forward bias the PN junction to fix a suitable DC operating point. Then, the AC signal is superimposed to the DC bias voltage.

Fig.2-41 shows how the AC signal changes the position of operating point. The positive half cycle moves the operating point upward and the negative cycle moves the operating point downward. Depending on the peak-to-peak value of the AC signal we get a highest operating point and a lowest operating point. By subtracting the voltages and currents of these two operating points we can calculate the change in voltage (ΔV_D) and the change in current (ΔI_D). Dividing ΔV_D by ΔI_D will give a resistance which is called **average AC resistance** or **average dynamic resistance** and is denoted as r_d . That is,

$$r_d = \frac{\Delta V_D}{\Delta I_D} \quad (2-38)$$

This dynamic or AC resistance will depend on the shape of the characteristic curve of the diode, at least on the slope of the portion of the curve along which the operating point oscillates. Although it

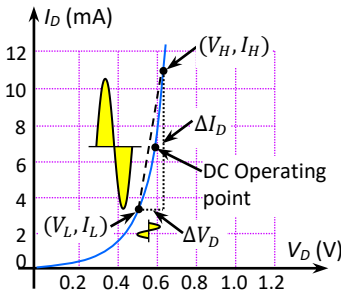


Fig.2-41: Dynamic resistance calculation from two points

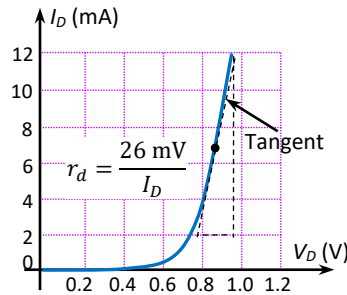


Fig.2-42: Dynamic resistance at a single point using tangent

seems that, r_d is the difference of the DC resistance at the highest and the lowest operating points, actually it is not true. This can easily be understood from Fig.2-41.

From the graph, $\Delta V_D = V_H - V_L$ and $\Delta I_D = I_H - I_L$

But,

$$r_d = \frac{\Delta V}{\Delta I} = \frac{V_H - V_L}{I_H - I_L} \left(\neq \frac{V_H}{I_H} - \frac{V_L}{I_L} \text{ or, } r_d \neq R_{DH} - R_{DL} \right) \quad (2-39)$$

Thus, we find that r_d is not the difference of the DC resistances at the two operating points. Actually, r_d is the reciprocal of the slope of the line connecting the two operating points. To calculate r_d corresponding to a single DC operating point, we have to consider the value of ΔV_D and ΔI_D to be very small so that the oscillation of the operating points shrinks to the DC operating point. That is, we can write $\Delta V_D = dV_D$ and $\Delta I_D = dI_D$. Thus,

$$r_d = \frac{dV_D}{dI_D} = \frac{1}{dI_D/dV_D} \quad (2-40)$$

Now, with the help of Calculus, we can calculate the value of r_d from Eq.(2-28)

$$I_D = I_S \left(e^{\frac{qV_D}{nkT}} - 1 \right)$$

or,

$$I_D = I_S e^{\frac{qV_D}{nkT}} - I_S$$

$$\text{or,} \quad \frac{dI_D}{dV_D} = \frac{d}{dV_D} \left(I_S e^{\frac{qV_D}{nkT}} - I_S \right)$$

$$\text{or,} \quad \frac{dI_D}{dV_D} = \left(\frac{q_e}{nkT} I_S e^{\frac{q_e V_D}{nkT}} - 0 \right)$$

$$\text{But we can assume,} \quad I_D \approx I_S e^{\frac{q_e V_D}{nkT}}$$

$$\therefore \quad \frac{dI_D}{dV_D} \approx \left(\frac{q_e}{nkT} I_D \right)$$

$$\text{or,} \quad \frac{1}{dI_D/dV_D} = r_d \approx \left(\frac{nkT}{q_e I_D} \right) = \left(\frac{nV_T}{I_D} \right)$$

Considering, $n = 1$, $k = 1.38 \times 10^{-23} \text{ J/K}$, $T = (273 + 27) \text{ K} = 300 \text{ K}$ and $q_e = 1.6 \times 10^{-19} \text{ C}$ we get,

$$r_d = \frac{(1.38 \times 10^{-23} \text{ J/K}) \times (300 \text{ K})}{1.6 \times 10^{-19} \times I_D}$$

$$\therefore \quad \boxed{r_d \approx \frac{26 \text{ mV}}{I_D}} \quad (2-41)$$

From this calculation we also find that the value of thermal voltage $V_T = 26 \text{ mV}$ as mentioned earlier. Using Equ.(2-41), we can calculate the value of r_d at any point of the I - V characteristic curve of the diode. As shown in Fig.2-42, r_d now gives the reciprocal of the slope of the curve at the DC operating point.

The AC resistance determined by Equ.(2-41) merely gives the resistance of a PN junction. But a practical diode has some additional resistances which are the bulk resistance of P-type semiconductor and N-type semiconductor, contact resistance of the connecting leads, etc. The value of this additional resistances range from 1 to 2 Ω and is denoted by r_B . Thus, the actual AC resistance of a diode will be,

$$\boxed{r_d \approx \frac{26 \text{ mV}}{I_D} + r_B} \quad (2-42)$$

Example 2-15

Two operating points are shown on the I - V curve of a diode. Calculate the DC resistance and AC resistance using Equ.(2-42) for these two operating points. Assume, $r_B = 2 \Omega$.

Solution:

Calculation of DC or static resistance R_D :

From the graph for operating point 1, $V_{D1} = 0.625 \text{ V}$ and $I_{D1} = 2 \text{ mA}$

Using Equ.(2-37),

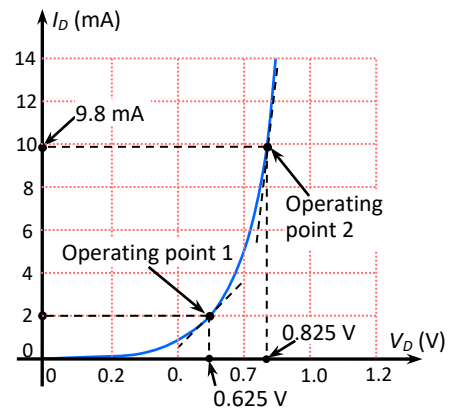


Fig.2-43: Two operating points on IV curve for Example 2-1

$$R_{D1} = \frac{V_{D1}}{I_{D1}} = \frac{0.625 \text{ V}}{2 \text{ mA}} = 312.5 \Omega \text{ (Ans.)}$$

Using Equ.(2-42),

$$r_{d1} = \frac{26 \text{ mV}}{I_{D1}} + r_B = \frac{26 \text{ mV}}{2 \text{ mA}} + 2 \Omega = 15 \Omega \text{ (Ans.)}$$

For operating point 2, $V_{D2} = 0.825 \text{ V}$ and $I_{D2} = 9.8 \text{ mA}$

Using Equ.(2-37),

$$R_{D2} = \frac{V_{D2}}{I_{D2}} = \frac{0.825 \text{ V}}{9.8 \text{ mA}} = 84.2 \Omega \text{ (Ans.)}$$

Using Equ.(2-42),

$$r_{d2} = \frac{26 \text{ mV}}{I_{D2}} + r_B = \frac{26 \text{ mV}}{9.8 \text{ mA}} + 2 \Omega = 4.7 \Omega \text{ (Ans.)}$$

Comments: Both DC and AC resistances decrease for higher operating point (high values of V_D and I_D). For a particular operating point, the AC resistance is always smaller than the DC resistance.

2.20 Capacitance in PN Junction

The terminal characteristics of a diode (in fact any device) depends on frequency. Although at low or medium frequency the variation of the performance with frequency can be neglected, at very high frequency we must be aware of the stray capacitance of the diode. **Stray capacitance** is defined as the capacitance which is automatically developed during the device fabrication. Actually this is unwanted capacitance. There are two types of capacitances in PN junction diode. These are **transition capacitance** and **diffusion capacitance**. Both types of capacitance are present in the forward and reverse bias condition. In reverse bias, transition capacitance is dominating and in forward bias diffusion capacitance is dominating.

2.20.1 Transition capacitance

We know that a parallel-plate capacitor consists of two conducting plates and a dielectric material between them (Fig.2-44). The capacitance of this capacitor is given by,

$$C = \frac{\epsilon A}{d} \quad (2-43)$$

At equilibrium condition of a PN junction, there exists a depletion layer between a P-type and an N-type semiconductor. As the depletion layer has no free carriers its resistance is very high. It can behave as a dielectric material as it can store charge (positive and negative ions). With a lot of majority carriers, both the P-type and an N-type material can work as conducting plates. Thus, the PN junction forms a parallel plate capacitor. The capacitance associated with this arrangement is called **transition capacitance** or **depletion layer capacitance**. This capacitance is also called simply **junction capacitance**. The value of this capacitance decreases with the reverse bias voltage and can be calculated as,

$$C_T = C_{T0} \left(1 + \frac{-V_D}{V_B} \right)^{-1/2} \quad (2-44)$$

Here, C_{T0} is the transition capacitance with zero bias, $-V_D$ is the applied reverse bias voltage and V_B is the barrier voltage of the PN junction.

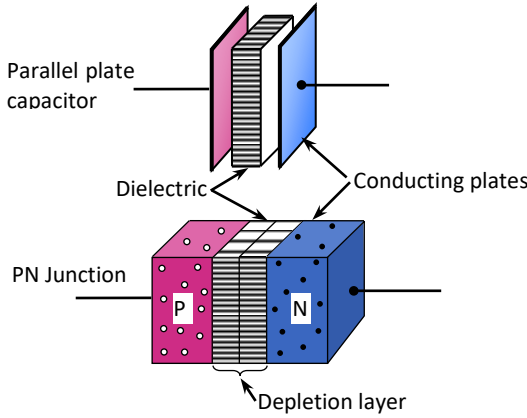


Fig.2-44: Resemblance of parallel plate capacitor and PN junction

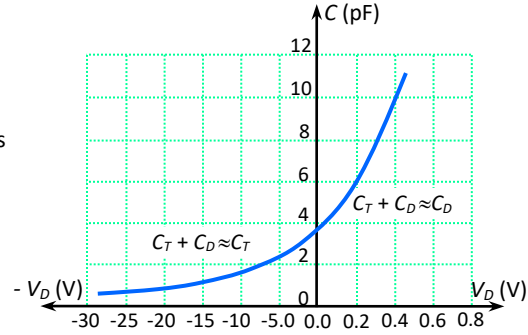


Fig.2-45: Variation of transition and diffusion capacitance

The value of this capacitance can also be calculated as using Equ.(2-43),

$$C_T = \frac{dQ}{dV} = \frac{\epsilon A}{d} = \frac{\epsilon A}{W} \quad (2-45)$$

Here, ϵ is the dielectric constant of Si, A is the cross-sectional area of the PN junction and W is the width of the depletion layer. Since the width of the depletion layer increases with the increase of reverse bias, the transition capacitance decreases with reverse bias voltage as shown in Fig.2-45. The depletion layer capacitance is essentially the capacitance of reverse biased PN junction. The value of transition capacitance is very low. The typical value for a low current diode is 4 pF.

Example 2-16

Consider a silicon PN junction at $T = 300$ K, with doping concentrations of $N_a = 10^{16} \text{cm}^{-3}$ and $N_d = 2 \times 10^{15} \text{cm}^{-3}$. Assume that $n_i = 1.5 \times 10^{10} \text{cm}^{-3}$ and the unbiased junction capacitance $C_{T0} = 1.0$ pF. Calculate the junction capacitance at $V_D = -1$ V and $V_R = -10$ V.

Solution:

Using Equ.(2-27),

$$V_B = V_T \ln \left(\frac{N_a N_d}{n_i^2} \right) = (26 \times 10^{-3} \text{V}) \ln \left[\frac{(10^{16} \text{cm}^{-3})(2 \times 10^{15} \text{cm}^{-3})}{(1.5 \times 10^{10} \text{cm}^{-3})^2} \right] \approx 0.636 \text{ V}$$

Using Equ.(2-44), the junction capacitance for $V_D = -1$ V can be calculated as,

$$C_T = (1.0 \text{ pF}) \left(1 + \frac{1 \text{ V}}{0.636 \text{ V}} \right)^{-1/2} \approx 0.625 \text{ pF [Ans.]}$$

For $V_D = -10$ V,

$$C_T = (1.0 \text{ pF}) \left(1 + \frac{10 \text{ V}}{0.636 \text{ V}} \right)^{-1/2} \approx 0.245 \text{ pF [Ans.]}$$

Comments: The magnitude of the junction capacitance is usually at or below the pico-Farad range, and it decreases as the reverse bias voltage increases.

2.20.2 Diffusion Capacitance

The capacitance effect which is produced due to the storage of charge carriers is called **diffusion capacitance**, C_D . This capacitance is essentially the capacitance of forward-biased PN junction. If a forward bias voltage is applied to a PN junction a forward current (I_F) will flow from P-side to N-side through the depletion layer. The forward current is due to the diffusion of majority carriers. If we suddenly reverse the polarity of bias voltage (reverse bias) the forward current I_F will stop immediately leaving some majority charge carriers in the depletion region. Now due to the reverse bias, the depletion layer will be widened. The charge carriers, left by the forward current, will start to flow back from the depletion layer in reverse direction as the minority carriers do. Thus, initially the reverse current will be comparatively high and gradually decreases to the level of reverse saturation current, I_S . The effect is similar to the discharge of a capacitor. Similarly, when we forward bias a PN junction at first majority carriers will be accumulated in the depletion region and then the forward current will start to flow. This effect is similar to the charging of a capacitor. This charging and discharging property of the PN junction is represented by a capacitance and called **diffusion capacitance**, C_D . The value of diffusion capacitance can be calculated by the following equation,

$$C_D = \frac{dQ}{dV} = \left(\frac{\tau}{V_B} \right) I_D \quad (2-46)$$

where, τ is the minority carrier life-time, V_B is the barrier voltage of the PN junction and I_D is the forward current level.

From this equation, it is clear that the diffusion capacitance increases with the increase in forward current, i.e., with the increase of forward bias voltage. But, as we know, the resistance level of a PN junction decreases with the increase of forward current. Hence, the time-constant ($\tau = RC$) of the PN junction remains almost constant which is very important. The value of diffusion capacitance is higher than that of transition capacitance. The typical value of a low current diode is 1 nF with the value of $I_D = 10$ mA.

Both the transition capacitance and the diffusion capacitance depend on the applied voltage. This dependency of the capacitance on voltage is used in varactor diode. The varactor diode is used in electronic circuit as a variable capacitor, whose capacitance can be changed by changing applied voltage [described in details in section 2-30].

2.21 Ideal Diode

Actually, in electronic circuit a diode is used as a switch that allows current only in one direction (forward direction). Although, a PN junction can behave as a one-way switch, in addition to switching property it has barrier voltage V_B , reverse saturation current I_S , junction capacitance ($C_T + C_D$) and some resistances (R_D, r_d). Actually these additional parameters are unwanted. So, an **ideal diode** is such a diode that provides only the one-way switching property, but will have no barrier voltage, reverse current, junction capacitance, and any resistance. In fact, an ideal diode will have no limitation like – reverse breakdown, forward current limitation etc. We cannot manufacture an ideal diode but we can only imagine it. The I - V characteristic curve of an ideal diode is given in Fig.2-46(a).

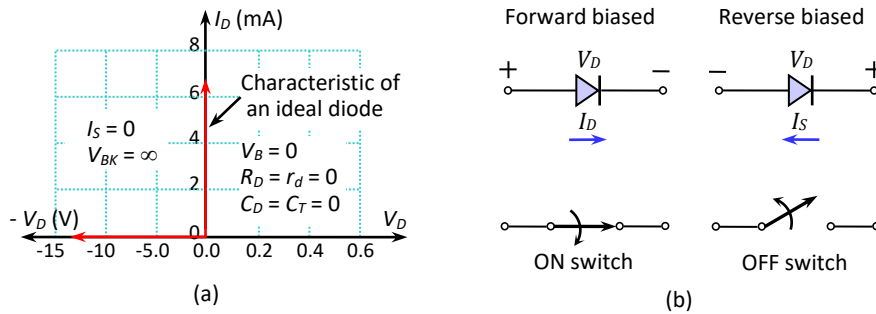


Fig.2-46: (a) Characteristics of an ideal diode, (b): Switch and ideal diode analogy

The operation of an ideal diode can be compared with that of a mechanical switch with the exception that the diode conducts current in only one direction and the switch conducts current in both directions. Mechanical switch can be used for both AC and DC voltage. On the other hand an ideal diode can be used as a switch only for DC. If the diode is forward-biased, it works like a closed (ON) switch and if it is reverse-biased the diode works like an open (OFF) switch as shown in Fig.2-46(b). However, for rectification purpose diodes are used with AC voltage.

2.22 Diode Approximation and Equivalent Circuit

We have already discussed the characteristics of an ideal diode and practical diode. It is not possible to manufacture an ideal diode. So, in electronic circuits we use practical diodes. As the parasitic components (barrier voltage, resistance etc.) of the practical diode are very small, in different circumstances we can neglect different parasitic components and approximate the operation of the practical diode and it can be represented by an equivalent circuit.

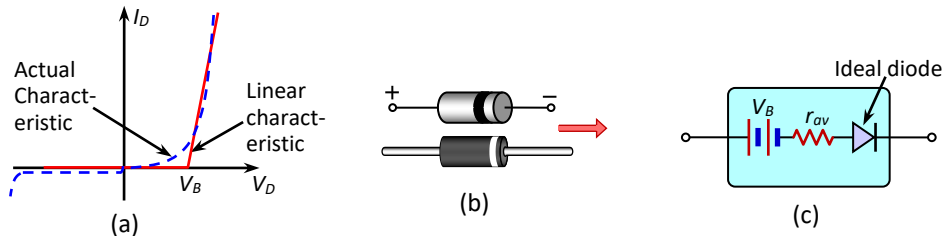


Fig.2-47: (a) Actual and linearized characteristics, (b) Actual diodes, and (c) Equivalent circuit of a diode

2.22.1 DC equivalent Circuit

At first, if we neglect the non-linearity of the characteristic curve, we can represent it by three straight lines- two in forward bias and one in reverse bias as shown in Fig.2-47(a). For this characteristic, the equivalent circuit of the diode can be represented by a barrier voltage V_B , a resistance r_{av} and an ideal diode D connected in series as shown in Fig.2-47(c). Here, r_{av} is the average resistance of the portion of the characteristic curve after the knee point.

We know that the value of r_{av} is very small, around 100Ω . Hence, in some operation, where the resistance value of the external circuit is very high compared to r_{av} , we can neglect it. In that case the equivalent circuit of the diode will consist of a barrier voltage V_B , and an ideal diode D .

connected in series as shown in Fig.2-48(b). The characteristic of this approximation will be as shown in Fig.2-48(a).

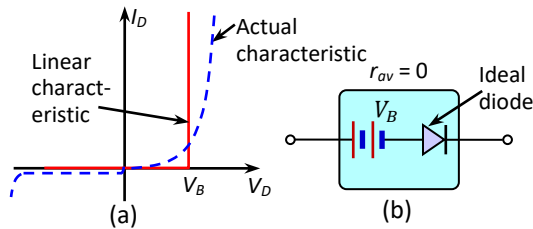


Fig.2-48: (a) Actual and linearized characteristics, and (b) Equivalent circuit of a diode

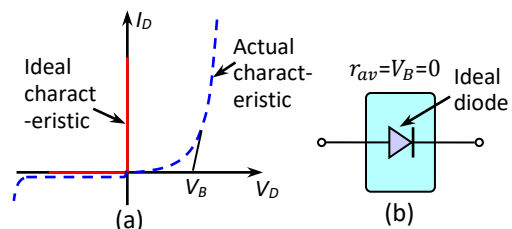


Fig.2-49: (a) Actual and linearized characteristics, and (b) Equivalent circuit of a diode

Even in some situations, where the external resistance is very high and the circuit operating voltage level is also very high compared to V_B , the operation of a practical diode can approximate to that of an ideal diode. Here, we neglect r_{av} and V_B . The characteristic curve and the equivalent circuit are shown in Fig.2-49(a) and (b), respectively.

In these three equivalent circuits, the capacitance effect of the diode has not been considered. That is, these three equivalent circuits are considered DC equivalent circuits.

2.22.2 Equivalent Circuit with Capacitance

The capacitance of a PN junction is very small. Hence, in normal operation we can neglect it. But, the capacitance becomes effective in very high frequency operation and during the ON-OFF transition time. Considering the junction capacitance, the equivalent circuit will be as shown in Fig.2-50. As, the capacitance becomes effective during the ON-OFF transition time, it limits the switching frequency of diode which is represented by a parameter called **reverse recovery time**.

2.23 Reverse Recovery Time

In many applications, diodes must switch rapidly between forward bias and reverse bias (i.e., from ON to OFF and from OFF to ON). Most diodes switch very rapidly from OFF state to ON state, however, they take comparatively longer time to switch from ON state to OFF state. This happens due to the capacitance effect of the diode. The situation is explained in Fig.2-51(a). Here, to change the state of a diode a square wave voltage pulse is applied to a diode. When the voltage changes from negative to positive, the diode becomes forward biased very quickly and the forward current I_F starts to flow immediately. But, when the voltage changes from positive to negative, the diode starts to conduct in reverse

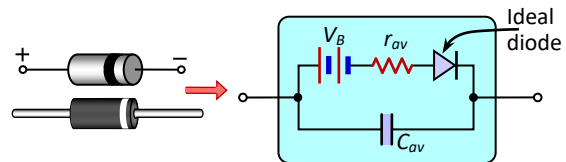


Fig.2-50: AC equivalent circuit of a diode considering r_{av} and V_B

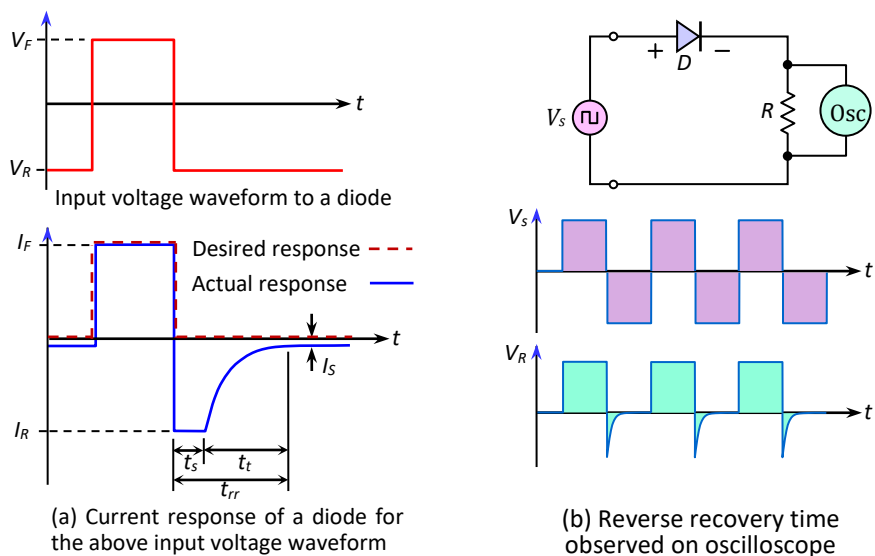


Fig.2-51: (a) Reverse recovery time (b) Effect of reverse recovery time on diode switching

direction instead of being OFF. The reverse current I_R initially equals the forward current, then it gradually decreases to the reverse saturation current I_S . The high level of reverse current occurs because at the instant of reverse bias, there are some charge carriers crossing the depletion region, and these must be removed. Actually, this is the reason of diffusion capacitance as described earlier. Thus, the time required for the reverse current to fall to the value of reverse saturation current (I_S) is called **reverse recovery time**, t_{rr} . Reverse recovery time has two components. The time required by the carrier to move to their majority side from the depletion layer is called storage time, t_s , and the time required by the current to fall to saturation level is called the transition time, t_t [see Fig.2-51(a)]. The sum of t_s and t_t is t_{rr} . Typical values of t_{rr} for switching diodes range from 4 ns to 50 ns.

If we want to observe the reverse recovery time on oscilloscope, we have to apply a very high frequency square wave to a diode in series with a resistor as shown in Fig.2-51(b). Due to the reverse recovery time of the diode, the voltage developed across the resistor will produce negative spikes. For instance, the 1N4148 has a t_{rr} of 4 ns. If this diode has a forward current of 10 mA and it is suddenly reverse biased, it will take approximately 4 ns for the reverse current to decrease to 2 nA (I_S of 1N4148 diode). Reverse recovery time is so short in small-signal diodes that we can neglect its effect at frequencies below 10 MHz or so. However, for frequencies above 10 MHz we have to take t_{rr} into account.

2.24 Silicon Diode versus Germanium Diode

Diodes are most commonly fabricated using Silicon (Si) and Germanium (Ge). As the atomic structures of Si and Ge are different, the Si-diode and Ge-diode show many different characteristics. The most common differences are tabulated here.

Table 2-3: Comparison of Si diodes and Ge diodes

Sl. No.	Si Diode	Ge Diode
---------	----------	----------

1.	The atomic number of silicon is 14.	The atomic number of germanium is 32.
2.	Electric distribution of Si is: 1s2 2s2 2p6 3s2 3p2	Electric distribution of Ge is: 1s2 2s2 2p6 3s2 3p6 4s2 3d10 4p2
3.	Si has no d electrons	Ge has d electrons
4.	Has comparatively smaller atomic radius	Has comparatively larger atomic radius
5.	The barrier voltage of Si diode is ≈ 0.7 V	The barrier voltage of Ge diode is ≈ 0.35 V
6.	In reverse bias condition, Si diode has low reverse current.	In reverse bias condition, Ge diode has high reverse current.
7.	Forward resistance is high: $500\ \Omega$ to $800\ \Omega$	Forward resistance is low: $200\ \Omega$ to $400\ \Omega$
8.	Reverse resistance is high: Several $M\Omega$	Reverse resistance is low: Several $k\Omega$
9.	Si diodes are a bit cheaper.	Ge diodes are a bit expensive.
10.	Temperature sensitive	Highly temperature sensitive
11.	For their large operating temperature range, they are widely used.	For their short operating temperature range, they are not widely used.

2.25 Diode Data Sheet

To select a suitable diode for a particular application, the data sheets or the specifications, provided by the manufacturer, must be consulted. A portion of the data sheet of 1N4000 series is shown in the Table 2-4. The maximum ratings are the maximum voltages, currents, etc., that can be applied without destroying the device. A practical data sheet includes many parameters. However, the meanings of the most common parameters are described below.

V_{RRM} : V_{RRM} is the **maximum recurrent peak reverse voltage** and also referred as **peak inverse voltage** or **DC blocking voltage**. This is the maximum reverse bias voltage that the diode can tolerate. If the reverse bias voltage applied to a diode exceeds this value the diode will be destroyed.

V_{RMS} : V_{RMS} is the **maximum RMS voltage**. 1N4000 diodes are mainly used to convert AC voltage into DC voltage. V_{RMS} is the maximum RMS AC voltage that can be rectified by the diode. If the applied AC voltage exceeds this value the diode will be destroyed.

I_O or I_F : I_O is the **maximum steady state forward current**. This is maximum DC or RMS current that may be passed continuously through the diode. This value is related to the power rating of the diode. For power diode this value will be very high.

I_{FSM} : I_{FSM} is the **maximum forward surge current**. This surge current is very much higher than the maximum forward current, I_O , and can pass through the diode not more than a specified period of time. It is a current that may be allowed to flow briefly when a circuit is switched ON.

Table 2-4: A portion of a diode data sheet

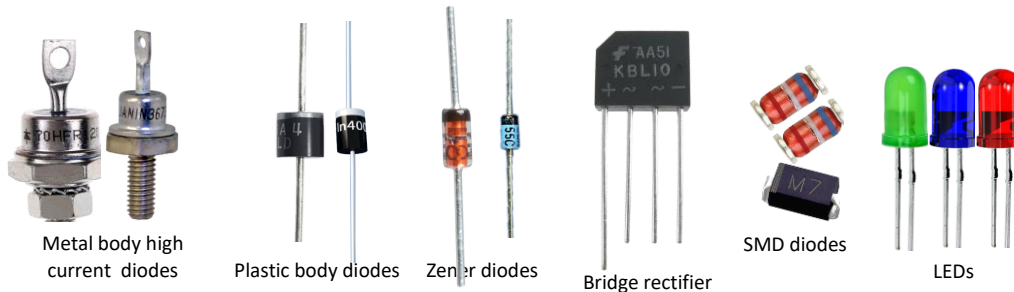
Characteristics	Symbo	1N 4001	1N 4002	1N 4003	1N 4004	1N 4005	1N 4006	1N 4007	Units
Maximum recurrent peak reverse voltage V_{RRM}		50	100	200	400	600	800	1000	Volts
Maximum RMS voltage V_{RMS}		35	70	140	280	420	560	700	Volts
Maximum DC blocking voltage	V_{DC}	50	100	200	400	600	800	1000	Volts
Maximum average forward rectified current at $T_a = 75^\circ\text{C}$	I_O	1.0							Amps
Peak forward surge current, 8.3 ms single half sine-wave superimposed on rated load	I_{FSM}	30.0							Amps
Maximum instantaneous forward voltage drop at 1.0 A	V_F	1.1							Volts
Maximum DC reverse current $T_a = 25^\circ\text{C}$ at rated DC blocking voltage $T_a = 125^\circ\text{C}$	I_R	5.0 50.0							μA
Typical junction capacitance (Note 1)	C_J	15							pF
Operating junction and storage temperature	T_J, T_{stg}	-65 to +125				-65 to +150			$^\circ\text{C}$

Notes: 1. Measured at 1.0 MHz and applied reverse voltage of 4.0 V

V_F : V_F is the **maximum instantaneous forward voltage drop**. This is the maximum voltage that will be dropped by a diode for a given forward current and temperature (at 25°C).

I_R : I_R is the **maximum reverse current** that will flow through the diode at the maximum DC blocking voltage.

P_M : P_M is the **maximum power dissipation** that a diode can withstand at 25°C . Actually, it is the product of V_F and I_F or $P_M = (0.7 \text{ V}) \times (I_F)$.

**Fig.2-52:** Photographs of different types of diodes

2.26 Photographs of Common Diodes

So far we have discussed about only one type of diode which is called rectifier diode. Beside this rectifier diode, there are many other types of diodes. Although, most of them are made of PN junction, some have other structures. The constructions and operation process of Zener diode, light emitting diode, pin diode, varactor diode etc. are described in the following headings. However, to get some idea about the physical shape and size, the photographs of some common varieties of diodes are given in Fig.2-52.

2.27 Zener Diode

As discussed earlier, for a very large reverse bias voltage the reverse current rapidly increases from reverse saturation current. However, this phenomenon is used to produce a constant voltage in devices called **Zener diodes**. There are two mechanisms that cause breakdown in reverse biased PN junction.

2.27.1 Zener breakdown mechanism

This type of breakdown takes place in the PN junctions where the doping concentration is very high. Due to the high doping concentration, the width of the depletion layer will be comparatively small as shown in Fig.2-53(a). The electric field strength of the depletion layer is, $E = V_R/W$ (W = width of the depletion layer). For small value of W , the electric field strength (or intensity) will be very high. The high intensity electric field causes electrons to break away the covalent bonds and leave from their parent atoms. With the breakdown of many such electrons, insulating depletion layer converts into a conductor. This breakdown of the depletion layer by electric field is called **Zener breakdown** according to the name of its inventor. The Zener breakdown occurs usually with reverse bias voltage less than 5 V.

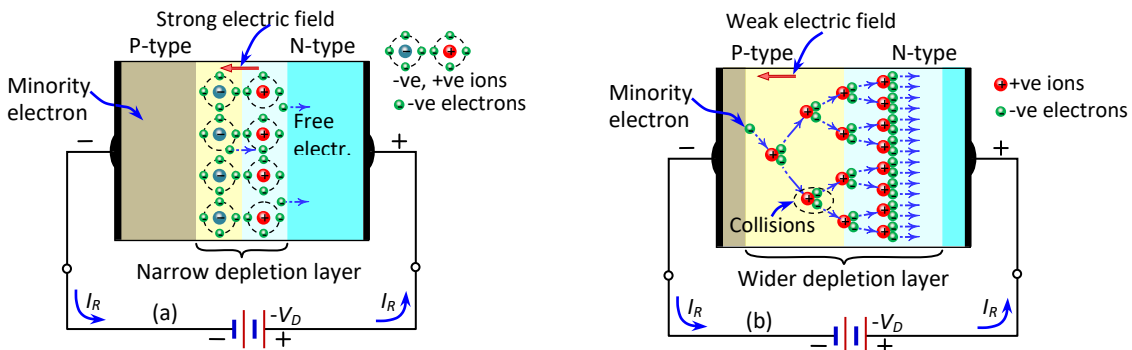


Fig.2-53: (a) Zener breakdown: Electrons detached by strong field,
(b) Avalanche breakdown: Electron multiplication by collisions

2.27.2 Avalanche breakdown mechanism

If the doping concentration of the PN junction is low the width of the depletion layer will be very large, when reverse biased as shown in Fig.2-53(b). In this case, the electric field intensity will not be sufficient to produce Zener breakdown. However, when the minority reverse-current flows, the carriers travel a comparatively long distance to achieve large kinetic energies. If these energetic carriers collide with the electrons of covalent bonds of the crystal, a high energy is transferred to the electrons and they become free. This knocking down of the electrons is called **ionization by collision**. The newly developed free electrons also collide with other atoms and produce more free electrons and so on [as shown in Fig.2-53(b)]. In this way, the carrier concentration in the depletion layer increases and the reverse current increases rapidly resulting in the junction breakdown. As this breakdown mechanism is similar to the avalanche of snow it is called **avalanche breakdown**. Avalanche breakdown is normally produced by reverse voltage levels above 5 V.

Although, any of these two types of breakdowns may take place in a breakdown diode, the device is called Zener diode, according to the name of its inventor **Clarence Melvin Zener** an American physicist.

2.27.3 Characteristics of Zener Diode

Zener diode has almost the same characteristic as the normal low power rectifier diode except the lower value of reverse breakdown voltage. The typical characteristics of a Zener diode are shown in Fig.2-54(a) and the symbol of a Zener diode is shown in Fig.2-54(b).

The forward characteristic is exactly same as the forward characteristic of a normal diode. The forward threshold voltage is $V_B = 0.7$ V for Si diode.

In the reverse bias condition very small amount of reverse current flows. But after the Zener breakdown voltage, V_Z , the reverse current increases rapidly. Actually a Zener diode is used at any point (between I_{ZK} to I_{ZM}) of the reverse characteristic curve after breakdown. The Zener diode is used in circuit to produce a constant voltage. The performance of a Zener diode as constant voltage source depends on the shape of the reverse characteristics. It will provide a constant voltage which is equal to the Zener breakdown voltage, V_Z . The variation of this voltage with current (I_Z) depends on a parameter called **dynamic impedance** of Zener diode r_Z (or Z_Z) which can be calculated as,

$$r_Z = \frac{\Delta V_Z}{\Delta I_Z} \quad (2-47)$$

As illustrated in Fig.2-54(a), r_Z defines how V_Z changes with the change of reverse current. The value of r_Z depends on the value of reverse current and has the maximum value at the knee point

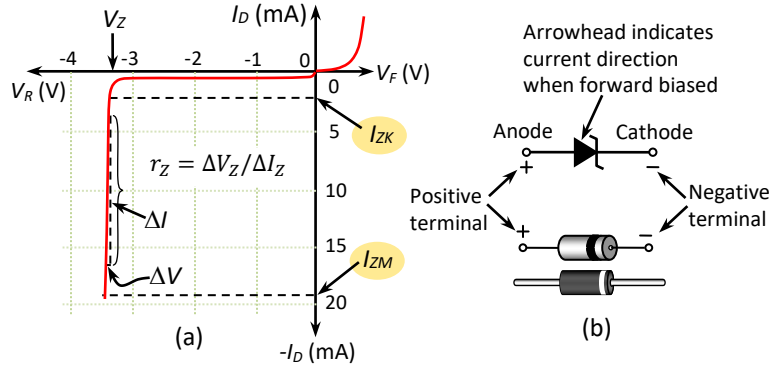


Fig.2-54: (a) Zener diode characteristics, (b) Symbol and practical diodes.

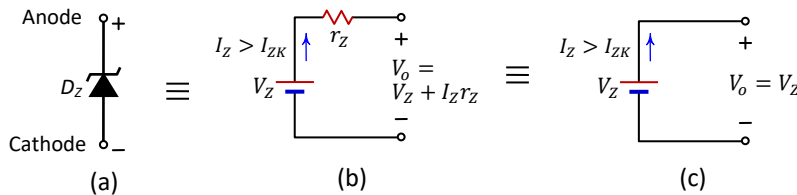


Fig.2-55: (a) Zener diode, (b) Equivalent circuit, and (c) Simplified equivalent circuit

of the curve. The Zener diode current may be set to any value between I_{ZK} and I_{ZM} . Here, I_{ZK} is the reverse current at the knee point of the reverse characteristic and I_{ZM} is the maximum Zener diode current which is limited by the power of a Zener diode.

In reverse bias condition, a Zener diode can be represented by an equivalent circuit as shown in Fig.2-55. According to this model, the output voltage of a Zener diode will be,

$$V_0 = V_Z + I_Z r_Z \quad \text{[for Fig.2-55(b)]} \quad (2-48)$$

$$V_0 = V_Z$$

[for Fig.2-55(c)]

(2-49)

Table 2-5: A portion of a Zener diode data sheet [Courtesy of Fairchild Semiconductor]

FAIRCHILD
SEMICONDUCTOR®

Tolerance = 5%

April 2009

1N4728A - 1N4758A

Zener Diodes

DO-41 Glass Case
COLOR BAND DENOTES CATHODE

Absolute Maximum Ratings* Ta = 25°C unless otherwise noted

Symbol	Parameter	Value	Units
P _D	Power Dissipation @ TL ≤ 50°C, Lead Length = 3/8"	1.0	W
	Derate above 50°C	6.67	mW/°C
T _J , T _{STG}	Operating and Storage Temperature Range	-65 to +200 °C	

*These ratings are limiting values above which the serviceability of the diode may be impaired.

Electrical Characteristics Ta = 25°C unless otherwise noted

Devices	V _Z (V) @ I _Z (Note 1)			Test Current I _Z (mA)	Max. Zener Impedance			Leakage Current		Non-Repetitive Peak Reverse Current I _{ZSM} (mA) (Note 2)
	Min.	Typ.	Max		Z _Z @ I _Z (Ω)	Z _{ZK} @ I _{ZK} (Ω)	I _{ZK} (mA)	I _R (μA)	V _R (V)	
1N4728A	3.135	3.3	3.465	76	10	400	1	100	1	1380
1N4729A	3.420	3.6	3.780	69	10	400	1	100	1	1260
1N4730A	3.705	3.9	4.095	64	09	400	1	50	1	1190
1N4731A	4.085	4.3	4.515	58	09	400	1	10	1	1070
1N4732A	4.650	4.7	4.935	53	08	500	1	10	1	970
1N4733A	4.845	5.1	5.355	49	7	550	1	10	1	890
1N4734A	5.320	5.6	5.880	45	5	600	1	10	2	810
1N4735A	5.890	6.2	6.510	41	2	700	1	10	3	730
1N4736A	6.460	6.8	7.140	37	3.5	700	1	10	4	660
1N4737A	7.125	7.5	7.875	34	4	700	0.5	10	5	605
1N4738A	7.790	8.2	8.610	31	4.5	700	0.5	10	6	550
1N4739A	8.645	9.1	9.555	28	5	700	0.5	10	7	500
1N4740A	9.500	10	10.50	25	7	700	0.25	10	7.6	454
1N4741A	10.45	11	1.550	23	8	700	0.25	05	8.4	414
1N4742A	11.50	12	12.60	21	9	700	0.25	05	9.1	380

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1N4728A – 1N4758A Rev. H3

www.fairchildsemi.com

Notes: 1. Zener Voltage (V_Z)

V_Z is measured with the device junction in the thermal equilibrium at the lead temp.(T_L) at 30°C ± 1°C and 3/8" lead length.

1N4728A – 1N4758A Zener Diodes

There are a number of Zener diodes produced by different manufacturing companies with different breakdown voltage and power ratings. To write the names and specifications of all of them are out of this book's scope. However, to get some idea about the datasheet of Zener diode, a portion of the Datasheet of 1N4728A manufactured by Fairchild Semiconductor is shown the following table.

Example2-17

As shown in Table 2-5, the Zener diode 1N4728A has a typical value of $V_Z = 3.3\text{ V}$ at $I_Z = 76\text{ mA}$, and $Z_Z = 10\ \Omega$. If 100 mA of current flows through this diode, calculate the output voltage i) Using the model of Fig.2-55(b), and ii) Using the model of Fig.2-55(c). iii) Calculate the power dissipated by the Zener diode for both the cases.

Solution:

i) 3.3 V is found when $I_Z = 76\text{ mA}$. So, V_o will increase for additional current. For the equivalent circuit of Fig.2-55(b), the output voltage can be calculated using Equ.(2-48).

$$V_o = V_Z + I_Z Z_Z = 3.3\text{ V} + (100 - 76)\text{ mA} \times 10\ \Omega = 3.54\text{ V [Ans.]}$$

ii) For the equivalent circuit of Fig.2-55(c), the output voltage can be calculated using the same equation assuming $r_Z = 0$.

$$V_o = V_Z + I_Z Z_Z = 3.3\text{ V} + (100 - 76)\text{ mA} \times 0 = 3.3\text{ V [Ans.]}$$

iii) The output power dissipated by the Zener diode will be,

For the exact model, $P_Z = 3.54\text{ V} \times 100\text{ mA} = 354\text{ mW [Ans.]}$

For the simplified model, $P_Z = 3.3\text{ V} \times 100\text{ mA} = 330\text{ mW [Ans.]}$

Comment: For lower values of I_Z , the effect of r_Z on the output voltage is very small. Thus, r_Z can be neglected in most of the practical applications.

2.28 Light Emitting Diode

Light emitting diode (LED) is a very common optoelectronics device. Although it is mainly used as an indicating device, nowadays high power LEDs are used as light source (LED tube lights, LED bulbs etc.) in our houses. LEDs are also used to manufacture today's very high quality televisions, computer monitors and mobile phone displays. LED is nothing but a PN junction of semiconductor. Semiconductor materials have a good capability to absorb and emit light in the visible range. In a forward biased PN junction light is emitted when the free electrons recombine with holes.

2.28.1 Construction

A cross-sectional view of an LED junction is shown in Fig.2-56(a). It is nothing but a PN junction. Single element semiconductors like Si, Ge, Ga etc. are indirect band-gap materials. They are used to manufacture rectifier diodes not the LEDs. In these single element PN junctions, visible light will not be emitted when electrons recombine with holes. On the other hand, alloys like gallium arsenide (GaAs), gallium phosphide (GaP), gallium arsenide phosphide (GaAsP) etc. are indirect band-gap materials and are suitable for LED manufacturing. Combinations of different materials and different components ratio produce light of different colors. LEDs made from GaAs emit invisible infrared light. GaP material provides either red light or green light, while red or yellow emission can be produced by using GaAsP. To make an LED an N-type epitaxial layer is grown upon a substrate, and the P-region is created by diffusion method. Charge carrier recombination occurs in the P-region, so the P-region is kept uppermost to allow the light to escape from the junction. The substrate is placed in a cup-type reflector which is made either aluminum or even gold. The reflector reflects as much light as possible toward the surface of the device and also works as the cathode connection as shown in Fig.2-56(b). The positive (anode) lead is connected to P-type material by a very narrow wire. The whole arrangement is encapsulated in an epoxy lens (plastic cover). The lens can be clear

or colored to identify the emitted light color when the LED is not energized. Some LEDs have glass particles embedded in the epoxy lens to diffuse the emitted light and increase the viewing angle of the device. In physical LEDs [Fig.2-57] the anode lead is longer than the cathode, but inside the plastic cover, the head of the cathode is larger than that of anode.

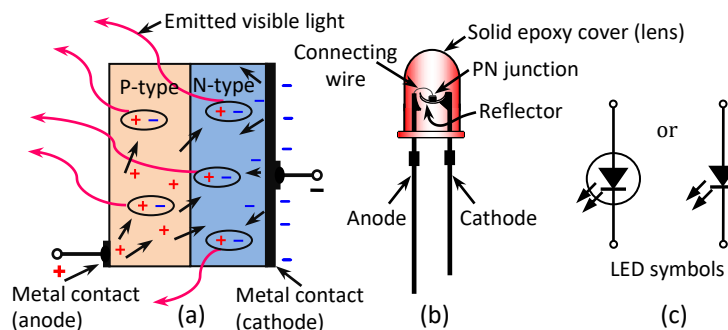


Fig.2-56: (a) Construction of LED using PN junction (b) Typical physical construction of LED (c) Symbols of LED

2.28.2 Operation

When an LED is forward biased with a suitable voltage (about 1.5 V or more, depending on the color) a forward current will flow. As the mobility of electrons is higher than that of holes mainly majority carrier electrons flow from N-side to the P-side. In the P-type material the free electrons recombine with holes which are the majority carriers of P-type materials. We know that the free electrons recite in conduction band and the holes recite in the valence band. Hence, when the free electrons come to valence band from conduction band they emit their extra energy which is equal to band-gap energy (E_g). In direct band gap-materials, this energy is emitted as photons or light. By selecting suitable materials and changing their combination ration the band-gap of the alloy can be changed. As the light color depends on energy, LEDs of different band-gap energy can produce light of different colors.

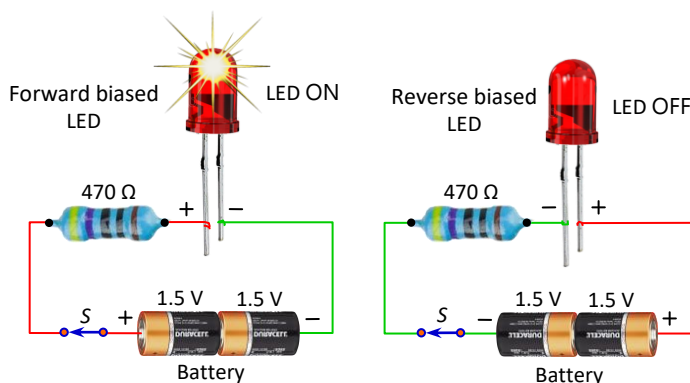


Fig.2-57: LED produces light when forward biased

LEDs of different colors need different forward voltages as shown in Table 2-6. The amount of light produced by the LED depends of the forward current, but the current should not exceed the

maximum rating of the LED. Typical value of forward current of normal low power LEDs is 1 mA to 50 mA.

Table 2-6: Some common materials used for LED manufacturing

Construction material	Color	Typical forward voltage (V)
GaAsP	Red	1.8
GaAsP	Orange	2.0
AlInGaP	Yellow	2.1
GaP	Green	2.2
GaN	White	4.1
GaN	Blue	5.0

2.29 Organic Light Emitting Diode (OLED)

An OLED is a device that consists of two or three layers of materials composed of *organic molecules* (contain carbon) or *polymers* that emit light with the application of currents. OLEDs emit light in a similar manner to LEDs, through a process called *electro phosphorescence*. But in general, OLEDs have very different properties from basic LEDs. They use molecular orbitals that are analogous to the valence and conduction bands of inorganic LEDs. The color of the light depends on the type of organic molecule in the emitting layer. The basic structure of a typical OLED is shown in

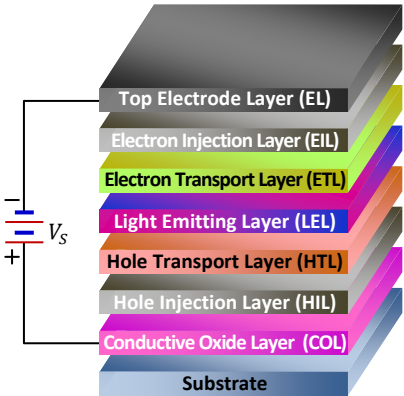


Fig.2-58: Construction of OLED

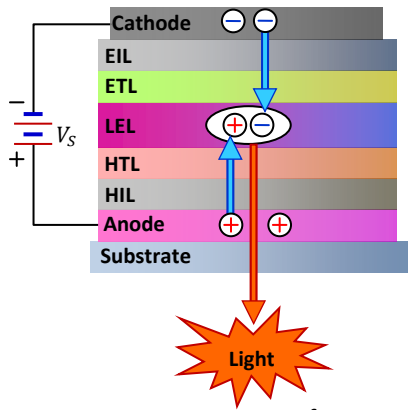


Fig.2-59: Operation of OLED

Fig.2-58. OLED devices consist of one to three functional organic layers including a hole transport layer, a light-emitting layer, and an electron transport layer, and all of these layers are situated between an anode and a cathode. In an OLED device, the light-emitting layer is excited by the recombination energy of electrons from the cathode and holes from the anode, and then the light-emitting layer emits light when returning to the ground state. One of the electrodes consists of transparent material in order to extract light from the light emitting layer. In addition, the light color depends on the chemical structure of light-emitting material, therefore it can be controlled by

material design. White color OLED lighting is fabricated by adjusting red, green and blue light-emitting materials to the appropriate ratio.

There are two different types of OLED. Traditional OLEDs use small organic molecules deposited on glass to produce light. The other type of OLED uses large plastic molecules called polymers. Those OLEDs are called light-emitting polymers (LEPs) or, sometimes, polymer LEDs (PLEDs). Since they're printed onto plastic (often using a modified, high-precision version of an inkjet printer) rather than on glass, they are thinner and more flexible.

The basic principle of operation of OLEDs and normal LEDs is identical. OLED emits light by conducting electricity to the organic materials that are sandwiched between the electrodes. On the other hand, LED emits light by conducting electricity to inorganic materials that are sandwiched between electrodes.

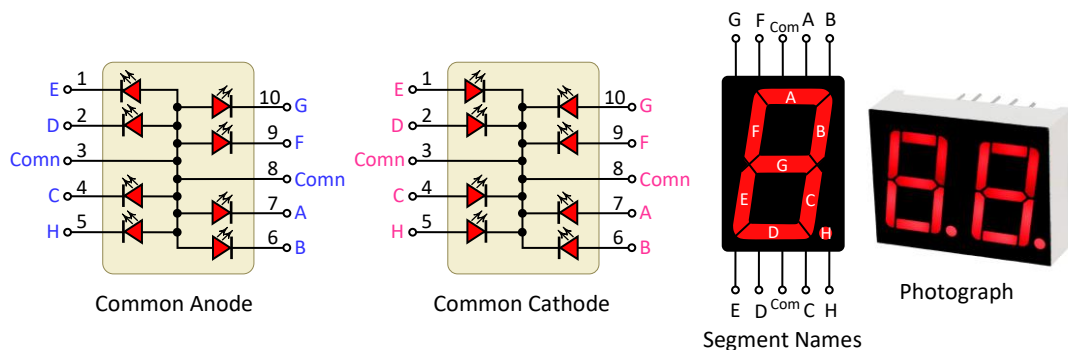


Fig.2-60: Construction of 7-segment display using LEDs

Electrons are provided to the emissive layer and removed from the conductive layer when there is current between the cathode and anode. This removal of electrons from the conductive layer leaves holes. The electrons from the emissive layer recombine with the holes from the conductive layer near the junction of the two layers. When this recombination occurs, energy is released in the form of light that passes through the transparent cathode material. If the anode and substrate are also made from transparent materials, light is emitted in both directions, making the OLED useful in applications such as heads-up displays.

OLEDs can be sprayed onto special substrates just like inks are sprayed onto paper during printing. Inkjet technology greatly reduces the cost of OLED manufacturing and allows OLEDs to be printed onto very large films for large displays like 80-inch TV screens or electronic billboards.

Applications of OLEDs

Currently, the major applications of OLEDs are in displays, but OLEDs are also used in lighting and are available at retailers for home lighting as well as street lighting. They tend to be less intense, so they are typically designed as panels. They are superior in efficiency and power consumption to standard LEDs and can be recycled at the end of their operating life. Since about 20% of the world's electricity production goes to lighting, saving energy for lighting can make a big difference in total energy use.



Fig.2-61: Various application of LEDs

Applications of Standard LEDs

Standard LEDs are used for indicator lamps and readout displays on a wide variety of instruments, ranging from consumer appliances to scientific apparatus. A common type of display device using LEDs is the seven-segment display. Combinations of the segments form the ten decimal digits as illustrated in Fig.2-60. Each segment in the display is an LED. By forward-biasing selected combinations of segments, any decimal digit and a decimal point can be formed. Two types of LED circuit arrangements are the *common anode* and *common cathode* as shown.

One common application of an infrared LED is in remote control units for TV, DVD, gate openers, etc. The IR LED sends out a beam of invisible light that is sensed by the receiver in our TV, for example. Also, IR light-emitting diodes are used in optical coupling, often in conjunction with fiber optics. More areas of applications include industrial processing and control, position encoders, bar graph readers, and optical switching.

2.30 Varactor Diode

The inherent property of a P-N junction diode is- *its internal capacitance varies with the variation of the applied voltage across it*. The varactor diode is also one kind of semiconductor diode that is used as a variable capacitor by changing the applied voltage. It is alternatively known as *varicap* or *voltacap* or *Voltage Variable Capacitor* (VVC) diode. Basically, varicap is used in reverse-biased and its mode of operation depends on its transition capacitance C_T . Reverse-biased junction of a diode behaves like capacitor.

2.30.1 Construction

If the reverse voltage across the diode is increased, the depletion layer widens, that decreases the junction capacitance. Hence, we can change diode capacitance by simply changing voltage. Silicon diodes which are optimized for this variable capacitance effect are called varactors. The construction, schematic symbols, and junction capacitance versus junction voltage curve for a varactor diode are shown in Fig.2-62. It consists of P-type and N-type semiconductor layers sandwiched together, with the N-type layer attached to a mesa structure (a portion of a large diode is etched to construct a specific device).

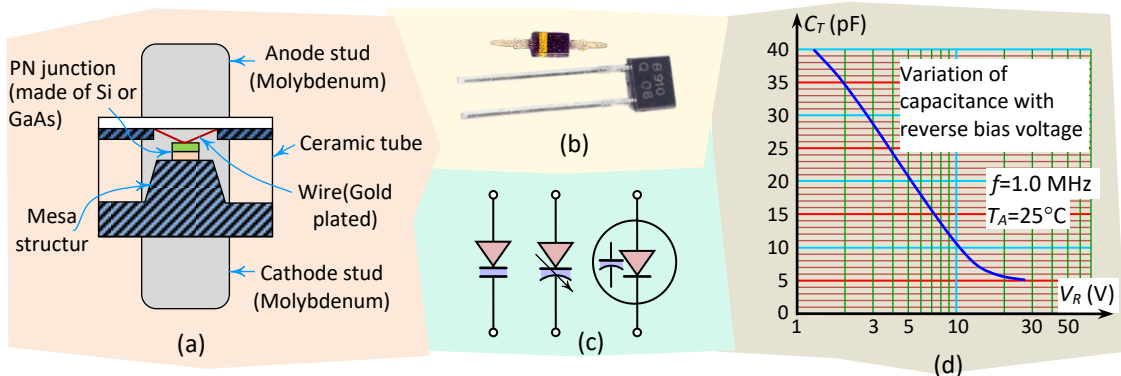


Fig.2-62: Varactor diode: (a) Construction (b) Photographs, (c) Schematic symbols, and (d) Typical characteristics

A gold plated molybdenum stud is connected to N-type layer by the mesa structure and it acts as cathode terminal. The P-type layer is connected to another gold plated molybdenum stud and acts as anode through a gold wire.

The entire arrangement is enclosed in a ceramic tube, except for some portion of the molybdenum studs. The P-type and N-type layers are made up of gallium or silicon arsenide depending on the type of application for which it is used. Silicon is used for low frequency applications and gallium arsenide is used for high-frequency applications. In varactor diode the concentration of impurities near the PN junction is very less and it gradually increases as we move towards the surfaces, as in step recovery diodes.

The application of a varicap in a typical tuning circuit is shown in Fig.2-63. Here C_1 and C_2 are two coupling capacitors, and behave as short circuit in the operating frequency range. The transistor current source pumps a fixed amount of current (few milliamperes) into the resonant LC tank circuit. The varactor diode is negatively biased by the control voltage (V_C) and acts as the capacitor (C) of the tank circuit. By varying the DC control voltage, we can vary the capacitance and hence the resonant frequency of the LC circuit. Depending on the value of LC , the tank circuit generates signal of different frequencies. The energy for the tank circuit is supplied by the transistor current source.

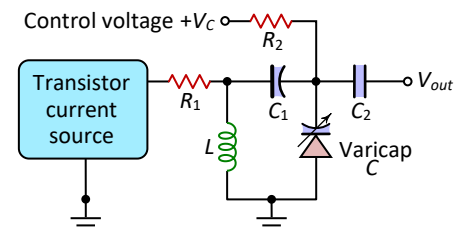


Fig.2-63: Tuning circuit using varactor

Applications

- Varactor diodes are widely used in frequency modulation
- Harmonic generators
- Tuning circuits- radios, TV tuners
- Automatic frequency controllers
- Frequency multipliers
- Active filters
- High frequency radios, etc.

2.31 Schottky Diode

The time, it takes to turn OFF a forward-biased diode is called the reverse recovery time t_{rr} [as describe in Section 2-23]. Due to this reverse recovery time, normal diode cannot be used for frequency above 10 MHz. To overcome this problem, German physicist **Walter H. Schottky** has invented the Schottky diode in 1914. A **Schottky diode** is formed by joining a doped semiconductor region (usually N-type) with a metal such as **gold**, **silver**, or **platinum**. Rather than a PN junction, there develops a metal-to-semiconductor (MS) junction, as shown in Fig.2-64. The forward voltage drop is typically around 0.3 V because there is no depletion region as in a PN junction diode. Similar to PN junction the MS junction represents a low resistance to current flow when it is forward biased (metal positive with respect to N-type silicon) and a high resistance when reversed biased. A depletion region and a barrier potential are established in a metal-semiconductor junction by a mechanism similar to

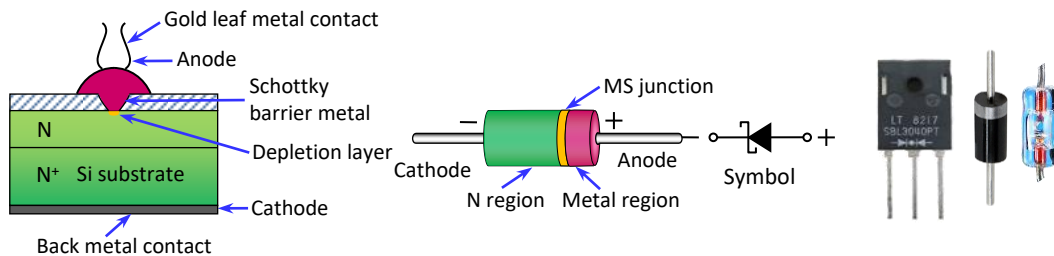


Fig.2-64: Construction and symbol of Schottky diode

that described for PN junction. The main difference is, the carrier diffusion consists of only electrons diffusion from semiconductor to metal. The electrons accumulated in the metal surface and the depletion region exists in the semiconductor side of the junction. Schottky diodes are majority-carrier device, in contrast to PN junction diodes which are minority-carrier devices (the current flows by the diffusion of minority carriers, i.e., electrons in the P-region and holes in the N-region). Since, minority carriers must be removed from the junction for a PN diode to be OFF, there is always a delay due to minority carrier lifetime when a PN diode is turned OFF (as described in Section 2-23). Having no minority carriers to be cleared from the junction, a Schottky diode can be turned OFF considerably faster than its PN-junction counterpart. Schottky diode is used in digital logic families due to its low voltage of operation and very high speed. Fig.2-64 represents the construction and symbol of Schottky diode.

Schottky diodes are high-current diodes used primarily in high-frequency and fast-switching applications. They are also called **hot-carrier diodes** or **Schottky barrier diodes**. The term hot-carrier is derived from the higher energy level of electrons in the N region compared to those in the metal region. A Schottky diode symbol is shown in Fig.2-64. The LS (stands for low-power Schottky) family of TTL logic is one type of digital integrated circuit that uses the Schottky diodes to reduce power loss and increase switching speed.

2.32 Step Recovery Diodes

Step recovery diode (SRD) is a type of semiconductor diode, specially designed for low reverse recovery time, t_{rr} . It is also known as a **voltage-dependent variable capacitor**, **snap off diode**, and **charge storage diode**. It works like a normal diode, i.e., it conducts current in forward biased condition and blocks current in reverse biased condition. The main difference is it can instantly switch from ON state to OFF state.

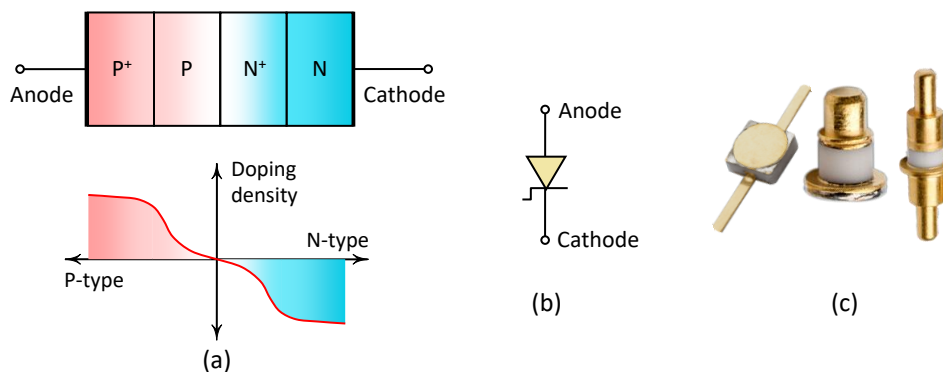


Fig.2-65: (a) Construction of step recovery diode, (b)Electrical symbol, and (c) Photographs

Step recovery diode has lower doping concentration than a conventional diode. Moreover, the doping concentration decreases near the junction and, thus, the density of the charge carrier is low around the junction. Therefore, very small (almost negligible) amount of charge is stored near the junction. It provides very fast switching speed from ON-state to OFF-state and hence the value of t_{rr} is small compared to the normal rectifier diodes. The SRD snaps open state instantly, and this phenomenon is called **reverse snap-off**. Therefore, it is also known as **snap off diode**. The values of t_{rr} are 10 ns to 100 ns.

Construction-wise step recovery diode has a similar structure as a conventional diode except for its doping concentration. It is a PN junction diode with a doping concentration that decreases gradually near the junction. The charge carrier density is maintained very low near the junction as compared to that near the terminals. Due to this unusual doping, it has different switching characteristics than a normal diode. The doping profile and symbol of SRD are given in Fig.2-65(a) and (b).

Step recovery diode works as an ordinary diode at low frequencies, i.e., they conduct when forward biased and do not conduct when reverse biased. The forward bias voltage of this diode is little bit higher than normal diode. It may be 1.3 V to 3.6 V. Moreover, the reverse bias voltage tolerance of this diode is also high. When it is switched from forward bias to reverse bias it changes state instantly.

The cut off frequency of step recovery diode is 200 GHz – 300 GHz. This high range of cut-off frequency enables the diode to be used as a multiplier for circuits operating at a frequency of about 10 GHz. Step recovery diodes can generate sharp pulses that makes it appropriate for using as harmonic generators. It is used in voltage controlled oscillators, in frequency synthesizers, in Comb Generator for generating multiple harmonics of its input signal, phase detector for detecting the phase in a communication system. They are also used in modulation of signal.

2.33 Small Signal Diodes

Small signal diode is a type of PN junction diode which is used for voltage signals. The junction area of this diode is very small, and hence, the stored charge is very small, that results in low capacitance. Small-signal diode is also called the **signal diode** or **switching diode**.

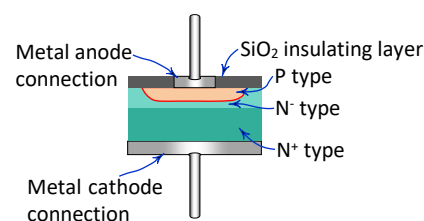


Fig.2-66: Construction of planar small signal diode

The basic construction for a silicon planar diode is to infuse a P-layer on top of an N-layer. The N-layer consists of two layers – low-doped N^- layer and a highly-doped N^+ layer as shown in Fig.2-66. The N^- and the P-layer form a PN junction. As the P-layer is infused onto the N^- base layer, the shape of the PN junction becomes curved. The curvature of the junction makes the parameters of the diode more difficult to predict accurately. Therefore, an improved small signal diode is made using a process as shown in Fig.2-67. After the P type layer is made, and while the diode is still part of a silicon wafer containing many identical components, a section of the PN block is etched away to form a diode, which is known as **Mesa Diode**. This device now has a linear small-junction, giving a lower junction capacitance and the linear PN junction gives more predictable characteristics. Here, also, the N-type layer is divided into N^- and a N^+ layers which give a better distribution of electric charge across the diode, helping to reduce the reverse potential across the junction and to give better reverse

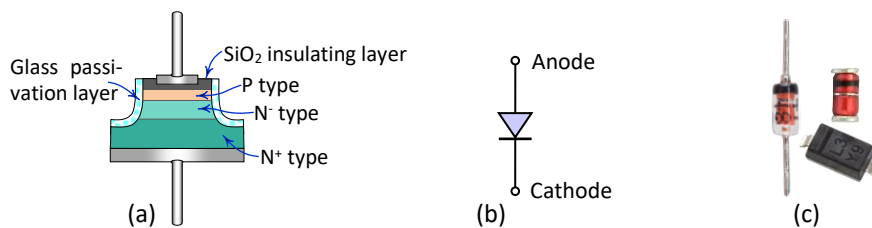


Fig.2-67: (a) Construction of small signal Mesa diode, (b) Electrical symbol, and (c) Photographs

voltage capability. The sides of the mesa are also coated with an insulating layer of either silicon oxide or glass to passivate the junction area. This passivation protects the junction from deterioration due to oxidization. Many types of signal diodes are available in both wire ended and surface mount (SMT) format [Fig.2-67(c)].

These diodes are used in circuits with high frequencies or fast pulses and have a very fast Reverse Recovery Time (t_{rr}), typically a few nanoseconds or less. They are physically smaller than dedicated power rectifiers and have lower maximum reverse voltage parameters.

2.34 Point Contact Diode

A **Point Contact Diode** is formed by touching a metallic wire with an N-type semiconductor to form a small area of contact. This forms a small point junction and hence the name **Point Contact** diode. It is widely used because such a small point junction possesses a small value of junction capacitance. Thus, the charge storage at the junction is low. Due to this, the switching ability of this diode is much better than a conventional diode.

As shown in Fig.2-68(a), it is constructed with a contact of an N-type semiconductor substrate and Tungsten or Phosphor Bronze wire which is called **cat-whisker**. The semiconductor used in the construction of point contact diode can be either Silicon or Germanium. But Germanium is used extensively because it possesses higher carrier mobility. The dimension of the semiconductor substrate is about 1.25 mm^2 and its thickness is 0.5 mm. One face of the semiconductor substrate is soldered to the metal base by the technique of radio frequency heating. The contact point of the cat-whisker is made very sharp so that the contact area is very small. The anode and cathode

terminals are connected through metallic contacts. The whole assembly is enclosed inside a glass or ceramic cover.

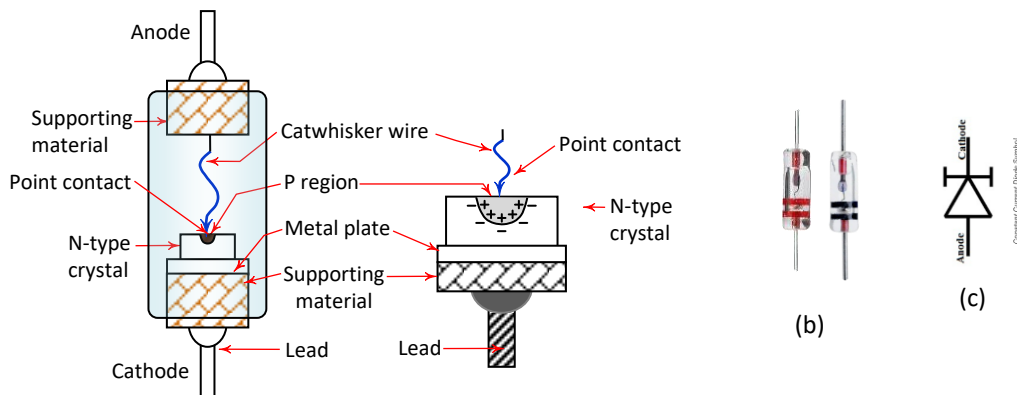


Fig.2-68: (a) Construction of point contact diode, (b) Photograph, and (c) Electrical

When forward bias is applied to point contact diode the current produced in the device is passed through the cat-whisker. So, the cat-whisker tungsten wire gets heated. Due to this heating, the wire undergoes deformation. Thus a small gap is deliberately left for the expansion of wire under the large current. The junction capacitance and diffused capacitance in this diode is very small, i.e., about 0.1 pF to 1.0 pF. This is because the area of contact between the wire and the N-type substrate is very small. Due to the small area of junction, the density of charge carriers near the junction is very low. Thus, the low charge storage imparts it the ability of fast switching.

The main applications of point contact diodes are – high frequency circuits, radio frequency mixers, video detector circuits, envelope detectors.

2.35 Transient Voltage Suppression Diode

The **Transient Voltage Suppression (TVS)** diodes are solid state PN junction devices. They are specifically designed in order to protect sensitive semiconductor devices or circuits in a PCB layout. They are used to prevent transient events associated with electrostatic discharge (ESD).

The constructions of the TVS diodes are same as the Zener diodes. The major difference between TVS diode and Zener diode is that TVS diodes are designed to suppress transient voltage while Zener diodes are designed to regulate voltage. The cross-sectional area of this diode is very large to increase the current capacity and, hence, power rating. The larger the cross-sectional area of the TVS diode, the higher is the absorbing capacity of transient currents.

As shown in Fig.2-69(b), the TVS diodes are used in parallel to the component/circuit to be protected from surge.

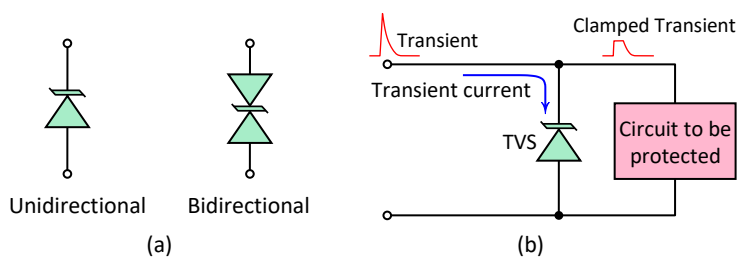


Fig.2-69: TVS diode: (a) Symbols and (b) Typical circuit connection

In normal condition, TVS diode will have no effect in the circuit because it will behave as open or high impedance. But, when the applied voltage exceeds its threshold voltage, the junction of the TVS diode breaks down due to the avalanche mechanism as described in Section 2.27.2. The diode becomes short momentarily and diverts the transient current from flowing through the load. Hence, the load is protected. Voltage across the circuit is limited to TVS diode clamping voltage. After the transient voltage vanishes, diode returns to high impedance state again. TVS diodes are available in wide range of voltages. Typically the diodes are available from 5 V to 440 V. There are two types of TVS diodes – *Unidirectional* and *Bidirectional*. Unidirectional diodes are used for DC voltage and the bidirectional diodes are used for AC voltage. The symbols of the TVS diodes are shown in Fig.2-69(a).

TVSs are used as board level protectors for semiconductors as well other sensitive parts. The applications of TVS diodes include MOS memory, AC power lines, signal and data lines, telecommunication equipments etc. from lightning surges as well as unwanted voltage transients.

Varistor diodes have a laminated structure, with breakdown voltage and capacitance being controlled by the number of layers and the interlayer design. On the other hand, TVS diodes are a combination of P-type and N-type semiconductors. They are silicon-based ESD protection components.

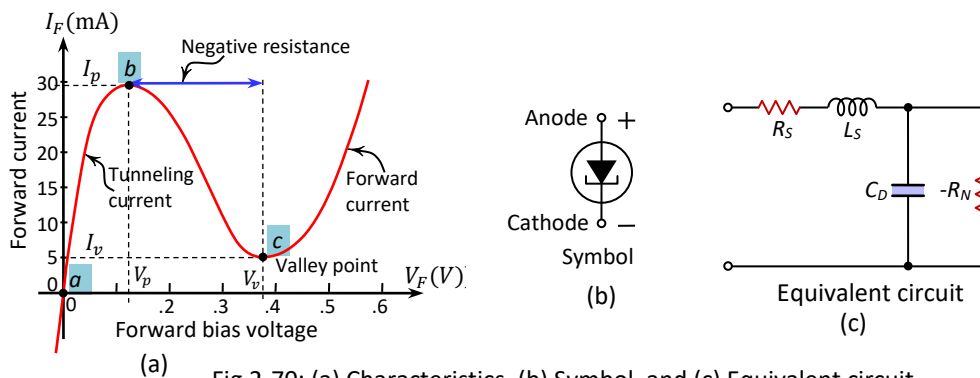


Fig.2-70: (a) Characteristics, (b) Symbol, and (c) Equivalent circuit

2.36 Tunnel Diode

An ordinary PN junction diode is doped by impurity atoms in the concentration $\sim 10^{15} \text{ cm}^{-3}$. With this order of doping, the width of the depletion layer is of the order of a micron. Now if the concentration is greatly increased to $\sim 10^{18} \text{ cm}^{-3}$ then the width of the depletion layer reduces to about $100 \text{ \AA}$ or 1/100 times the width of the depletion layer of an ordinary PN junction diode. This

width of $100 \text{ \AA}$ is extremely small, being only about $1/5$ of the wavelength of the visible light. This type of diode with very small depletion layer is called a **tunnel diode** because due to this extremely thin depletion layer many carriers can **tunnel** through the junction. This heavily doping also reduces the reverse breakdown voltage to a very small value (approximately zero) with the result that the diode appears to be broken down for any reverse voltage. Since in tunnel diode the tunneling action of the carrier takes place at the velocity of light, it is a useful circuit element even at microwave frequencies. Also, its small physical size and hence the associated stray capacitance and lead inductance are so low that they can be properly used at very high frequencies.

Tunnel diodes are usually fabricated with germanium, gallium-arsenide (GaAs). Fig.2-70(a) represents the tunnel diode forward current-voltage characteristics and Fig.2-70(b) shows the symbol. It is clear from the curve that the diode starts conducting from zero voltage, immediately with the application of the forward bias. As the forward bias voltage increases, the

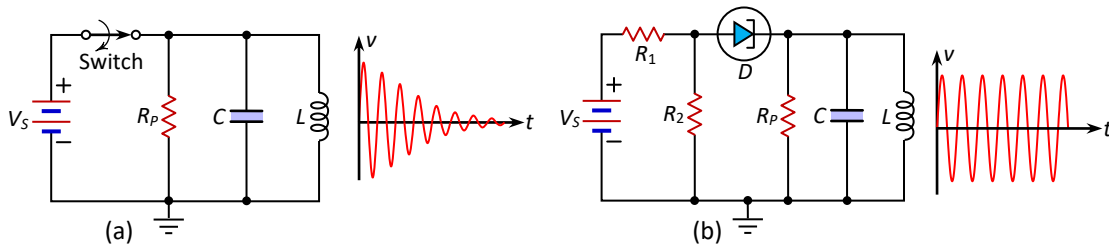


Fig.2-71: (a) Damped oscillation in a parallel resonant circuit (b) Undamped oscillation in the same circuit with tunnel diode

current rises quickly and reaches a peak value, I_p , when the forward voltage is V_p (point *b*). When forward voltage is increased further, the current starts decreasing till it achieves its minimum value, called the **valley current**, I_v , corresponding to valley voltage V_v (point *c*). For further increase in voltage (greater than V_v) the diode current increases again as in ordinary junction diodes. In the region *bc*, immediately to the right of *b* [Fig.2-70(a)], current decreases with the increase in forward voltage. This characteristic is called the **negative resistance** characteristics between peak current I_p and valley current I_v . This negative resistance is equal to the reciprocal of the average slope of the characteristics in this region and is determined from the relation

$$R_n = -\frac{dv}{di} \quad (2-50)$$

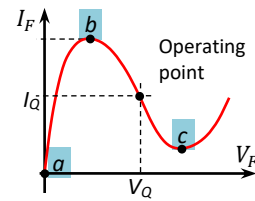


Fig.2-72: Operating point for oscillator

When a tunnel diode operates in the negative resistance region it can be represented by an equivalent circuit as in Fig.2-70(c). In the equivalent circuit, L_S is due to lead inductance significant at very high frequencies. R_S is due to bulk resistance of the device, lead and ohmic contacts at lead-semiconductor junction. C_D is due to depletion layer capacitance associated with the forward biased junction. $-R_N$ is negative resistance component of the diode.

A parallel resonant circuit consists of a capacitance, inductance, and resistance in parallel, as in Fig.2-71(a). R_P is the equivalent parallel resistance of the series resistance of the coil. When the power switch is turned ON a damped sinusoidal output is developed. The damping is due to the

resistance of the tank circuit. When current flows through this resistance, some energy is lost ($I^2 R_P$ loss) and causes decaying of the wave.

If a tunnel diode is placed in series with the tank circuit and biased at the center of the negative-resistance portion of its characteristic curve, as shown in Fig.2-72, a sustained (constant amplitude) oscillation will result in. This is because the negative-resistance characteristic of the tunnel diode counteracts the positive-resistance characteristic of the tank resistance. In the circuit of Fig.2-71(b), resistors R_1 and R_2 are used to bias the tunnel diode. The tunnel diode is only used at very high frequencies.

2.37 Photodiode

It is one kind of light sensor that converts light energy into electrical energy (voltage or current). **Photodiode** is a type of semiconductor device that consists of a P-type and an N-type layer, i.e., with a PN junction. Between the P-type and N-type materials, sometimes an intrinsic layer is present. This type of structure is called **PIN** (**P-type Intrinsic N-type**) diode. Its construction and characteristics are shown in Fig.2-73.

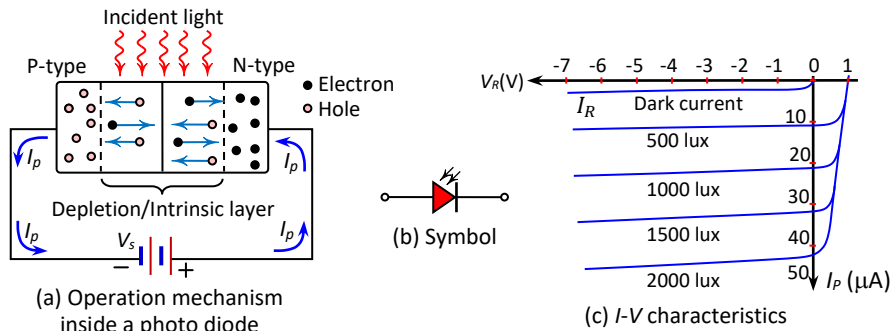


Fig.2-73: Photo diodes: (a) Operation, (b) Symbol, and (c) I-V characteristics

Photodiode materials are made of generally silicon (Si), germanium (Ge), and indium gallium arsenide (InGaAs), indium-gallium-arsenide-phosphide (InGaAsP), etc. Internally, a photodiode has optical filters, built in lens, and an exposed surface through which light can enter into the PN junction [Fig.2-74]. When surface area of photodiode increases, it results in slow response, but the sensitivity increases. The photographs of some photodiodes are shown in Fig.2-74.

The photodiode accepts light energy as input to generate electric current or voltage. It is also called as **photodetector**, **photo sensor** or **light detector**. Photodiode is an active device and normally used in reverse biased condition. With a suitable reverse voltage, the reverse current is directly proportional to the light intensity incident on it. As the current is generated due to the incident light, it is called photo-current, I_p . The I-V characteristic of this device is shown in Fig.2-73(c).

A reverse-biased PN junction has a small amount of reverse current I_R due to thermally-generated electron-hole pairs. In silicon, I_R is in the range of nano-amperes. But, for photodiode, the number of these minority carriers is increased by light energy. The characteristic curves of Fig.2-73(c) show that for a given reverse voltage, I_p increases with increase in the level of illumination. By changing the illumination level, reverse current can be changed. In this way, reverse resistance of the diode can be changed by a

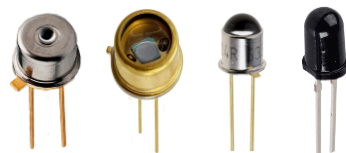


Fig.2-74: Photographs of photodiodes

factor of nearly 20. This mode of operation is called the **photoconductive mode**.

Photoconductive Mode

In this operation, trans-impedance amplifiers (current to voltage converters) are used as preamplifiers for photodiodes, as shown in Fig.2-75. This amplifier converts the current of photo diode into voltage. Moreover, it provides a suitable voltage level (generally small voltage) for the photodiode in the input side. In case of PIN structure, the existence of intrinsic region enhances the area for radiation absorption. Hence, this process increases the sensitivity of this device. The photodiode can work in two other modes

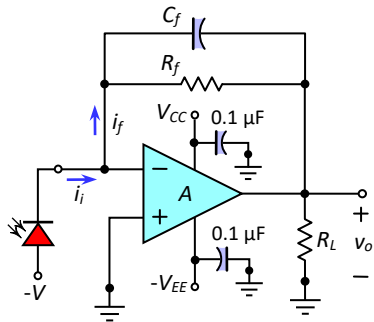


Fig.2-75: Application of photodiode as photoconductive mode

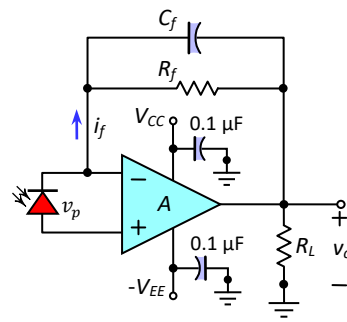


Fig.2-76: Application of photodiode as photovoltaic mode

Photovoltaic Mode

This is otherwise called as **zero bias mode**. When a photodiode operates in low frequency applications and ultra-level light applications, this mode is preferred. When photodiode is irradiated by a flash of light, voltage is produced. This voltage will have a very small dynamic range and it has a non-linear characteristic. When photodiode is configured with Op-Amp in this mode, there will be a very less variation with temperature. The circuit arrangement is shown in Fig.2-76.

Avalanche Diode Mode

In this mode, the diode operates at a high reverse bias condition. It allows avalanche multiplication (as described in Section 2-27-2) to each photo-produced hole-electron pair. Hence, this operation produces internal gain within the photodiode. The internal gain increases the device response. The circuit arrangement for this mode will be exactly same as in Fig.2-75.

Applications of photodiode include i) Optical communication ii) Light detector iii) Switching etc.

2.38 PIN Diode

A PIN diode is a 3-layer device in which an intrinsic layer (undoped semiconductor) is sandwiched between a P-type and an N-type regions which results in a PIN structure. The intrinsic region offers very high resistivity of the order of $0.1 \Omega \cdot \text{m}$. Along with high resistance a PIN diode also possesses a lower value of capacitance due to the wider intrinsic layer between the P and N layers.

2.38.1 Working Principle of PIN Diode

A PIN diode is almost similar to a normal PN junction diode except the presence of intrinsic region, as shown in Fig.2-77. When no external bias voltage is applied, a depletion layer is produced in the same way as in normal PN junction diode. The depletion layer is wider on the I-region compared to N-region because of high doping of N region.

If the diode is forward biased, the width of the depletion layer decreases and current flows through the device due to the movement of the majority carriers. If the diode is reverse biased, the width of the depletion layer increases. For a particular reverse voltage the entire I-region becomes depleted. This particular voltage is known as **swept out voltage**. Usually, its value is -2 V . In this condition, very small amount of reverse current flows through the device due to the minority carriers.

2.38.2 Some Important Features of PIN Diode

Low capacitance: As we already discussed that a PIN diode offers a lower value of capacitance due to the larger distance between P and N regions.

High breakdown voltage: Due to the presence of intrinsic region, PIN diode exhibits a higher value of breakdown voltage. This is because a higher voltage is required to destroy the thick depletion region.

Sensitive to light: The depletion region is responsible for generating energy when radiation falls on its surface. The existence of an intrinsic region enhances the area for radiation absorption. Hence, PIN diodes are widely used as photodetectors.

2.38.3 Applications of PIN diode

As already discussed, PIN diodes are used as photodetectors, radio frequency switches, voltage rectifiers. PIN diode possesses a faster response time, and hence, they these are widely used in microwave applications.

2.39 Gunn Diode

Gunn diode is a high frequency microwave semiconductor device. It is fabricated with N-type semiconductor material only, not a PN junction device. In 1963, **J. B. Gunn** invented this device.

Even though it has no junction, it is called a diode, because it has two terminals (anode and cathode) attached to it. The Gunn diode is made up of a single piece of N-doped semiconductor alloy- such as GaAs, InP, etc. As shown in Fig.2-78(a), two thinner N^+ layers are made on opposite ends of the active material (N-layer) to connect the anode and cathode leads. Moreover two Au-Si alloy layers are used as contact mounts. Generally, the top contact mount is connected to gold-coated-lead by Au wire and the bottom Au-Si layer works as a buffer layer above the substrate. The entire assembly is enclosed by ceramic walls.

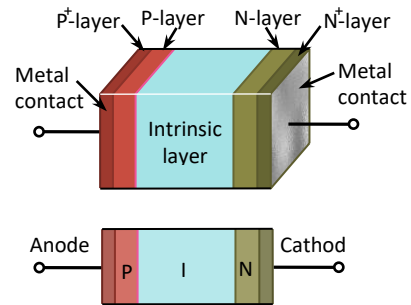


Fig.2-77: Constructional blocks of a PIN diode

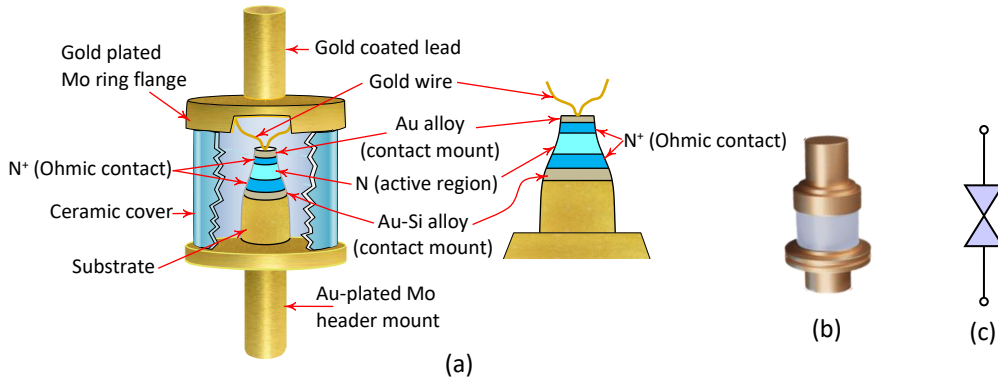


Fig.2-78: (a) Construction of Gunn diode, (b) Photograph, and (c) Electrical symbol

It is an active two-terminal solid-state device that shows **Gunn effect**. Its operation can be explained on the basis of i) domain formation, ii) two-valley theory of **Ridley–Watkins–Hilsum** (RWH), or iii) the transfer electron mechanism.

As shown in Fig.2-79, there are two valleys in the conduction band of the N-type GaAs. At room temperature, the valence band is full of valence electrons, and the conduction band is partly filled. The forbidden energy gap between the valence band and the conduction band is about 1.43 eV. Electrons in the lower valley exhibit a small effective mass (m_1) and very high mobility, μ_1 . Electrons in the upper valley exhibit a large effective mass (m_2) and very low mobility, μ_2 . The two valleys are separated by a small energy gap, $\Delta E \approx 0.36$ eV. For a lower applied potential difference across the material, a low electric field will be produced. For this electric field, electrons in the lower valley, moves with a higher mobility μ_1 , and hence, the material behaves Ohmically. For a certain higher potential difference, the electric field reaches a certain

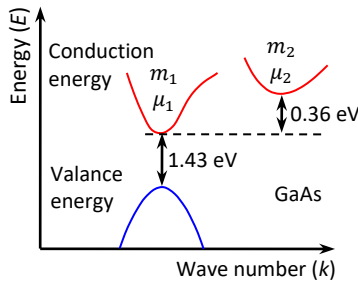


Fig.2-79: Energy band diagram of GaAs

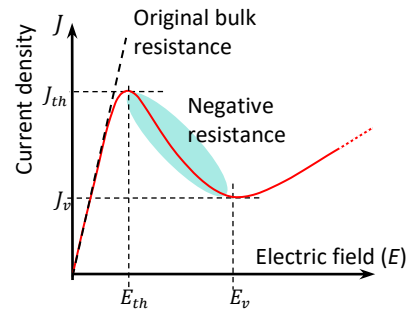


Fig.2-80: Characteristics of Gunn diode

threshold value (E_{th}), and the electrons are swept from the lower valley to the upper valley where the mobility of the electrons is low (μ_2). Low mobility of electrons results in low current. If the rate of transfer of electrons from the lower valley to the upper valley is very high, the current will decrease with an increase in voltage as shown in Fig.2-80. This portion of the I - V characteristics is called the negative resistance region. The average electron mobility, μ is given by

$$\mu = \frac{n_1\mu_1 + n_2\mu_2}{n_1 + n_2} \quad (2-51)$$

where

- n_1 is electron density in lower valley
- n_2 is electron density in upper valley
- μ_1 is mobility of electron in lower valley
- μ_2 is mobility of electron in upper valley
- $n_1 + n_2$ is total carrier concentration

The importance of the Gunn diode lies in its dynamic negative resistance characteristic [Fig.2-80]. In any N-type semiconductor, the following relations govern current, field and drift velocity.

$$v_d = \mu E \quad (2-52)$$

$$\therefore \frac{d(v_d)}{d(E)} = \mu \quad (2-53)$$

$$J = nq\mu E \quad (2-54)$$

$$\therefore \frac{d(J)}{d(E)} = nq\mu \quad (2-55)$$

From Equ.(2-54), we find that the current density is directly proportional to the mobility (μ). Therefore, the current density decreases when the charge carriers move from lower valley to the higher valley, where the mobility of charge carriers is low.

2.39.1 Domain Formation in Gunn Diode

When bias voltage (V_B) is applied across the diode terminals, an electric field (E_0) is established across the N-type GaAs, as shown in Fig.2-81. As E_0 increases, electron drift velocity increases up to a certain value (E_{th}). This is because the drift velocity (v_d) is directly proportional to the applied electric field.

However, with a further increase in bias voltage, the mobility of electrons decreases, which causes the velocity to decrease, and thereby slows down the electrons. This slowing-down results in a traffic jam, and more electrons accumulate to a layer. This layer is called **domain** or **charge layer**. This charge layer produces an electric field (E) in such a direction that it decreases the original field (E_{0i}) on the left (E_{0f}) and increases it to the right (E_{0f}). As a result, the accumulated charge suddenly pushed toward the anode on the right, forming a current pulse. When this pulse moves to the anode, E_0 goes back to the original value, and starts another domain and again another current pulse.

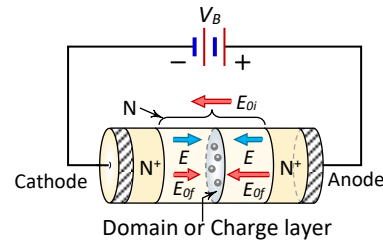


Fig.2-81: Domain formation in Gunn diode

In this way, N-type GaAs releases the power that it has acquired during the domain formation in the direction to the anode. In favorable conditions, this release of power at anode will be utilized by the noise signal of a particular frequency, depending on the Gunn diode configuration, and gets amplified. The amplification of noise energy results in sustained oscillations of a particular frequency. The frequency of oscillation depends on the length of the GaAs piece and the concentration of electrons or doping concentration.

Bias voltage and other parameters

The value of biasing voltage for the Gunn diode depends on the active layer (N-layer) width. For GaAs as the active material, the value of E_{Th} is 3.3 kV/cm and that of E_v is 11 kV/cm [see Fig.2-80]. For proper operation, the device must be biased in between these two points. Let us assume a typical value of applied electric field, say $E_B = 6$ kV/cm. Therefore, for $L = 5 \mu\text{m}$ the required bias voltage should be,

$$V_B = 6 \text{ kV}/(1 \text{ cm}) \times (5 \mu\text{m}) = 6000 \text{ V}/(10^{-2} \text{ m}) \times (5 \times 10^{-6} \text{ m})$$

or,

$$V_B = 3 \text{ V}$$

Similarly, for $L = 50 \mu\text{m}$, the bias voltage should be, $V_B = 30 \text{ V}$.

The current density at 6 kV/cm is 400 A/cm². Therefore, for $3 \times 10^{-8} \text{ m}^2$ area, $I = 120 \text{ mA}$.

Applications

Gunn diode is used in i) Radar transmitters as a low power oscillator, ii) Oscillators with pulse signals, iii) Transponders in telemetry and air traffic control industries, iv) Broadband linear amplifier v) Sequential logic circuits and fast combinational circuits vi) Microwave receivers as low and medium power oscillator vii) Parametric amplifier as pump sources, viii) Local oscillator in the microwave frequency range of 1 GHz to 100 GHz.

2.40 Shockley Diode

Shockley diode is a four-layer, PNPN device. It conducts when it is forward biased and stop conducting when it is reverse biased. The major difference between conventional diode and Shockley diode is- it starts conducting when the forward voltage exceeds **break-over voltage**. Shockley diode operates in two states: either ON or OFF. Thus, it is used as a switching device. According to its construction, it is called PNPN diode. It is a two-terminal device that is why it is called as diode.

As shown in Fig.2-82, it is constructed by sandwiching four layers of semiconductor– P-type, N-type, P-type and one more N-type layer. Due to sandwiching four layers together, three PN junctions (J_1 , J_2 , and J_3) are formed.

When forward voltage is applied across the diode, that is, anode is connected to higher potential and the cathode is connected to lower potential, junctions J_1 and J_3 become forward biased. But junction J_2 is reverse biased. So, the applied voltage is almost entirely dropped across this reverse biased junction J_2 . When the applied voltage exceeds a threshold value, J_2 breaks down and the device behaves as (almost) short circuit.

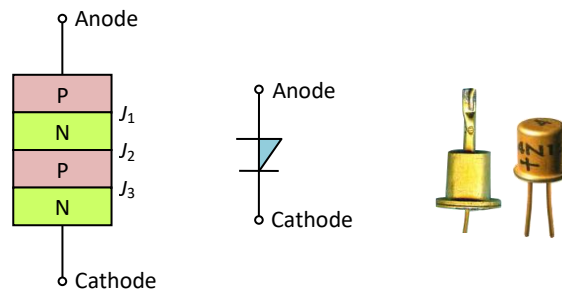


Fig.2-82: Shockley diode: (a) Construction (b) Symbol, and (c) Photographs

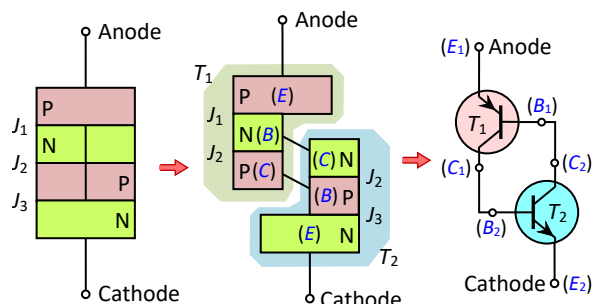


Fig.2-83: Two-transistor analogy of Shockley diode

On the other hand, for reverse bias condition, J_1 and J_3 become reverse biased and J_2 is forward biased. Here also for a certain reverse bias voltage, the diode gets catastrophic break down. So, the diode is not operated in this breakdown condition.

The operation of a Shockley diode can be understood by using a **two-transistor model**. The four layers constitute internally two transistors T_1 (PNP) and T_2 (NPN) connected as shown in Fig.2-83. Junction J_1 is the $B-E$ junction of transistor T_1 and junction J_3 is the $B-E$ junction of transistor T_2 . Junction J_2 is common to both the transistors and forms the $B-C$ junction of them.

The I - V characteristic curve of this device is shown in Fig.2-84. When the applied voltage is greater than the threshold value, J_1 breaks down and emitter current (I_{E1}) flows in transistor T_1 . Due to this I_{E1} , I_{C1} is also produced. This I_{C1} works as the base current (I_{B2}) of transistor T_2 . Due to this I_{B2} , I_{C2} is produced which is also the base current of T_1 . Thus, one transistor assists the other to increase their currents. Finally, both the transistors become saturated, and the voltage across J_2 decreases but current increases. This portion of characteristic curve (shown by shaded area) is called negative resistance region, as the voltage decreases with the increase in current. After being saturated, the current increases from a particular limit, which is called the **holding current**, I_H . To keep the diode ON, the holding current is the minimum current. The diode will be OFF, if it is reverse biased or the forward current falls below the holding current ($I_F > I_H$).

Shockley diodes are used as triggering devices and as relaxation oscillators. They are mainly used to trigger silicon controlled rectifier (SCR).

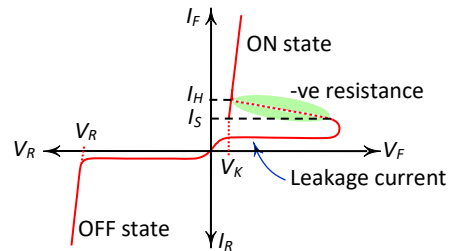


Fig.2-84: I - V characteristics of Shockley diode

2.41 Liquid Crystal Display

Liquid crystal display (LCD) is a flat, thin display device that has replaced the older CRT displays. Although, nowadays LED and OLED displays have become popular, LCDs are used for small devices like— calculators, watches, multimeters etc. as dot matrix and 7-segment designs. Several organic compounds exhibit optical properties of a crystal though they remain in liquid form. These materials are used in LCD displays. Depending on the behavior of the liquid crystals, there are two types of LCDs

- Field effect type and
- Dynamic scattering type

2.41.1 Field Effect Type

Field effect LCDs are of two types- transmissive and reflective. Their constructions are same, except a reflective used in the second type. For both types, nematic type liquid crystal is layered between glass-plates with transparent electrodes deposited on the inner faces as shown in Fig.2-85.

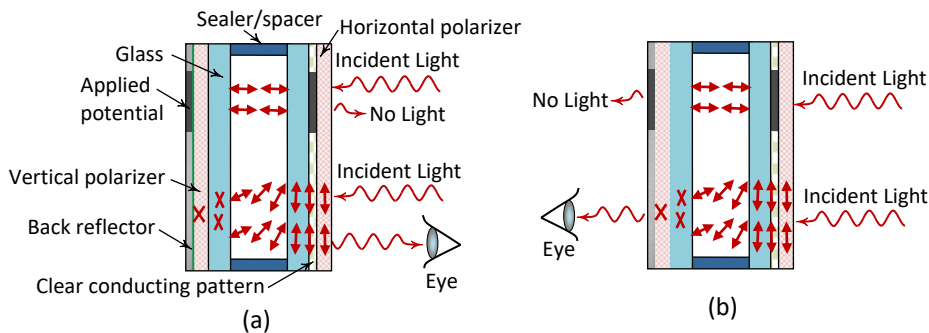


Fig.2-85: Construction of LCD (a) Reflective and (b) Transmissive

Two thin optical polarizers, one with vertical polarization and the other with horizontal polarization, are placed on each glass sheet. As shown in Fig.2-86(a), grooves are made on the inner faces of the glass plates. Vertical grooves are made on the glass attached to vertical polarizer and horizontal grooves are made on the glass attached to horizontal polarizer.

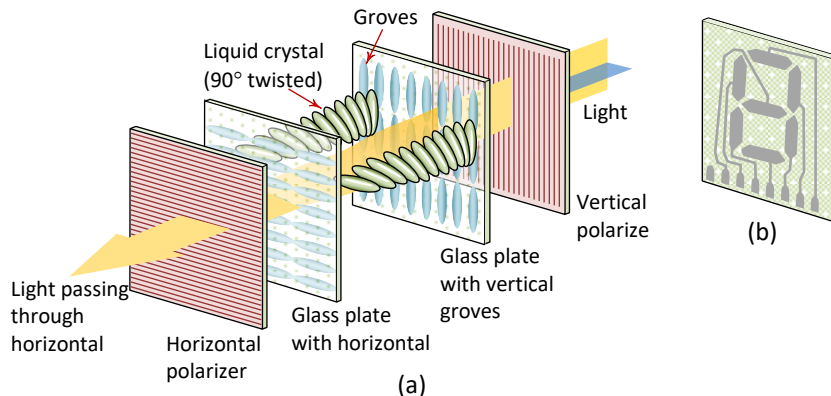


Fig.2-86: Transmissive LCD with no applied potential (a) 3-D Construction, (b) Front transparent 7-segment contacting layers design

Without any applied potential, the rod-like atoms of liquid crystal align with the vertical and horizontal grooves of the glass plates and become 90° twisted. Light becomes vertically polarized passing through the vertical polarizer. When light passes through the liquid crystal it becomes 90° twisted or horizontally polarized. This horizontally polarized light easily comes out through the horizontal polarizer. Thus, the segment of LCD looks bright without any applied potential.

On the other hand, when a potential is applied to the transparent electrodes [as shown in Fig.2-86(b)], the rod-like atoms align parallel to the electric field, developed by the applied potential, (becomes perpendicular to the glass plates) as shown in Fig.2-87. Now, the 90° twisting of the liquid crystal does not exist. Therefore, the vertically polarized light is not rotated and try to come out of the horizontally polarizer. Due to the mismatch of the polarizations, light cannot pass through the horizontal polarizer. So the segment, where potential difference is applied, looks dark. The operation described so far needs a source of light at opposite to viewing face of the LCDs and this variety is called **transmissive LCD**. The main advantage of this LCD is that it can be used without any ambient light.

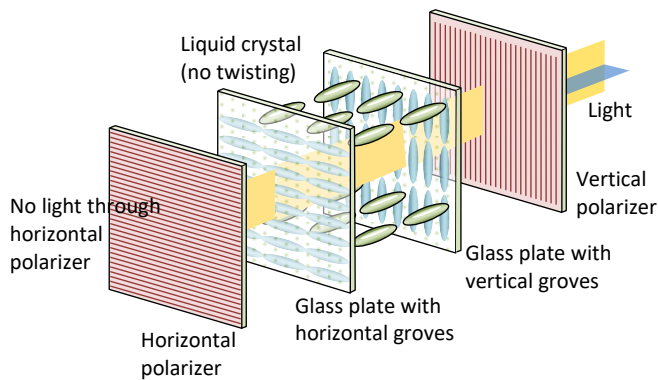


Fig.2-87: 3-D Construction of transmissive LCD with applied potential

On the contrary the reflective variety of LCD works with ambient light, so, it cannot be used at dark places- like at night. The construction of this LCD is shown in Fig.2-85(a). Here, the LCD does not have light source of its own. In place of the light source, a reflecting plate is used after the vertical polarizer. This plate also works as the back electrode. Light falls on the LCD and get reflected from the reflector, if no field is applied. But if electric field is applied, 90° twisting of the crystal atoms breaks and light does not come out. So the LCD looks dark.

2.41.2 Dynamic Scattering Type

The construction of the dynamic scattering LCD is similar to that of reflecting LCD. Only the difference is, here, there is no polarizing sheet. Fig.2-88 shows the construction of this type of LCD. It consists of two glass plates with the nematic liquid crystal fluid in between the plates. The back plate is coated with thin transparent layer of conductive oxide; whereas front plate has a photoetched conductive coating with seven segment pattern or dot-matrix pattern (or any other desired pattern) as shown in Fig.2-88. With the pure nematic liquid crystal, impurities (or ions) are added to increase its conductivity.

Working Principle

In the normal state, without application of electric field, all the molecules in the nematic liquid crystal are parallel to the glass plate (lower portion of Fig.2-88). So, the incident light will simply pass through the liquid and the liquid-crystal structure will appear clear or transparent.

When the electric field is applied (upper portion of Fig.2-88), the dipoles of molecules rotate into alignment in the direction exactly perpendicular to the field direction produced by the applied voltage. Now, the ions are pulled by the electrodes and hence the positive ions will go towards negative potential and vice versa and a turbulence is produced. When the applied voltage is greater than a threshold level (for commercial units the threshold level is usually between 6 V and 20 V), the turbulence becomes high and the crystal atoms achieve completely random orientations.

So, the light falling on them is scattered and hence the assembly appears dark. When the field is switched OFF, the molecules are rearranged and the assembly becomes transparent again.

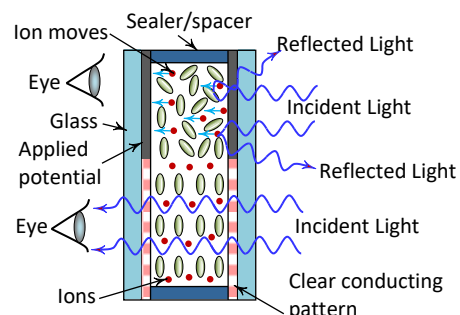


Fig.2-88: Construction and operation of dynamic scattering LCD

A digit on an LCD display may have the segment appearance shown in Fig.2-86(b). The darker areas are actually a clear conducting surface connected to the terminals below for external control. On the back side there is also a conducting plate. In normal number display system, 7 segments are made and controlling ON/OFF of the segments 0 to 9 can be generated. On the other hand, alphanumeric displays have many dots that can be controlled individually [Fig.2-89]. Here, the characters are displayed by controlling a matrix of 5×7 dots.

The main advantages of LCD are: Low power, Good contrast, and Low cost. On the other hand, the main disadvantages of LCD are: Slow operational response, Large size, Small life span for DC applications etc.

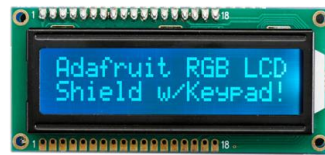


Fig.2-89: Photograph of an LCD

2.42 Solar Cell

The **solar cell**, also called **photovoltaic cell**, is simply a semiconductor device that absorbs light energy from the sun (or any other sources) and converts it into electrical energy. In 1839, French scientist **Edmond Becquerel** discovered the photovoltaic effect at the young age of 19. Willoughby Smith first described the effect of light on selenium during the passage of an electric current in 1873. But, the first solar cell was invented by **Charles Fritts** in 1883, where he coated a thin layer of selenium (a semiconductor material) with an extremely thin layer of gold. The resulting cells had a conversion efficiency of only about 1%. This invention led to the launching of a movement for producing solar energy. Solar cells were first used in a prominent application on the Vanguard satellite in 1958, as an alternative power source to the primary battery power source.

2.42.1 Construction

A simple conventional solar cell (first gen Si cell) structure is depicted in Fig.2-90. When an N-type semiconductor and a P-type semiconductor are fabricated in a single Si wafer, a PN junction is developed. This is typically achieved through diffusion or implantation of specific impurities (dopants) or via a deposition process.

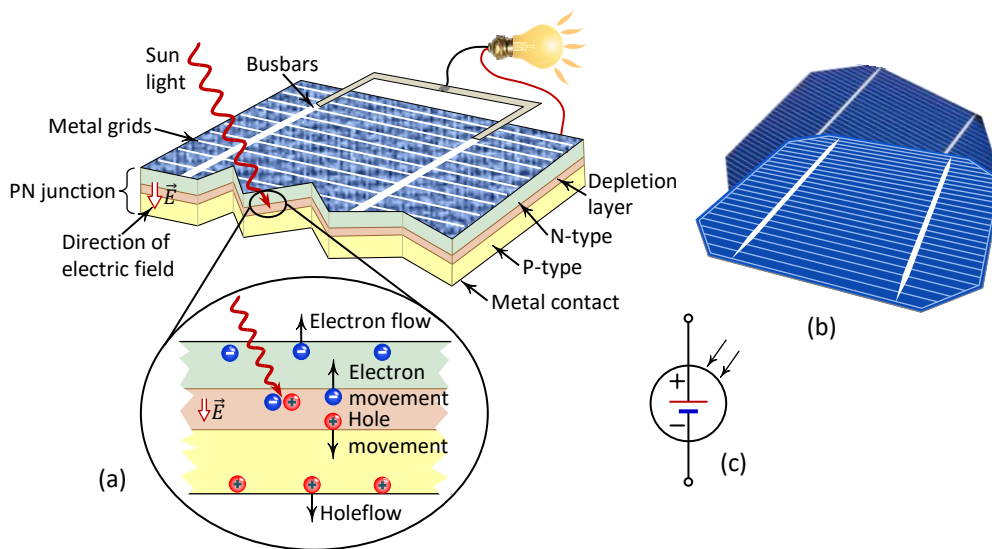


Fig.2-90: Si solar cell: (a) Construction, (b) Photographs, and (c) Symbol

A metallic grids arrangement with two or three bus bars on top of the cell forms the -ve electrical contact. Another contact is the metallic layer on the back side of the cell. To minimize the reflection loss, an antireflective coating (metal oxide mainly TiO_2) is produced on top of the cell. For this purpose, texturing (pyramid-like structures) is also produced on this side. Like the normal PN junction, a depletion layer with an electric field is produced in this device. This electric field is the key factor for its operation.

2.42.2 Operation

Sunlight is incident on the front face (N-type layer) of the solar cell. The photons of light collide with electrons and produce **hoe-electron pairs**. These hole-electrons are separated by the electric field of the depletion layer of the cell and develop the electric current. Only photons with sufficient energy $h\nu > E_g$, can generate hole-electron pairs, that is, photons with energy greater than the semiconductor band gap (E_g) will contribute to the energy conversion process. Thus, the spectral nature of sunlight is an important consideration in the design of efficient solar cells. Due to the electric field, the holes move to the P-type layer (backward) and the electrons move to the N-type layer (upward). When these charges accumulate on the P and N layers, a potential difference is produced. Therefore, if a load is connected across the cell, electrons flows through the load and then to the P side and the conventional current to the opposite direction. This flow of current produces power in the load. The potential difference produced in a solar cell is approximately equal to the barrier voltage of the PN junction. Hence, it depends on the energy pap (E_g) of the material.

2.42.3 Solar Cell I – V Characteristics

Solar cell is actually a current source. Its output current depends on the intensity of incident solar radiation. The higher the incident solar radiation, the larger is its output current. For a particular solar radiation, the variation of output voltage and current is shown in the I - V characteristics in Fig.2-91(a). From this graph, it is clear that, with the increase in output current the output voltage decreases very slowly up to a current level. But after this level, the output voltage decreases very rapidly with very small increase in current. If the output terminal is open, i.e., $I = 0$, the output voltage of the cell is maximum and it is called the **Open Circuit Voltage** (V_{OC}). On the other hand, if the output terminals of the cell are shorted, that is, for the maximum current the output voltage will be zero. This maximum current is called the **Short Circuit Current** (I_{SC}). At the short circuit and the open circuit condition the output power of the cell will be zero. At the knee point of the curve, there is a point where the product of current and voltage is the maximum. This point of the curve is called the **Maximum Power Point** (MPP).

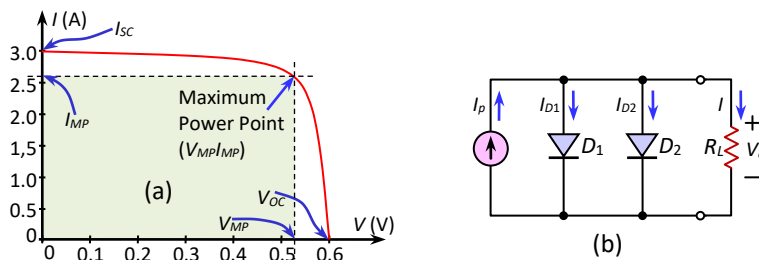


Fig.2-91: (a) Characteristics, (b) Equivalent circuit Construction and

The equivalent circuit of the solar cell is shown in Fig.2-91(b). Here, D_1 is an ideal diode with $n = 1$, and D_2 is a practical diode with $n = 2$ in the Shockley's equation $[I_D = I_S(e^{qV/nkT} - 1)]$. The rest of

the parameters are same as described for PN junction diodes. From this equivalent circuit the value of the output current (I) can be calculated as,

$$I = I_p - I_{D1} - I_{D2}$$

or,

$$I = I_p - I_{o1}(e^{qV/kT} - 1) - I_{o2}(e^{qV/2kT} - 1) \quad (2-56)$$

The quality of Si solar cell is measured from various parameters. One important parameter is the **Fill Factor** (FF). It is defined as the power of the solar cell at the maximum power point to the product of V_{OC} and I_{SC} . Mathematically,

$$FF = \frac{P_{MP}}{I_{SC}V_{OC}} = \frac{V_{MP}I_{MP}}{I_{SC}V_{OC}} = \frac{P_{out}}{I_{SC}V_{OC}} \quad (2-57)$$

The value of FF is 1 for ideal solar cell but for practical solar cell it may vary from 0.5 to 0.9. Another important parameter is the **efficiency** of the solar cell. It is defined as the ratio of the output power (at MPP) to input power. The output power at the MPP is $V_{MP}I_{MP}$ and input power on a cell is insolation ($Watt/m^2$) times the area of the cell in meter. Mathematically,

$$\eta = \frac{P_{out}}{P_{in}} = \frac{V_{MP}I_{MP}}{I_{rad} \times A} \quad (2-58)$$

Here, I_{rad} is the solar radiation in W/m^2 and A is the area of the cell in m^2 . There are various types of solar cells. The Si solar cell may be of three types. These are mono-crystalline, poly-crystalline and amorphous. Presently, the maximum efficiency of these three types of Si solar cells are 25 %, 18 %, and 13 %, respectively.

Sample Questions:

1. What do you mean by an electrical circuit and an electronic circuit?
2. Explain the Bohr's atom model.
3. Calculate the energy of an electron in a single hydrogen atom.
4. What is eV? Convert 1 eV into Joule.
5. What are valence electrons? How these electrons control the electrical properties of materials?
6. What is energy band? How energy band is developed in a material?
7. Explain conductor, semiconductor and insulator in terms of energy band diagrams.
8. Give the classifications of semiconductor materials.
9. Explain how N-type semiconductor is made.
10. Explain how P-type semiconductor is made.
11. What do you mean by drift and diffusion of charge carriers?
12. Explain hole-movement mechanism.
13. What is PN junction? Explain how depletion layer is developed in a PN junction.
14. Explain how the barrier voltage is developed in a PN junction.
15. Why does the majority carrier diffusion not continue to produce equal carrier concentration in both sides of a PN junction?
16. Explain the four current components of a PN junction in equilibrium. Why the net current is zero?

17. Draw the I - V characteristic curve of a diode and explain. What is knee voltage?
18. What is thermal voltage? Calculate the value of thermal voltage at $T = 300$ K.
19. Explain the effect of temperature on the performance of a PN junction diode.
20. Explain different resistances of a PN junction diode.
21. What is surface leakage current? Why this current is produced? Explain the reason.
22. Explain transition and diffusion capacitance.
23. What are the properties of an ideal diode? Explain with characteristic curve.
24. Draw the high frequency equivalent circuit of a PN junction diode and explain.
25. Explain with suitable diagrams the different equivalent circuits of a diode.
26. What is reverse recovery time of a diode? How this time limits the frequency of switching of a diode?
27. Give a comparison between a silicon diode and a germanium diode.
28. In diode datasheets values of different parameters are given. Write the names of 5 important parameters and explain their meanings.
29. What is Zener diode? Explain Zener and avalanche breakdown mechanisms.
30. Draw the characteristic curve of a Zener diode and explain. What is dynamic impedance of a Zener diode and how this parameter affects the operation of a Zener diode as a constant voltage source?
31. Explain the construction and operation of an LED. Why the blue LED needs very high value of forward bias voltage?
32. Draw the exact and approximate equivalent circuit of a reverse biased Zener diode and explain.
33. Explain the parameters: V_Z , I_{ZK} , I_{ZM} , P_D in the context of a Zener diode.
34. Write the names of some materials used for LED fabrication with their colors and bias voltages.
35. Describe the construction and operation of a PN junction LED.
36. Describe the construction and operation of an OLED.
37. Mention some applications of LEDs and OLEDs.
38. What is a varactor diode? Describe the construction and operation of a varactor diode.
39. Draw the diagram of a tuning circuit using a varactor diode and explain.
40. What are the applications of varactor diodes.
41. What is Schottky diode? Describe their construction and operation principle.
42. What are the uses of Schottky diodes?
43. What is step recovery diode? Explain the construction and operation of a step recovery diode.
44. Write some applications of step recovery diodes.
45. What is small signal diode? Explain the construction and operation of a small signal diode.
46. What is point contact diode? Explain the construction and operation of a point contact diode.
47. What is transient voltage suppression diode? Describe their construction and operation.
48. With the help of I - V characteristics, explain the operation of a tunnel diode.
49. Draw the equivalent circuit of a tunnel diode and explain.
50. Explain with diagram, how undamped oscillations can be produced in a resonant circuit using tunnel diode.

51. What is photo diode? With suitable diagram describe the construction and operation process of a photo diode. Draw the characteristic curves of this diode.
52. With suitable circuit diagrams, explain different modes of operations of a photo diode.
53. What is PIN diode? Why is the name PIN? Describe its construction and operation process of a PIN diode.
54. Write some important properties of a PIN diode. What are the applications of PIN diodes?
55. What is Gunn diode? Describe the construction and operation of a Gunn diode.
56. Draw the I - V characteristics of a Gunn diode and explain.
57. Explain with diagram the domain formation mechanism in a Gunn diode.
58. Write some applications of Gunn diodes.
59. What is Shockley diode? Why is it called a diode? Describe its construction and operation procedure.
60. Using two-transistor analogy explain how the negative resistance is developed in a Shockley diode.
61. What is LCD (liquid crystal display)? Give the classifications of LCDs.
62. Describe the construction and operation of field effect type LCD.
63. Describe the construction and operation of dynamic scattering type LCD.
64. What is solar cell? Explain the construction and operation of a solar cell.
65. Explain the carrier separation mechanism in a solar cell.
66. Draw the I - V characteristics of a solar cell and explain.
67. Define the terms: V_{OC} , I_{SC} , FF , efficiency and maximum power point of a solar cell.
68. Draw the equivalent circuit of a solar cell and derive the equation for load current.

Exercise

1. The electron in a hydrogen atom has a kinetic energy of 13.6 eV. Calculate the potential energy and the net energy of the electron. [Ans. $PE = -27.2$ eV, $E_n = -13.6$ eV]
2. The hydrogen atom has only one electron. Its potential energy is -27.2 eV. Calculate the kinetic energy and the ground state energy of the hydrogen atom. [Ans. $KE = -13.6$ eV, $E_n = -13.6$ eV]
3. In a situation, an electron acquires a speed of 10^5 ms^{-1} . Calculate its kinetic energy in Joules and eV unit at that situation. [Ans. 4.55×10^{-21}], 2.84 eV]
4. Calculate the intrinsic carrier concentration in Si at a temperature of 27°C . [Ans. $1.5 \times 10^{10} \text{ cm}^{-3}$]
5. Calculate the intrinsic carrier concentration in Ge at a temperature of 27°C . [Ans. $24.03 \times 10^{12} \text{ cm}^{-3}$]
6. A Si crystal is doped by arsenic (As) with a doping concentration of 2 ppm. If the crystal contains $5 \times 10^{28} \text{ m}^{-3}$ atoms and $n_i = 1.5 \times 10^{16} \text{ m}^{-3}$. Calculate the number of electrons and holes in the crystal. [Ans. $n_e = 10^{23} \text{ m}^{-3}$, $n_h = 2.25 \times 10^9 \text{ m}^{-3}$]
7. At 300 K temperature, a sample of intrinsic silicon has electron density of $1.5 \times 10^{10} \text{ cm}^{-3}$ and hole density of $1.5 \times 10^{10} \text{ cm}^{-3}$. One of every 2×10^6 atoms of this sample is replaced by a phosphorous atom. Assuming all the donor electrons are in the conduction band, calculate the ratio of the charge carrier density in the doped silicon to that for the intrinsic silicon. Given: density of silicon, $\rho = 2.3290 \text{ g/cm}^3$, Avogadro's number, $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$, and Molar mass of Si, $M = 28.09 \text{ g/mol}$. [Ans. $n_{pe}/2n_i = 1.66373$]
8. In a P-type silicon, the majority hole density is $5 \times 10^{21} \text{ holes/m}^3$. The intrinsic electron density of silicon is $n_i = 1.5 \times 10^{16} \text{ electrons/m}^3$. Find the (minority carrier) electron density

of the P-type silicon sample. Also calculate the ratio of minority carrier and majority carriers. [Ans. $n_e = 4.5910^{10}$ electrons/m³, $n_e:n_h = 1:(1.852 \times 10^{-11})$]

9. In a process of N-type germanium preparation, pentavalent impurity atoms have been mixed in such a level that its electron density becomes 8.5×10^{22} electrons/m³. The intrinsic electron density of germanium is $n_i = 2.4 \times 10^{19}$ electrons/m³. Find the (minority carrier) hole density of the developed P-type silicon sample. Also calculate the ratio of minority carrier and majority carriers. [Ans.]
10. Calculate the values of conductivity (σ) and resistivity (ρ) of an intrinsic silicon semiconductor material.
11. Calculate the values of conductivity (σ) and resistivity (ρ) of an intrinsic germanium semiconductor material
12. Calculate the diffusion current density for Si semiconductor at $T = 300$ K. Assume the electron concentration varies linearly from $n_e = 10^{10}$ cm⁻³ to $n_e = 10^{15}$ cm⁻³ over a distance from $x = 0$ to $x = 5$ μ m. Assume that the diffusion constant for electrons is $D_e = 30$ cm²/s. [Ans. $J_e = 9.6$ A/cm²]
13. Calculate the diffusion current density for Ge semiconductor at $T = 300$ K. Assume the electron concentration varies linearly from $n_e = 10^{10}$ cm⁻³ to $n_e = 10^{15}$ cm⁻³ over a distance from $x = 0$ to $x = 5$ μ m. Assume that the diffusion constant for electrons is $D_e = 30$ cm²/s. [Ans. $J_e =$ A/cm²]
14. In a circuit, charge is changing with time according to $Q(t) = (0.05 \text{ C}) \times [\sin(2\pi 100/\text{s} \cdot t)]$. Calculate the instantaneous current at $t = 0$ s, and $t = 100$ s. [Ans. 31.416 A and -30.746 A]
15. Suppose a silicon bar is made using the P-type silicon sample of Ex.7. If the motilities of electrons and holes are $\mu_e = 0.14$ m²/(V · s) and $\mu_h = 0.05$ m²/(V · s), respectively, calculate the conductivity and resistivity of the silicon bar. [Ans.]
16. Calculate the potential barrier of a silicon PN junction, at junction temperature $T = 300$ K. Consider that the doping concentration in P-type region is $N_a = 10^{15}$ cm⁻³ and that in N-type region is $N_d = 10^{16}$ cm⁻³ and the intrinsic carrier concentration in silicon is $n_i = 1.5 \times 10^{10}$ cm⁻³. [Ans.]
17. Calculate the potential barrier of a germanium PN junction, at junction temperature $T = 300$ K. Consider that the doping concentration in P-type region is $N_a = 10^{16}$ cm⁻³ and that in N-type region is $N_d = 10^{17}$ cm⁻³ and the intrinsic carrier concentration in silicon is $n_i = 2.4 \times 10^{19}$. [Ans.]
18. APN junction diode is operated at a temperature of $T = 300$ K. Assuming $I_s = 10^{-12}$ A, and $n = 1$, determine the diode current for i) $V_D = +0.72$ V and ii) $V_D = -0.72$ V. [Ans. 0.963 A, -10^{-12} A]
19. Determine the diode voltage (V_D) and current (I_D) for the circuit shown in Fig. 2-35. Consider a diode with a given reverse-saturation current of $I_s = 10^{-13}$ A. [Ans. 0.63633 V, 4.26 mA]

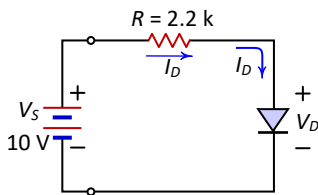
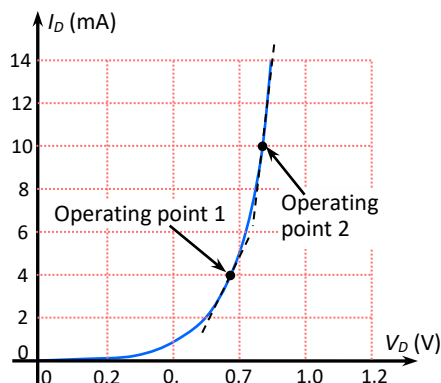
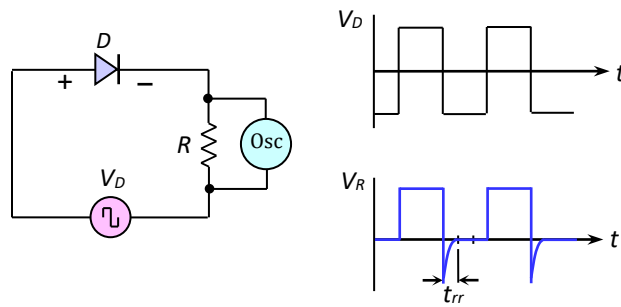


Fig.2-: Circuit for Ex.16

Fig.2-43: Two operating points on I - V curve for Ex.20

20. At ambient temperature of 25°C , the barrier voltage of a Si diode is 0.75 V . What will be barrier-voltage when the junction temperature is i) 100°C , and ii) 20°C ? [Ans. i) 0.5625 V , ii) 0.7625 V]
21. A silicon diode has a saturation current of 7 nA at an ambient temperature of 25°C . What will be the value of the saturation current at 100°C ? [Ans. $1.257\text{ }\mu\text{A}$]
22. Given a Si diode with a reverse current of $5\text{ }\mu\text{A}$ at 25°C and $80\text{ }\mu\text{A}$ at 100°C , calculate the surface leakage current. [Ans.]
23. If the surface-leakage current is 2 nA for a reverse voltage of 20 V , what is the surface-leakage current for a reverse voltage of 30 V , and 80 V ? [Ans. 80 nA]
24. As shown in the following Fig.2-, by applying a DC voltage a DC operating point is set for a diode circuit. When an AC signal is applied in the circuit the operating point oscillated between, P_h and P_L . Calculate the average AC resistance faced by the AC signal. Also calculate the DC resistance and AC resistance using Eq.(1.17 for the three operating points. Assume, $r_B = 2\text{ }\Omega$.
25. Consider a silicon PN junction at $T = 300\text{ K}$, with doping concentrations of $N_a = 10^{16}\text{ cm}^{-3}$ and $N_d = 2 \times 10^{15}\text{ cm}^{-3}$. Assume that $n_i = 1.5 \times 10^{10}\text{ cm}^{-3}$ and the unbiased junction capacitance $C_{T0} = 1.0\text{ pF}$. Calculate the junction capacitance at $V_D = -1\text{ V}$ and $V_R = -10\text{ V}$. [Ans. 2-16 of book]
26. As shown in the following Figure, a square wave voltage V_D is applied to a diode circuit. The voltage developed across the resistor R is observed on an oscilloscope. If the value of t_{rr} is $1\text{ }\mu\text{s}$, calculate the frequency of the applied waveform.



27. As shown in Table 2-5, the Zener diode 1N4732A has a typical value of $V_Z = 4.7\text{ V}$, and $Z_Z = 08\ \Omega$. If 12 mA of current flows through this diode, calculate the output voltage i) Using the model of Fig.2-55(b), and ii) Using the model of Fig.2-55(c). iii) Calculate the power dissipated by the Zener diode for both the cases.
28. A solar cell with $FF = 0.9$, produces $V_{OC} = 0.65\text{ V}$ and $I_{SC} = 4.2\text{ A}$ when illuminated by solar radiation of 1000 W/m^2 . If the area of the cell is $10\text{ cm} \times 10\text{ cm}$, calculate the efficiency of the cell. [Ans. $\eta = 18.72\%$]